Aggregate Outcomes of Nonlinear Prices in Supply Chains*

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Abstract

We develop a general equilibrium model of nonlinear pricing in supply chains with roundabout production: sellers offer price-quantity schedules to price discriminate across heterogeneous buyers. Under CES demand and Pareto-distributed productivity, the optimal contract is a buyer-sector-specific two-part tariff. Sellers collect margins through allocative markups, which distort quantities, and flat fees, which distort entry. In the universe of Chilean firm-to-firm transactions, unit prices fall with quantity in a convex schedule, consistent with the model. Despite lower entry, nonlinear pricing closes over a third of the welfare gap between uniform pricing and efficiency by lowering allocative markups that compound along supply chains. Relative to uniform pricing, optimal policy shifts from revenue toward entry subsidies, and policies designed for uniform pricing forgo 5% of welfare. The shift and the loss grow as buyers become more homogeneous, because sellers collect more of their margins through flat fees.

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1 Introduction

Price discrimination has long been recognized as pervasive in firms' pricing strategies: "one of the most prevalent forms of marketing practices" (Varian, 1989). Yet its economy-wide extent and its aggregate welfare and policy implications remain largely unexplored in supply chains, where firm-to-firm transactions far exceed final sales in value and distortions such as markups propagate and compound (Jones, 2011).

In this paper, we develop a tractable general equilibrium framework to study the aggregate welfare and policy implications of price discrimination in supply chains, through both nonlinear pricing (second-degree) and buyer-sector-specific pricing (third-degree). Firms are linked through arbitrary input–output linkages, and firms both set and face nonlinear prices. Sellers can condition contracts on their buyers' sector but not on their productivity. Under two assumptions, CES demand and Pareto-distributed firm productivities, the optimal contract is a single buyer-sector-specific two-part tariff: a flat fee and a marginal price. Moreover, because each seller's revenue inherits the functional form of its buyers' revenue, the model remains tractable even with roundabout production.

A buyer-sector-specific two-part tariff implies that average unit prices fall with the quantity purchased in a convex schedule that differs across buyer sectors. We find these patterns in the universe of Chilean firm-to-firm trade, nearly 1.5 billion transactions in 2024. Within seller-product-day cells, unit prices decline with quantity and flatten at large quantities, consistent with second-degree price discrimination, and the schedule differs across buyer sectors, mainly for small purchases, consistent with third-degree price discrimination.

Nonlinear pricing changes how market power distorts the economy relative to uniform pricing. Under uniform pricing, each seller charges all its buyers the same price per unit, so it can extract surplus only by raising the price of every unit, which distorts quantities traded. Nonlinear pricing gives the seller a second instrument, the flat fee of the two-part tariff, that extracts surplus without affecting buyers' marginal choices. This changes the distortions arising from market power along two margins. On the intensive margin, markups on marginal prices that govern quantities, which we call allocative markups, are lower than under uniform pricing. This limits the accumulation of markups along the supply chain and increases welfare by reducing deadweight losses. On the extensive margin, flat fees redistribute profits along the supply chain and distort firm entry, with an ambiguous effect on welfare. The net effect of nonlinear pricing is ultimately a quantitative question.

We calibrate the model to Chilean data and obtain three results. First, nonlinear pricing raises welfare relative to uniform pricing. Banning price discrimination, so that firms charge a single price per unit, lowers welfare from 84% to 75% of the efficient level: allocative markups rise and their deadweight losses compound along the supply chain, more than offsetting the gains in entry from eliminating flat fees. Nonlinear pricing thus closes more than a third of the gap between

uniform pricing and the efficient allocation. Second, nonlinear pricing reshapes the optimal policy mix of uniform revenue and entry subsidies. Relative to the optimum under uniform pricing, the optimal revenue subsidy falls from 43% to 29% because allocative markups are lower, while the optimal entry subsidy rises from 2% to 17%, because flat fees distort entry. Applying the uniform-pricing optimum to the nonlinear-pricing economy costs about 5 percentage points of efficient welfare.

Third, how far prices depart from uniform pricing, and hence our results, depends on how heterogeneous buyers are, because heterogeneity limits how much surplus sellers can extract through a common flat fee. When buyers differ widely in scale, sellers rely on marginal prices, and market power mainly distorts quantities. As buyers become more homogeneous, marginal prices approach marginal cost and sellers collect their margin through flat fees, which increasingly tax buyers' profits and distort entry. The optimal policy mix therefore shifts from the revenue subsidy toward the entry subsidy, consistently attaining 97–99% of efficient welfare, while the cost of applying the uniform-pricing prescription grows, until it leaves welfare below its no-policy level. Without policy, welfare relative to the efficient allocation is hump-shaped in buyer homogeneity: unlike in partial equilibrium, marginal prices closer to marginal cost do not always raise welfare, as flat fees distort entry.

We start from the canonical screening problem in partial equilibrium, in which a monopolist with constant marginal cost faces heterogeneous buyers whose types it does not observe and offers a menu of quantities and transfers.[1] When buyer productivity is Pareto distributed, the optimal schedule is a single two-part tariff: a flat fee and a constant marginal price.[2] The marginal price governs allocations, while the flat fee extracts rents (Wilson, 1993; Armstrong, 1996). Moreover, when buyers' revenue functions are homogeneous in quantity, the seller's revenue inherits the same functional form as its buyers', which is what allows us to embed nonlinear pricing in supply chains where firms both charge and pay nonlinear prices.

We examine whether average unit prices fall with quantity in a convex schedule and converge to the marginal price, using the population of formal firm-to-firm transactions in Chile. For each product transacted, we observe the unit price and quantity, together with the identities of the buyer and the seller and the timestamp of the trade. Within seller-product-day cells, unit prices fall with quantity: the largest transactions in quantity get a 30% discount per unit relative to the smallest quantity traded, and the decline flattens at larger quantities. This decreasing and convex schedule characterizes sectors accounting for 93% of transactions.

This schedule appears across seller industries and varies with the buyer's industry, mainly for small purchases. This evidence is consistent with a combined pricing scheme: price-quantity

[1] See Mussa and Rosen, 1978; Maskin and Riley, 1984; Tirole, 1988, ch. 3. Because buyers in our setting are firms, the buyer's gross utility in the canonical framework corresponds to its gross revenue.

[2] A similar result also appears in Bergemann et al. (2023). More generally, any concave nonlinear tariff can be implemented as a menu of two-part tariffs, in which options with a higher flat fee carry a lower marginal price (Tirole, 1988, ch. 3).

menus within buyer sectors (second-degree price discrimination) that differ across buyer sectors (third-degree). Quantity discounts strengthen when we absorb persistent buyer-specific prices, including those arising from buyer market power, and survive proxies for bargaining, repeated relationships, and fixed per-order costs. We do not target these discounts in the calibration of the model developed below. Disciplined by profit shares, input-output transactions, and the upper tail of the firm-size distribution, the model's average discount is of similar magnitude as the data's (28% in the model against 31% in the data).

Guided by the partial-equilibrium result and the evidence, we develop a multisector general equilibrium model with arbitrary input–output linkages, in which firms both charge and pay nonlinear prices. Productivity is Pareto distributed within each sector, and sellers observe their buyers' sector but not their productivity, so they combine second-degree price discrimination within sectors with third-degree price discrimination across them. Preferences and technology generate CES demand and constant returns to scale, and firms are atomistic, so each seller takes aggregate prices and its buyers' marginal valuations as given. Under these assumptions, the optimal contract remains a single buyer-sector-specific two-part tariff, and each seller's revenue inherits the functional form of its buyers' revenue. This recursivity closes the model along the supply chain, even with roundabout production. Conditional on this contract structure, we prove the steady-state equilibrium exists and is unique under parameter restrictions satisfied in our calibration.

We compare the nonlinear-pricing equilibrium with two benchmarks: a ban on price discrimination, under which each seller charges a single quantity-invariant price to all its buyers and earns the standard CES markup, and the efficient allocation, which combines the ban with sector-specific output subsidies that restore marginal-cost pricing conditional on entry. Because demand is CES, uniform pricing distorts the economy only through markups that compound along the supply chain, so output subsidies alone implement the efficient allocation, including entry (Baqaee and Farhi, 2020a). Meanwhile, nonlinear pricing distorts the economy along two margins: allocative markups on marginal prices (intensive) and entry through flat fees (extensive).

The model summarizes how nonlinear pricing differs from uniform pricing in a single statistic, the screening elasticity ξ, which measures how homogeneous buyers are in scale: it combines the dispersion of productivity in the buyer's sector with the substitutability of the buyer's own output, which maps productivity differences into differences in scale. For $\xi > 1$, the screening elasticity determines both how far allocative markups fall below the uniform-pricing markup and how much of each seller's margin is collected through flat fees.

The allocative markup takes the CES form, but evaluated at an effective elasticity $\sigma\xi$. The seller's elasticity of substitution σ measures how easily buyers substitute across varieties of the seller's sector, which is the only determinant of markups under uniform pricing. Because $\xi > 1$, the allocative markup is always below the uniform-pricing markup.[3] Since the screening elasticity

[3]The condition $\xi > 1$, standard in models with Pareto-distributed firm size, ensures that the employment distribution has a finite mean, so that the labor market can clear.

belongs to the buyer's sector, the same seller charges different marginal prices to different buyer sectors, so allocative markups are defined by pairs of seller and buyer sectors rather than by sellers alone, and are higher for buyer sectors that are more dispersed in scale.

Nonlinear pricing leaves the margin a seller earns on its sales unchanged at the usual CES share $1/\sigma$ of revenue. The seller's elasticity of substitution determines how large the margin is, and the screening elasticity determines how it is collected: the more homogeneous buyers are, the larger the share collected through the flat fee rather than the marginal price. As buyers become more homogeneous, information rents shrink, so sellers also capture a larger share of the surplus each transaction creates.

Flat fees are inframarginal, so they do not affect input choices, but they are not neutral in general equilibrium. A firm's profits fall short of its CES margin by the flat fees it pays to its own suppliers, so flat fees act as a tax on entry whose size varies across sectors with the importance of intermediate inputs. This distortion operates through profits rather than marginal prices, so output subsidies, which restore efficiency under uniform pricing, cannot undo it.

We calibrate the model to Chilean firm-level balance sheets and firm-to-firm transactions. Elasticities of substitution are disciplined by profit shares and input–output elasticities by firm-to-firm payments, both corrected for the flat fees implied by the model; screening elasticities by the concentration of employment among the largest firms. We then use the calibrated model to answer three questions.

Our first question is whether nonlinear pricing raises welfare relative to uniform pricing, the equilibrium under a ban on price discrimination, and through which channels. A ban lowers welfare from 84% to 75% of the efficient level: relative to uniform pricing, price discrimination reduces the aggregate welfare cost of market power, closing more than a third of the gap, a share that is stable across robustness exercises.[4] The gain comes from lower marginal prices, which shrink the deadweight loss on every transaction and its compounding along the supply chain. The variety margin offsets about a third of this gain, mainly through lower retail entry: retailers pay fees out of an unchanged gross margin, so fewer of them break even. Almost the entire remaining gap to efficiency arises on firm-to-firm transactions: removing markups between firms alone raises welfare to 99% of the efficient level.

Our second question is how a planner who cannot ban price discrimination should set uniform revenue and entry subsidies. Under uniform pricing, markups act as a tax on production along the supply chain, so the optimal policy is a large revenue subsidy (Edmond et al., 2023); with CES demand, correcting markups also restores efficient entry, so the entry subsidy is close to zero (Baqaee and Farhi, 2020a). Under nonlinear pricing, the revenue subsidy falls, because marginal prices are already closer to marginal cost and part of the subsidy is paid on flat fees, which raises its fiscal cost without lowering marginal prices. The entry subsidy rises, because flat fees tax

[4] In log welfare, the share ranges from 35% to 43% (Appendix E).

entry, a distortion the revenue subsidy cannot correct. Applying the uniform-pricing optimum to the nonlinear-pricing economy therefore errs on both instruments, at a cost of about 5 percentage points of efficient welfare: its revenue subsidy pushes marginal prices below marginal cost for two-thirds of markup exposure, and its entry subsidy is too small. Somewhat larger revenue subsidies leave welfare below its no-policy level, whereas under uniform pricing they would still raise it.

Our third question is how these results depend on buyer heterogeneity, which we vary through the screening elasticity ξ while holding all other parameters fixed.[5] The cost of applying the uniform-pricing optimum rises steeply as buyers become more homogeneous, eventually leaving welfare below its no-policy level, whereas a planner who accounts for nonlinear pricing attains 97–99% of efficient welfare across the whole range. Because buyer heterogeneity can be measured from the concentration of employment among the largest firms, policymakers can assess how much the pricing assumption matters for policy design. Welfare without policy, relative to the efficient allocation, peaks at an intermediate degree of buyer homogeneity, above our calibrated value. In partial equilibrium a weaker information friction makes screening less distortionary, but with free entry along a supply chain weaker information friction eventually lowers welfare, because flat fees collect a growing share of sellers' margins and tax entry more heavily.

Taken together, our results show that the welfare cost of market power in supply chains depends not only on how large sellers' margins are, but also on how sellers collect them. Treating firm-to-firm prices as uniform misreads how market power distorts the economy and the policies that correct it. Margins collected through marginal prices distort quantities and call for revenue subsidies, while margins collected through flat fees distort entry and call for entry subsidies.

Related Literature. We extend the classic screening problem with one seller and many buyers to a general equilibrium economy with arbitrary input–output linkages, in which firms both set and face nonlinear prices. We relate to three strands of literature. The first is the literature on price discrimination and screening. Our model builds on classic results (Dupuit, 1844; Mussa and Rosen, 1978; Maskin and Riley, 1984; Wilson, 1993) and on surveys of price discrimination under imperfect competition (Varian, 1989; Stole, 2007). Empirical work in industrial organization estimates nonlinear tariffs and the markups they imply in specific markets (Miravete and Röller, 2004; Miravete, 2005). Recent work documents input price dispersion across buyers and its allocative consequences (Burstein et al., 2024). In partial equilibrium the welfare effects of third degree price discrimination are ambiguous and a necessary condition for welfare to rise is that output increases (Schmalensee, 1981; Varian, 1985; Aguirre et al., 2010). In general equilibrium with free entry, flat fees add a further channel. Meleshchuk (2018) embeds two-part tariffs in a trade model with a sin-

[5]Under uniform pricing, ξ affects neither welfare relative to the efficient allocation nor the optimal policy: with CES demand and atomistic firms, productivity dispersion scales welfare in the decentralized and efficient allocations by the same factor. Both relative welfare and the optimal policy under uniform pricing are therefore invariant to ξ.

gle vertical layer, in which intermediate producers screen final-good producers; there, fees reduce the number of final-good producers that operate and induce excess entry of intermediate producers, raising optimal input taxes. We show that in supply chains with roundabout production, where firms both pay and charge nonlinear prices, flat fees act as a tax on entry whose size varies with each sector's reliance on intermediate inputs, while allocative markups compound along the supply chain.

Our closest predecessor is Bornstein and Peter (2025), who bring nonlinear pricing into macroeconomic analysis in general equilibrium. They show that, for firms selling to heterogeneous final consumers, markup heterogeneity across firms need not reflect misallocation under nonlinear pricing, which instead arises across consumers within firms. We build on their insight that average prices are a poor guide to the distortions of market power, but study price discrimination between firms, which adds buyer-sector-specific pricing to quantity discounts, and in which firms both charge and pay nonlinear prices along supply chains with roundabout production. Because buyers within a sector face a common marginal price, distortions across buyers of the same seller arise across buyer sectors rather than within them. Beyond these, markups compound along the supply chain and flat fees tax entry.[6] Empirically, whereas they use retail micro data, we document quantity-dependent and buyer-sector-specific schedules in the population of firm-to-firm invoices.

The second is the literature on production networks and the aggregate effects of distortions, which studies how wedges propagate and compound through input–output linkages (Jones, 2011; Carvalho and Tahbaz-Salehi, 2019; Bigio and La'o, 2020; Baqaee and Farhi, 2020b), and how markups shape optimal entry (Baqaee and Farhi, 2020a). Most of these frameworks treat the markup on each link as a single price wedge that both governs allocations and determines profits, so markup dispersion, across firms or across buyers of the same product, is read as misallocation (Hsieh and Klenow, 2009; Baqaee and Farhi, 2020b; Afrouzi et al., 2023; Burstein et al., 2024). Under nonlinear pricing, market power splits into an allocative markup on marginal prices, which propagates through supply chains, and flat fees, which do not distort input choices but redistribute profits along the supply chain and distort entry. Markups measured from average unit prices are then dispersed even among buyers that face the same marginal price, and only allocative markups distorts quantities traded between firms. On the other hand, inframarginal markups reshape firms profit functions with general equilibrium effects on entry. Our welfare accounting builds on Baqaee and Farhi (2020b).

The third is the literature on the aggregate welfare costs of market power, starting from Harberger (1954) and revived by the evidence on rising markups and profits (Hall, 2018; Autor et al.,

[6]Both distortions require buyers to be firms. If the two-part tariff in our model were applied to sales to final consumers, with inelastic labor supply and a common screening elasticity and substitution elasticities across sectors, it would implement the efficient allocation: a common marginal markup leaves relative prices undistorted, and because the seller's margin remains the CES share $1/\sigma$ of revenue and flat fees are paid by consumers rather than firms, entry is efficient.

2020; Barkai, 2020; De Loecker et al., 2020). In heterogeneous-firm uniform pricing models, markups entail large welfare costs, which can be largely offset by uniform subsidies (Edmond et al., 2023). Under CES demand, the market level of entry is efficient in a one-sector economy (Dixit and Stiglitz, 1977), and departures from CES generate excess or insufficient entry (Mankiw and Whinston, 1986; Dhingra and Morrow, 2019). We show that price discrimination between firms distorts entry even under CES demand and changes both the size of the cost of market power and how to correct them. Because sellers extract surplus through flat fees, they distort quantities less, which reduces the deadweight losses that compound along the supply chain. But flat fees introduce a new distortion on entry, which shifts the optimal policy mix away from revenue subsidies and toward entry subsidies.

2 Optimal Nonlinear Price Characterization in Partial Equilibrium

We present a framework of optimal nonlinear pricing in partial equilibrium, deriving a result that serves as the building block for a general equilibrium supply chain model, in which firms charge and pay nonlinear prices. We consider a standard screening problem (e.g., Spence, 1977; Maskin and Riley, 1984; Laffont and Tirole, 1993; Tirole, 1988) and derive results under two additional assumptions: Pareto-distributed buyer types and isoelastic buyer revenue functions.

A seller with constant marginal cost $c > 0$ faces a continuum of buyers with privately observed productivity types $z \in [\underline{z}, \infty)$, drawn from $\mathbb{F}(z)$ with density $f(z)$. The seller chooses a direct mechanism, i.e., measurable functions $q : [\underline{z}, \infty) \to \mathbb{R}_+$ and $T : [\underline{z}, \infty) \to \mathbb{R}$, assigning to each reported type z a quantity $q(z)$ and a transfer $T(z)$. Under nonlinear pricing, the transfer $T(z)$ need not equal price times quantity, as under uniform pricing, where it would be $p\,q(z)$, and may include, for example, a flat fee and a per-unit component. Given quantity q, a buyer of type z generates gross revenue $zR(q)$ and obtains net surplus $\Pi(z) = zR(q(z)) - T(z)$.[7] The seller's problem is

$$\begin{aligned} \max_{(q,T)} \ \Pi_{\text{seller}} &= \int_{\underline{z}}^{\infty} \Big[T(z) - c\,q(z)\Big] f(z)\,dz \qquad (1) \\ \text{s.t.} \quad (IR) \quad & \Pi(z) \equiv zR\Big(q(z)\Big) - T(z) \geq 0, \\ (IC) \quad & \Pi(z) \geq zR\Big(q(\tilde{z})\Big) - T(\tilde{z}), \qquad \forall\, z, \tilde{z} \in [\underline{z}, \infty) \end{aligned}$$

Individual rationality (IR) and incentive compatibility (IC) ensure participation and truthful revelation. We normalize the outside option to zero, so IR is $\Pi(z) \geq 0$. In the optimum, the lowest-type IR binds ($\Pi(\underline{z}) = 0$). Under standard single-crossing and concavity conditions, as shown in

[7] In the canonical screening framework, $zR(q)$ is the buyer's gross utility, while $zR(q) - T$ is the buyer's net utility. In our application, buyers are firms, so we interpret $zR(q)$ as gross revenue and $zR(q) - T$ as profit net of the payment to the seller. As in Mussa and Rosen (1978), q can be read as quality-adjusted quantity; in the empirical part of the paper we document screening along the quantity dimension because it is directly observed in invoices.

Appendix A, the seller's problem can be rewritten as a pointwise optimization[8]:

$$\max_{\{q(z)\}} \Pi_{\text{seller}} = \int_{\underline{z}}^{\infty} \Big[\phi(z, q(z)) - c\, q(z)\Big] f(z)\, dz$$
$$\text{with} \quad \phi(z,q) \;=\; \Big(z \;-\; \frac{1}{h(z)}\Big) R(q), \qquad h(z) \;=\; \frac{f(z)}{1-\mathbb{F}(z)}$$

where $\phi(z,q)$ is the virtual surplus and $h(z)$ is the hazard rate. The virtual surplus, which represents the seller's effective revenue from serving type z, consists of two components. The first term, $zR(q)$, is buyer z's total revenue. The second term, $-\frac{1}{h(z)}\frac{\partial[zR(q)]}{\partial z} = -\frac{1}{h(z)}R(q)$, captures the truth-telling cost, i.e., the additional rents the seller must leave to higher types to prevent them from mimicking type z.[9]

The seller's pointwise first-order condition for the contract targeted to type z is,

$$\frac{\partial}{\partial q}\Big[\phi(z,q) - cq\Big]\Big|_{q=q(z)} = 0 \quad \Longrightarrow \quad \phi_q(z, q(z)) = \left(z - \frac{1}{h(z)}\right) R'(q(z)) = c \tag{2}$$

Now, we impose two further assumptions sequentially. First, buyer types are distributed according to a Pareto distribution with tail parameter κ. Under Pareto types, the inverse hazard rate is proportional to z, which implies that the seller's first-order condition reduces to a constant markup over marginal cost.

Lemma 1 (Optimal two-part tariff under Pareto types)**.** *Consider the screening problem in* (1)*. Suppose buyer types are Pareto distributed with tail parameter $\kappa > 1$ and lower bound $\underline{z}$, so that $h(z) = \kappa/z$. Assume further that $R'(0) = \infty$, and that expected buyer surplus is finite under the candidate allocation. Under these assumptions, the optimal nonlinear price schedule takes the form of a two-part tariff $\{F, p^{\text{NLP}}\}$:*

$$T(z) = F + p^{\text{NLP}} q(z), \qquad p^{\text{NLP}} = \frac{\kappa}{\kappa - 1} c \qquad \Pi(\underline{z}) = 0$$

Here F is a flat fee chosen so that the lowest type's participation constraint binds. We refer to p^{NLP} as the marginal price, to distinguish it from the average unit price $T(z)/q(z)$. See Appendix A.1 for the proof.

The Pareto assumption pins down the form of the optimal price schedule. To characterize the seller's revenue function, we further assume that buyers' revenue functions are homogeneous in the quantity transacted with the seller, so buyer type shifts demand for the seller's good without altering its curvature. Specifically,

$$R(q) = q^{\frac{\sigma-1}{\sigma}}, \qquad \sigma > 1 \tag{3}$$

[8]Formally, the problem is pointwise subject to monotonicity of the allocation. Under our assumptions, the virtual surplus $\phi(z,q)$ is increasing in z, which implies that the pointwise optimizer is monotone. Therefore, the monotonicity constraint is automatically satisfied and ironing is unnecessary; see Myerson (1981).

[9]The hazard rate summarizes how many buyers lie above z [i.e., $1 - \mathbb{F}(z)$] relative to the density at z [i.e., $f(z)$], and thus measures how easy it is to enforce truth-telling.

where σ is the curvature parameter (the demand elasticity faced by the seller). Under Pareto types, this specification implies $q(z) \propto z^{\sigma}$, so finiteness of output and of the seller's objective requires the tail condition $\kappa > \sigma$.

Corollary 1 (Propagation of revenue curvature)**.** *Under the assumptions of Lemma 1, the homogeneous revenue specification in* (3)*, and the tail condition $\kappa > \sigma$, the seller's aggregate revenue function inherits the same curvature as the buyers' revenue function. Let $Q \equiv \int q(z)\, d\mathbb{F}(z)$ denote the seller's total output. Then*

$$R^{seller}(Q) = AQ^{\frac{\sigma-1}{\sigma}}$$

for some constant $A > 0$. See Appendix A.2 for proof and characterization of A.

The corollary follows because, under the homogeneous revenue specification, both revenue collected through the per-unit charge and revenue collected through the flat fee scale with total output as $Q^{\frac{\sigma-1}{\sigma}}$.

Two remarks follow. First, exclusion is not optimal. Under the homogeneous revenue specification, raising the exclusion cutoff from $\underline{z}$ to $z_0 > \underline{z}$ increases the flat fee charged to the remaining buyers, but it also shrinks the mass of participating buyers. Under Pareto types, the flat fee scales as $F(z_0) \propto z_0^{\sigma}$, while the mass of participating buyers scales as $1 - \mathbb{F}(z_0) \propto z_0^{-\kappa}$. Hence total flat-fee revenue scales as $z_0^{\sigma-\kappa}$, which is decreasing under the tail condition $\kappa > \sigma$. The same is true for variable-profit revenue. Therefore, raising the exclusion cutoff strictly reduces the seller's profit.

Second, the quantities allocated $q(z_i)$ are determined by the marginal price p^{NLP}; the flat fee F only redistributes surplus but does not change q. Seller profit from transacting with type z_i has two components: a variable-profit rectangle $(p^{\text{NLP}} - c)\, q(z_i)$ and a flat-fee component F, which is set by the lowest-type participation constraint (Figure 1a). The deadweight loss is the area between the demand curve and marginal cost c, over the range of quantities from $q^{\text{NLP}}(z_i)$ to the efficient level $q^*(z_i)$. Changing F does not affect this area, while changing the marginal price does. Under the tail condition $\kappa > \sigma$, the marginal price under nonlinear pricing is lower than under the uniform-pricing benchmark, where the price would be a constant markup $\frac{\sigma}{\sigma-1}$ over marginal cost. As a result, the associated deadweight loss is smaller. Nonlinear pricing allows the seller to extract surplus partly through the flat fee and therefore requires a smaller distortion in the marginal price.

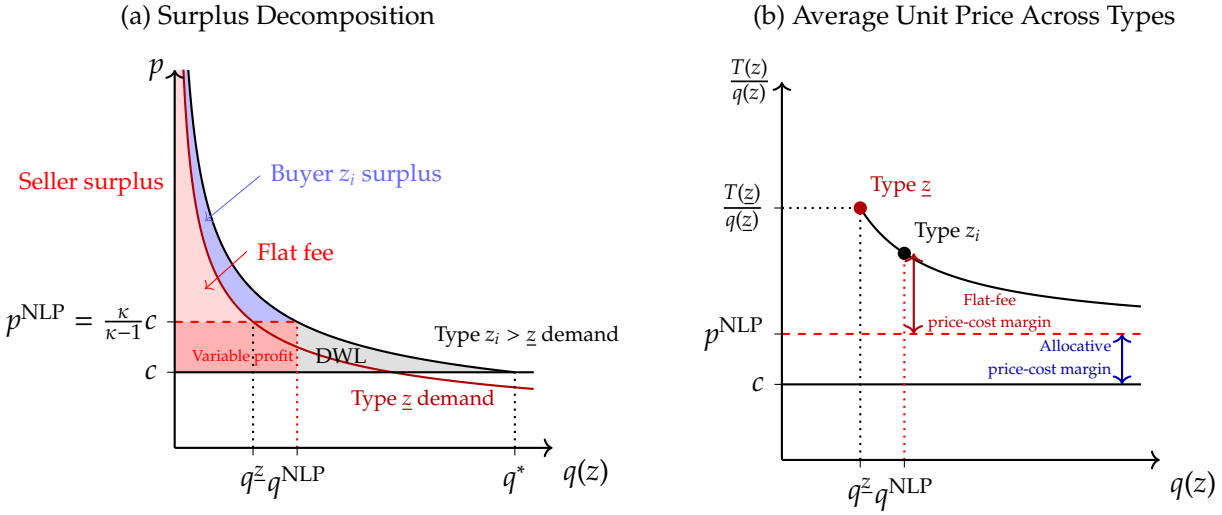
We define the total markup as the ratio of the average unit price, $T(z)/q(z)$, to marginal cost. We then decompose this markup into two components. The first component, which we call the allocative markup, is given by the ratio of the marginal price to marginal cost and equals $\frac{\kappa}{\kappa-1}$. We refer to it as allocative because it alters the quantity allocated, generating in this case a deadweight loss (Figure 1a). The second component is the flat-fee markup, which does not affect the allocation

but instead redistributes surplus from buyer to seller:

$$\underbrace{\mu_{\text{tot}}}_{\text{Total Markup}} := \frac{T(z)/q(z)}{c} = \underbrace{\frac{\kappa}{\kappa-1}}_{\text{Allocative}} + \underbrace{\frac{F}{q(z)c}}_{\text{Flat-fee}}$$

It follows that the total markup paid decreases with the quantity purchased, $q(z)$, as illustrated in Figure 1b. Because the flat fee F is identical across buyer types, spreading this fixed amount over more units makes the average unit price fall with quantity: $T(z)/q(z) = p^{\text{NLP}} + F/q(z)$. For small purchases, the flat-fee share $F/q(z)$ is large and the average unit price sits well above the marginal price; as quantity grows, $F/q(z)$ becomes negligible and the average unit price converges to the marginal price p^{NLP}, which governs quantities.

Figure 1: Surplus and Average Unit Prices



Notes. Both panels use isoelastic inverse demands, as implied by $zR(q)$ with $R(q) = q^{(\sigma-1)/\sigma}$, drawn for $\sigma = 2$ and $\kappa = 2.5$; the lowest type's demand is a scaled version of type z_i's. Panel (a): the marginal price pins down quantity; the flat fee, equal to the lowest type's surplus at p^{NLP}, is a lump sum that redistributes surplus without affecting q. Variable profit is the rectangle $(p^{\text{NLP}} - c)q^{\text{NLP}}$; efficient and two-part-tariff quantities of type z_i are labeled q^* and q^{NLP}, and the deadweight loss is the area between type z_i's demand and c from q^{NLP} to q^*. Panel (b): the average unit price $T(z)/q(z) = p^{\text{NLP}} + F/q(z)$ equals $\frac{\sigma}{\sigma-1}p^{\text{NLP}}$ for the lowest type and declines with quantity toward p^{NLP}. At any given quantity, the total price-cost margin is the vertical distance between $T(z)/q(z)$ and c. This margin decomposes into a flat-fee price-cost margin, $T(z)/q(z) - p^{\text{NLP}}$, and an allocative price-cost margin, $p^{\text{NLP}} - c$.

Implementability in supply chains and implied prediction. In Section 4, we build a multisector general equilibrium framework in which firms both pay nonlinear prices upstream and charge nonlinear prices downstream. In general equilibrium both revenue and marginal-cost functions become endogenous equilibrium objects, rather than primitives as in this section, shaped by nonlinear pricing. Our main result is that, when firm heterogeneity within a sector follows a Pareto

distribution, the equilibrium nonlinear contracts take the form of two-part tariffs, as in Lemma 1. Moreover, as in Corollary 1, the revenue functions are recursive, which allows the model to close consistently and firms to both pay and charge nonlinear prices along the supply chain network.[10]

If pricing is equivalent to a two-part tariff, total payment $T = F + pq$ implies an average unit price $T/q = F/q + p$ that is strictly decreasing and convex in q, with a horizontal asymptote at p (Figure 1b). In the next section, using firm-to-firm transactions at the product level for Chile, we examine whether average unit prices display this pattern.

3 Evidence on Nonlinear Pricing in Supply Chains

We use the population of firm-to-firm transactions in Chile for the year 2024 to document whether features of observed pricing are consistent with the nonlinear pricing structure pattern suggested by the partial equilibrium model. This exercise is not intended to identify a particular contractual mechanism: different pricing practices may coexist across transactions and relationships. Rather, its purpose is to establish that the empirical patterns motivating our general equilibrium model are pervasive in firm-to-firm transactions. We interpret the subsequent empirical evidence as equilibrium relationships that our model can rationalize.

Three main findings emerge. First, unit prices vary with transaction quantity and buyer characteristics, inconsistent with uniform pricing. Second, average unit prices decline with quantity, following a convex schedule that flattens at higher quantity purchased, a pattern consistent with nonlinear prices implemented as a two-part tariff. Third, these schedules vary across seller industries and, within the same seller, product, and day, across buyer industries for small purchases. Together, these findings indicate that the average contract combines quantity-based and buyer-group-based pricing, while allowing other mechanisms to operate in particular transactions and relationships.

Data description. We use data from the population of Chilean firm-to-firm value-added tax invoices collected by the Chilean Internal Revenue Service.[11] For each transaction-specific invoice, we observe seller and buyer IDs, a free-text product "detail", and for each item of the transaction, the corresponding price and quantity. These transaction records can be merged with firms'

[10] Our baseline assumes that all market power is on the seller side. A simple reduced-form way to allow for buyer market power is to include bargaining over the constant term in the incentive-compatible transfer schedule. We abstract from any form of buyer market power in the remainder of the paper.

[11] This study was developed within the scope of the research agenda conducted by the Central Bank of Chile (CBC) in economic and financial affairs of its competence. The CBC has access to anonymized information from various public and private entities, by virtue of collaboration agreements signed with these institutions. To secure the privacy of workers and firms, the CBC mandates that the development, extraction, and publication of the results should not allow the identification, directly or indirectly, of natural or legal persons. Officials of the CBC processed the disaggregated data. All the analysis was implemented by the authors and neither involved nor compromised the Chilean IRS. The information contained in the databases of the Chilean IRS is of a tax nature originating in self-declarations of taxpayers presented to the Service; therefore, the veracity of the data is not the responsibility of the Service.

accounting variables, including total revenue, employee headcounts, labor costs, materials costs, and capital expenditure.

We work at the most granular level and keep the full economy-wide universe of transactions available for 2024, without industry exclusions. Our unit of observation is each invoice line item between two tax IDs.[12] The "detail" field is often seller-specific (e.g., blue paint, brand XX, 3 gallons), so we treat products as seller-product pairs. Shipping appears as a separate invoice line, in which case it is not included in the unit price of the product.[13] Our approach complements Burstein et al. (2024), who use the same administrative source and document price-dispersion facts in a manufacturing subsample; here, we exploit the complete data available across all industries and retain maximum granularity to study nonlinear pricing in supply chains.

We perform three minimal data-cleaning steps to limit measurement error. First, we keep transactions with positive prices and quantities and nonmissing product detail. Second, we keep firms that reported positive sales in at least one month during 2024. Third, to limit the influence of extreme price observations, we compute the maximum-to-minimum price ratio within each seller-product-day cell containing at least two transactions and drop all transactions in cells with ratios above the 99th percentile of this distribution. These filters retain 98% of the value transacted. The final sample contains 679,679 seller IDs and 1,414,427 buyer IDs that traded 66,208,080 distinct products across 1.48 billion transactions in 2024.

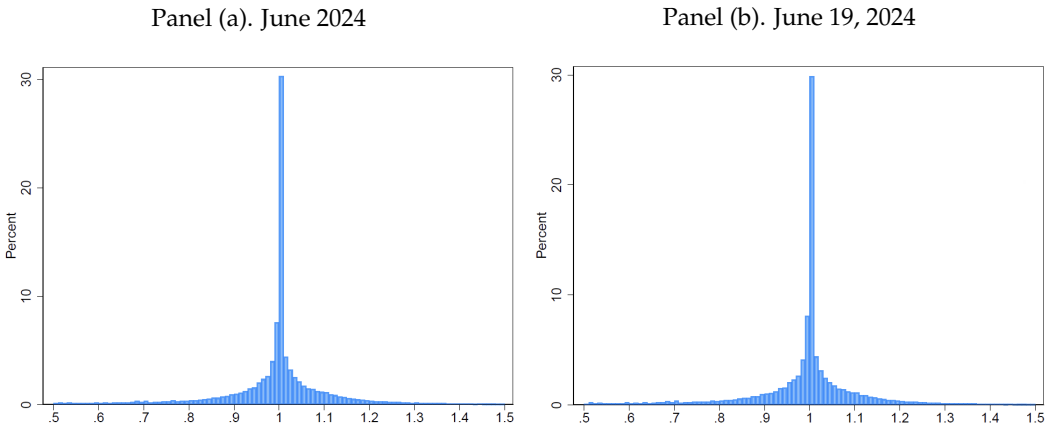
Price determinants. We begin by quantifying within-seller-product price dispersion. We observe substantial price variation for a given seller j and product g (the "detail" variable in the invoice) within a month. We construct a price-dispersion measure $\tilde{p}_{igjt} = p_{igjt}/\bar{p}_{jg\tau}$, where $\bar{p}_{jg\tau}$ is the mean price of seller-product pair (j, g) over period τ. We first set τ equal to June 2024 and then repeat the exercise using June 19, 2024, the day with the largest number of transactions that month. The daily exercise limits the role of aggregate price movements and demand and supply shocks within the month. The variance of $\ln \tilde{p}^{\tau}_{igjt}$ is 0.47 monthly and 0.49 daily, while 29% of transactions show no price dispersion in both samples.

Figure 2 displays the distributions of $\tilde{p}^{\tau}_{igjt}$. The similarity of the daily and monthly distributions suggests that aggregate price movements and common supply and demand shocks occurring within the month are unlikely to account for most of the observed price dispersion. Seller-product pairs accounting for 71% of transactions display within-period price variation and are therefore inconsistent with uniform pricing.

[12] We keep tax IDs which report sales, either to other firms or to final consumers.

[13] Separately reported shipping charges may mechanically generate a declining relationship between unit price and quantity when transportation costs increase less than proportionally with quantity.

Figure 2: Price Dispersion



Notes: This figure reports the distribution of the demeaned price for the month of June 2024 and for the day with more transactions in that month, the 19. We exclude seller-product pairs with only one transaction.

These findings motivate a flexible variance-accounting exercise that uses adjusted R^2 to assess how much within-seller-product-day price variation is associated with transaction quantity, observable buyer characteristics, and their interaction. To net out common shocks, we first estimate

$$\ln p_{igjt} = \beta_0 + \Psi_{jgd} + \epsilon_{igjt}$$

where Ψ_{jgd} are seller-product-day fixed effects and t indexes the time stamp within the day. The residual ϵ_{igjt} captures price differences across buyers purchasing the same product from the same seller on the same day. We then project ϵ_{igjt} using alternative fixed-effect sets S

$$\epsilon_{igjt} = \beta_0 + \Psi_S + \nu_{igjt} \tag{4}$$

where Ψ_S includes, in turn, (i) seller-product-specific quantity fixed effects, (ii) buyer-group fixed effects based on sector, size, and region (528 groups), and (iii) interactions between quantity and buyer group, allowing the relationship between quantity and prices to differ across groups. Comparing the resulting adjusted R^2 values shows how much residual price variation is associated with quantity, buyer characteristics, and their interaction. This exercise does not separately identify second-degree from third-degree price discrimination, because quantity choices and buyer types may generate observationally equivalent price patterns.[14] Our objective is therefore more limited: to assess whether the data contain patterns consistent with quantity-based pricing, buyer-based pricing, or a combination of the two.

[14]For example, if large buyers purchase greater quantities and pay lower unit prices, the pattern could reflect a common quantity-discount schedule, buyer-specific prices, or both. Transaction data alone do not distinguish these mechanisms without additional variation or structural assumptions.

Table 1 reports the adjusted R^2 from the daily specifications. Because all specifications use the same sample, the results compare how much of the same residual price variation is associated with alternative dimensions of pricing. Quantity fixed effects explain 34% of residual price variation, while buyer-group fixed effects explain 29%. Thus, transaction quantity accounts for a larger share of within-seller-product-day price variation than observable buyer characteristics.

Quantity-by-buyer-group fixed effects explain 55%, approximately 21 percentage points more than the quantity dimension alone. The result shows that allowing the price-quantity relationship to vary across observable buyer groups captures considerably more of the observed price variation. The data are therefore consistent with price-quantity schedules that vary across buyer groups.

Appendix B.1 repeats the exercise on the same transactions with seller-product-month rather than seller-product-day fixed effects. The wider cells contain repeated transactions between the same seller, product, and buyer, allowing us to include seller-product-month-by-buyer fixed effects. These fixed effects capture persistent buyer-specific price differences within a month, such as those arising from bargaining, individualized pricing, or buyers who regularly place large orders and therefore pay lower unit prices. They explain 48% of residual price variation, exceeding the explanatory power of quantity or observable buyer groups alone. Buyer identity therefore contains relevant information beyond sector, size, and region. Quantity-by-buyer-group fixed effects account for 64% of residual variation, indicating that allowing the price-quantity relationship to vary across observable buyer groups captures additional features of the data beyond persistent buyer-specific price levels. The same appendix repeats the exercise separately for manufacturing and retail and wholesale sellers, the two seller industries with the largest transaction volumes, and shows that the pattern is not confined to a single seller industry.

Table 1: Price Residual Determinants

	(1)	(2)	(3)
Adjusted R^2	0.34	0.29	0.55
S = Quantity	✓		
S = Buyer Group		✓	
S = Quantity × Buyer group			✓
N	271M	271M	271M

Notes: This table reports adjusted R^2 values from OLS regressions of price residuals ϵ_{igjt} on different specifications S of Equation (4), where residuals are obtained after controlling for seller-product-day fixed effects. All specifications use the same sample of 2024 firm-to-firm transactions. Buyer groups are defined by combinations of 11 sectors, 3 size categories, and 16 regions.

Nonlinear prices. We then examine whether observed unit prices vary with transaction quantities. We estimate

$$\ln p_{igjt} = \beta_1 \ln q_{igjt} + \Psi_{jgd} + \Psi_S + \varepsilon_{igjt} \tag{5}$$

where p_{igjt} and q_{igjt} are the unit price[15] and quantity for buyer i, product g, seller j, at transaction time t on day d. Ψ_{jgd} are seller-product-day fixed effects, while Ψ_S varies across specifications to absorb buyer and buyer-group heterogeneity. Seller-product-day fixed effects absorb all price determinants common to transactions of the same seller-product on a given day, including common cost and supply conditions. The coefficient β_1 is therefore identified from differences in prices and quantities across transactions within seller-product-day cells.[16]

A remaining concern is that buyers who purchase larger quantities may also obtain lower prices for reasons unrelated to a common quantity schedule, including persistent buyer characteristics or bargaining power. We therefore introduce buyer and buyer-group controls to assess whether the negative price-quantity relationship survives after absorbing such differences. All specifications are estimated on a common sample, so that differences across columns reflect changes in the conditioning set rather than changes in sample composition.

Table 2: Price-Quantity Coefficient Estimates

	(1)	(2)	(3)	(4)
$\ln q_{igjt}$	−0.036	−0.070	−0.054	−0.030
	(0.002)	(0.003)	(0.002)	(0.002)
Seller × product × day FE	Yes	Yes	Yes	–
Buyer FE	–	Yes	–	–
Buyer-group FE	–	–	Yes	–
Seller × product × day × buyer-group FE	–	–	–	Yes
Observations	422M	422M	422M	422M
Adjusted R^2	0.9645	0.9667	0.9654	0.9749

Notes: The table reports OLS coefficients from regressions of log unit price on log transaction quantity with high-dimensional fixed effects, Equation (5). Buyer groups are defined by the interaction of 11 sectors, 3 size categories, and 16 regions. All regressions are estimated on the same sample of Chilean firm-to-firm transactions in 2024, defined as the intersection of the estimation samples across the four specifications. Standard errors are clustered by seller-product and reported in parentheses.

[15] Unit price is the observed price per unit reported in the invoice; we do not otherwise transform or impute it.

[16] We estimate schedules at the transaction level because seller-product-day fixed effects then absorb supply and demand shocks most finely. In the model, the relevant unit is the buyer–seller relationship over a period, which may comprise several orders; for the quantitative results in Section 5.2, what matters is that part of each relationship's payments is inframarginal. Footnote 20 explains that the schedule is similar at the buyer–seller–product–month level.

Column 1 conditions on seller-product-day fixed effects and yields a coefficient of −0.036, implying that a doubling of transaction quantity is associated with approximately a 2.5% lower unit price. Adding buyer fixed effects in column 2 strengthens the coefficient to −0.070, indicating that persistent differences across buyers do not seem to account for the negative relationship. Replacing buyer fixed effects with buyer-group fixed effects in column 3 gives a coefficient of −0.054. Finally, column 4 absorbs seller-product-day-by-buyer-group fixed effects, so identification comes only from transactions within the same seller-product-day and buyer group. The coefficient remains negative at −0.030, approximately 83% of the column 1 magnitude, showing that the negative price-quantity relationship persists even within narrowly defined groups of buyers.[17]

Across specifications, therefore, larger transactions are associated with lower unit prices after conditioning on seller-product-day and buyer heterogeneity.[18] The estimates from Equation (5) establish the presence of quantity discounts but compress this relationship into a single coefficient and do not characterize the shape of the underlying price-quantity schedule. Is the price-quantity profile convex, flattening at high quantities, as implied by the two-part tariff of the model in Section 2? We turn to this next.

Nonlinear quantity discounts. Because products are measured in different units and trade at different scales, we express transaction quantities relative to the median quantity traded for each seller-product. Let q_{jg}^{med} denote the median transaction quantity for seller-product (j, g) in 2024 and define

$$\widetilde{q}_{igjt} \equiv \frac{q_{igjt}}{q_{jg}^{\text{med}}}, \qquad x_{igjt} \equiv \log \widetilde{q}_{igjt}$$

Thus, $x_{igjt} = 0$ corresponds to the seller-product median. This normalization removes differences in units and scale while preserving proportional distances in quantity: the same value of x has the same proportional interpretation for every seller-product.

We partition the pooled 1st–99th percentile support of x_{igjt} into 20 equal-width intervals and assign each transaction to its corresponding normalized-quantity bin. We then estimate

$$\ln p_{igjt} = \beta_0 + \sum_{b=2}^{20} \beta_b \mathbb{1}\{B_{igjt} = b\} + \Psi_{jgd} + \varepsilon_{igjt} \tag{6}$$

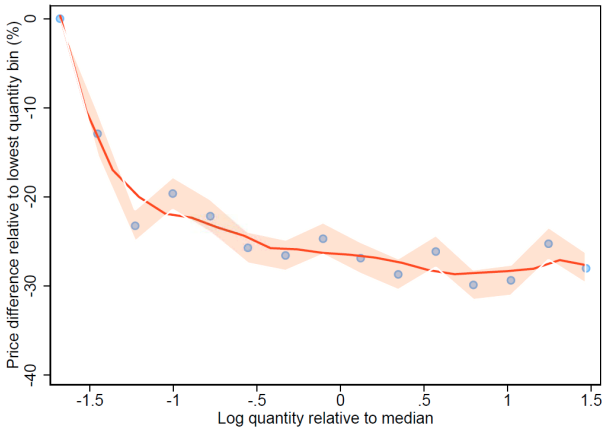
[17] We do not estimate buyer-specific slopes. Within a seller-product-day, the same buyer rarely purchases multiple distinct quantities, so slopes specific to a buyer and cell are not identified; pooled buyer slopes would rely on buyers trading across many seller-product-days and would be imprecise for most buyers.

[18] We repeat the same exercise from column 1 for each 1-digit seller sector in the economy; we show the results in Appendix B.2. The coefficient is negative in ten of the eleven sectors, the exception being Utilities, where it is small and positive. The coefficient is also close to zero, though negative, in Financial Services, while the largest coefficient in absolute value is observed in Construction, at −0.157. In utilities and financial services, invoice unit prices are not informative about pricing structure: utility tariffs in Chile are regulated and combine fixed and usage-dependent charges, and the main price of financial services, interest, is largely not recorded on VAT invoices.

where Ψ_{jgd} are seller-product-day fixed effects and the lowest normalized-quantity bin is the omitted category.[19] Identification continues to come exclusively from price and quantity variation within seller-product-day cells.

Figure 3 reports the resulting price schedule.[20] The figure displays bins 1–15, from the lowest estimated bin through approximately $x = 1.5$, that is, quantities from approximately 0.19 to 4.35 times the seller-product median, more than a twenty-fold range in transaction quantity. The remaining five bins, which extend to 13.4 times the median, lie between 1 and 4 log points above bin 15, small relative to the 33 log-point decline up to that bin, and their confidence intervals overlap with that of bin 15. All five remain precisely estimated discounts of 25 to 27% relative to bin 1. We omit them to avoid overcrowding the figure; all 20 bins, together with the distribution of normalized quantities, are reported in Appendix B.3.

Figure 3: Prices by Normalized Quantity



Notes: The figure reports percentage price differences across normalized-quantity bins relative to the lowest bin, from an OLS regression of log unit price on bin indicators with seller-product-day fixed effects, Equation (6). Quantity is normalized by the seller-product median, $x_{igjt} = \log(q_{igjt}/q_{jg}^{\text{med}})$, and bins are equal-width intervals in x_{igjt}. The red line is a smoothed fit through the estimated bin effects and the shaded region reports 95% confidence intervals based on standard errors clustered by seller-product. The figure displays bins 1–15, with midpoints ranging from $x = -1.679$ to $x = 1.471$. Bins 16–20 remain part of the estimation and are reported in Appendix B.3.

The estimated schedule is nonlinear and convex. Unit prices fall as quantity increases from the

[19] Under the two-part tariff in Lemma 1, average unit price is $T(z)/q(z) = p^{\text{NLP}} + F/q(z)$. Writing $x = \log q$, its logarithm is $\ln(T(z)/q(z)) = \ln\left(p^{\text{NLP}} + Fe^{-x}\right)$, which is strictly convex in x for any $p^{\text{NLP}}, F > 0$. Hence the model implies that log average unit price is convex in log quantity, not only that the price level is convex in the quantity level. This property is unaffected by expressing quantity relative to the seller-product median, since $\log(q/q_{jg}^{\text{med}})$ differs from $\log q$ only by a constant shift for each seller-product.

[20] The model's two-part tariff applies to a buyer's total purchases from a seller over a period, whereas Figure 3 relates invoice unit prices to individual order quantities. We therefore also aggregate transactions to the buyer-seller-product-month level and relate the monthly average unit price, total payments divided by total quantity, to normalized monthly quantity within seller-product-month cells. The schedule is similar: above the median, monthly average prices are 26 to 30 log points below the lowest bin, compared with 29 to 36 log points at the transaction level (Appendix Table A3). Larger purchases over the period are thus associated with lower average payments per unit.

lower part of the normalized-quantity distribution toward the seller-product median. The decline then becomes progressively smaller: beyond the median, additional increases in quantity are associated with comparatively small additional reductions in unit prices. This shape is consistent with the decreasing and convex price-quantity schedule implied by Lemma 1. Appendices B.4 and B.5 provide two complementary checks. First, an alternative within-seller-product rank normalization, which requires only an ordering of quantities and not cardinal comparability across products, yields a strongly monotone schedule: prices decline as transactions move toward the upper end of each seller-product's observed quantity support.

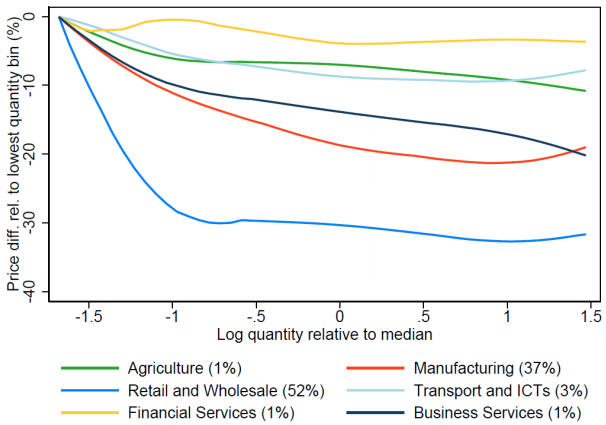
Second, estimating the schedule separately by seller industry with a two-piece linear specification in normalized log quantity, with a kink at the median (Appendix B.5), shows that it is decreasing and convex in industries accounting for 93% of transactions. Prices also decline with quantity in the remaining industries, except utilities and financial services, where they are not significantly decreasing below the median.

Heterogeneity across seller industries. We next examine whether the nonlinear price-quantity schedule differs across seller industries. We reestimate Equation (6) separately by 1-digit seller industry, retaining the same seller-product normalization and normalized-quantity bins used in the aggregate exercise. Figure 4 reports the six seller industries that together account for approximately 95% of transactions in the estimation sample (we do not include industries with fewer than 1% of transactions).

The magnitude of quantity discounts varies across seller industries. Retail and Wholesale exhibits the steepest schedule among the sectors shown, followed by Manufacturing and Business Services, while Agriculture and Transport and ICTs display more moderate declines. Financial Services is comparatively flat, consistent with the near-zero linear coefficients reported earlier. At the same time, the schedules generally preserve the aggregate pattern that most of the price decline occurs at lower normalized quantities, although the extent of subsequent flattening differs across sectors. Appendix B.4 shows that the broad pattern of seller-industry heterogeneity is also present under the alternative within-seller-product rank normalization.

Heterogeneity across buyer industries. Sellers within a given industry transact with buyers operating in different markets and facing different demand conditions. We therefore hold the seller industry fixed and examine whether the price-quantity schedule varies across buyer industries. We focus on Manufacturing and Retail and Wholesale, which together account for approximately 90% of transactions in the estimation sample. Within each seller industry, we restrict the sample to seller-product-day cells with transactions to at least two buyer industries and estimate a single regression with bin-by-buyer-industry effects and seller-product-day fixed effects, so that buyer industries are compared within the same seller, product, and day. All schedules are expressed

Figure 4: Prices by Normalized Quantity, by Seller Industry



Notes: The figure reports smoothed price-quantity schedules estimated by OLS separately for each 1-digit seller industry using Equation (6), with the same normalization, bins, and fixed effects as in Figure 3. The y-axis reports percentage price differences relative to the lowest normalized-quantity bin. Industry labels report each sector's share of transactions in the estimation sample, shown in parentheses. Confidence intervals are omitted for readability. Standard errors of the bin effects, clustered by seller-product, are below 3 log points for Manufacturing and for Retail and Wholesale, which together account for 89% of transactions, and below 9 log points in all industries shown. Of the 114 non-base coefficients, 97 are statistically different from zero at the 5% level; the exceptions are bins 2 and 3 in Agriculture, Manufacturing, and Transport and ICTs, and 13 of the 19 bins in Financial Services, where the schedule is imprecisely estimated.

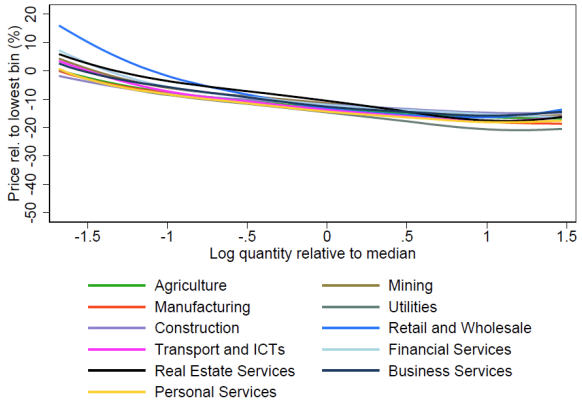
relative to the lowest bin of buyers in the seller's own industry, so that both levels and slopes are comparable across buyer industries.

Figure 5 shows that within the same seller, product, and day, buyer industries pay different prices mainly for small purchases. Among Retail and Wholesale sellers, prices in the lowest bin are between 23 and 43% lower for buyers outside Retail and Wholesale, but the schedules converge at about 0.4 times the median ($x \approx -1$) and remain close thereafter. Among Manufacturing sellers, Retail and Wholesale buyers pay more for small purchases, and the schedules again converge below the median and decline in parallel. Sellers thus charge similar prices on large purchases across buyer industries but different prices on small ones, where flat fees weigh most in the average price. This pattern is consistent with two-part tariffs whose flat fees differ across buyer industries.

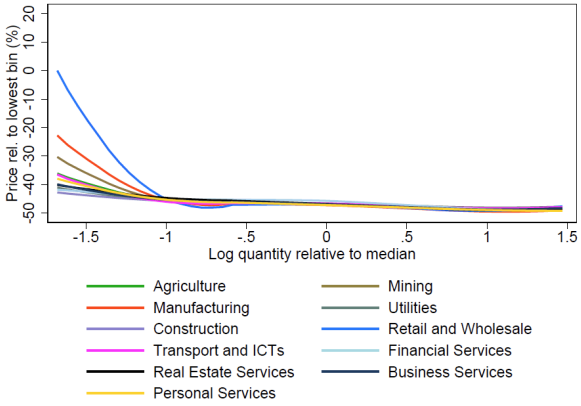
Alternative explanations. We interpret quantity discounts as evidence of quantity-based price discrimination by buyer industry; nevertheless, the preceding results do not rule out alternative mechanisms. We next assess alternative interpretations of the price-quantity patterns documented above: persistent buyer heterogeneity, bilateral bargaining, repeated relationships and learning, and cost-side scale economies. Because our ultimate interest is the aggregate welfare implications of pricing schedules, not every alternative threatens our interpretation. For the welfare efficiency gains we quantify in Section 5.2, what matters is whether sellers collect part of their margin in-

Figure 5: Prices by Normalized Quantity, by Buyer Industry

Panel (a). Manufacturing Sellers



Panel (b). Retail and Wholesale Sellers



Notes: The figure reports smoothed price-quantity schedules by 1-digit buyer industry, conditional on the seller industry, using the normalization and bins of Figure 3. Each panel is estimated in a single regression with bin-by-buyer-industry effects and seller-product-day fixed effects, on seller-product-day cells with transactions to at least two buyer industries. Panel (a) restricts the sample to Manufacturing sellers and Panel (b) to Retail and Wholesale sellers. The y-axis reports percentage price differences relative to the lowest bin of buyers in the seller's own industry. Confidence intervals are omitted for readability. The underlying bin-by-buyer-industry coefficients are precisely estimated: standard errors clustered by seller-product have a median of 2.0 and a maximum of 5.8 log points, and 199 of the 219 non-base coefficients are statistically different from zero at the 5% level. All exceptions are in the five lowest quantity bins, where prices are close to the reference level.

framarginally, so that marginal prices lie below average prices. Efficient bilateral bargaining and volume-based relational contracts share this feature and would deliver qualitatively similar allocative gains, although they might split rents between buyers and sellers differently, which, as we show later, matters for firm entry. The remaining concerns are that discounts reflect buyer-specific prices or costs rather than a pricing schedule.

With buyer fixed effects (Appendix B.6), so that we compare orders of the same buyer, the price-quantity relationship becomes larger in magnitude. Proxies for bargaining power barely change it over most of their range: moving across the interquartile range of the buyer's number of suppliers or the seller's dependence on the buyer shifts the elasticity by a few percent, and the gradient is steeper where the seller depends heavily on the buyer. Repeated trading and relationship tenure attenuate it slightly, consistent with sellers learning about their buyers, but leave it strongly negative.[21]

Cost-based explanations face two facts. First, separately invoiced shipping is excluded from unit prices, and quantity discounts are larger, not smaller, among the highest-priced products,

[21] Why would quantity-based screening persist between buyers and sellers who trade intensively? We discuss this in footnote 26.

where fixed per-order costs should matter least (Appendix B.6).[22] Second, for the same seller, product, and day, schedules differ across buyer sectors for small purchases. A cost-based explanation would therefore require each seller's per-order costs to differ systematically by buyer sector, which we view as unlikely to account for the pattern. Because seller-product-day fixed effects absorb cost and demand shocks common to a seller-product on a given day and hold the unit of measure fixed, and because unit prices are reported on the invoice rather than computed from values and quantities, neither common shocks nor mechanical measurement error are likely to drive the pattern.

The alternative mechanisms can therefore coexist with the documented pattern, but do not appear quantitatively capable of accounting for the average price-quantity relationship. Finally, we do not target these quantity discounts in the calibration. Their shape and size are nonetheless consistent with the model we develop below: prices decline and flatten with quantity, as under a two-part tariff, and the calibrated model, disciplined by profit shares rather than prices, produces an average discount of similar magnitude (Section 5.1).

Taking stock. The evidence points to systematic nonlinear pricing in firm-to-firm transactions. Within seller-product-day cells, unit prices decline with normalized quantity and then flatten, while their magnitude varies across seller industries and, for small purchases, across buyer industries within the same seller, product, and day. Several familiar forces can coexist with this pattern, but do not appear quantitatively capable of accounting for the average price-quantity relationship. Guided by these findings, in the next section we develop a multisector supply-chain general equilibrium model with heterogeneous firms and endogenous nonlinear prices.

4 A General Equilibrium Model of Nonlinear Pricing in Supply Chains

We develop and characterize a general-equilibrium supply-chain model in which firms simultaneously pay and charge nonlinear prices endogenously. We show that the optimal contract takes the form of a two-part tariff: a constant marginal price combined with a flat fee, extending Lemma 1 to a general-equilibrium setting in supply chains. This framework provides closed-form sufficient statistics for welfare analysis under different pricing assumptions and counterfactual policies.

4.1 Environment and Notation

There are two firm layers, $\ell \in \{u, r\}$, defined by their position with respect to final demand. Upstream firms (layer u) sell to retailers and other upstream firms, and source inputs from upstream

[22] Following Bornstein and Peter (2025), we re-estimate Equation (5) restricting the sample to seller-products in the top 1% of average unit price, where fixed per-order costs represent a negligible fraction of unit value; the price-quantity gradient remains strongly negative (Appendix Table A5, column 6).

suppliers. There is roundabout production with an arbitrary number of firm linkages upstream. Retailers (layer r) purchase inputs from upstream firms and sell exclusively to the representative final consumer.[23] There is a finite set of sectors $\mathcal{S}$, common to both firm layers. We use $s \in \mathcal{S}$ for buyer sectors and $s' \in \mathcal{S}$ for seller sectors. In each (ℓ, s), there is a continuum of firms that differ in their productivity z. Productivity z is Pareto-distributed within (ℓ, s) with lower bound $\underline{z}_s^\ell > 0$ and tail parameter κ_s^ℓ; the support is $z \in [\underline{z}_s^\ell, \infty)$.[24]

When a firm appears as a buyer, we index it by i; when it appears as a seller, we index it by j. The mass of firms in (ℓ, s) is N_s^ℓ. We treat N_s^ℓ as endogenous under free entry; details follow. We evaluate the economy in a steady state and omit time subscripts for brevity.

Market structure. Retailers sell to the representative consumer at uniform per-unit prices[25] while sourcing inputs from upstream firms at nonlinear prices. Upstream firms likewise purchase inputs from other upstream firms at nonlinear prices and sell their own variety to both retailers and other upstream firms. Sellers j observe the buyer's layer and sector pair (ℓ, s) but not the idiosyncratic buyer productivity z_i; they know only its (Pareto) distribution.[26] They can condition nonlinear contract menus on the buyer's layer and sector, but not on z_i, implying third-degree price discrimination across (ℓ, s) and second-degree price discrimination within each (ℓ, s).[27]

Preferences. The representative consumer owns all firms and inelastically supplies one unit of labor (L=1). Let P_Y be the final-good price index, and take the wage as the numeraire, $w \equiv 1$.[28] Final demand is Cobb–Douglas across retail sectors with a within-sector CES aggregator over

[23] This two-layer structure is motivated by Chilean administrative data: 80% of firms sell at least 90% of their sales to a single destination, final consumers or other firms, and these firms account for 73% of total sales (Appendix B.7).

[24] We impose the moment restrictions on κ_s^ℓ required for all equilibrium aggregates to be finite.

[25] For welfare effects of nonlinear pricing on final demand, see Bornstein and Peter (2025).

[26] A natural concern is that in a dynamic setting, repeated transactions may attenuate information frictions and move the environment relatively closer to personalized pricing or first-degree price discrimination. Repeated interaction need not undo screening: when the privately observed component of buyer type is permanent, the optimal long-term contract under commitment repeats the static one; without commitment, buyers anticipate that revealed types will be exploited and under-reveal (Laffont and Tirole, 1988). Consistent with this, quantity discounts barely attenuate with relationship tenure or intensity (Appendix B.6). In our framework, sellers observe the buyer's sector, but the same logic can be applied more granularly if sellers also observe a time-invariant firm-specific component of buyer heterogeneity. In that case, the object we call a "sector" can be interpreted as a firm type, while each transaction remains subject to a privately observed buyer-level demand shock. We adopt the sector-level formulation because it is the natural level at which we take the model to the data. If the screened component is a transitory shock, however, the relevant tail is that of its conditional distribution rather than of firm employment; Section 5.5 reports how results vary with ξ.

[27] Sustaining layer- and sector-specific nonlinear contracts requires the absence of zero-cost arbitrage or resale; secondary markets may fail to emerge due to repackaging costs, regulation, or other frictions.

[28] In the unsubsidized equilibrium, free entry implies zero aggregate net profits after accounting for entry expenditures. Hence household disposable income equals labor income, $wL = 1$. Therefore, normalizing $w = 1$, welfare is $1/P_Y$.

retail varieties

$$Y = \prod_{s \in \mathcal{S}} Y_s^{\theta_s}, \qquad \sum_{s \in \mathcal{S}} \theta_s = 1 \tag{7}$$

$$Y_s = \left(\int_{j \in \mathcal{R}_s} y_j^{\frac{\sigma_s^r - 1}{\sigma_s^r}} \, dv_s^r(j) \right)^{\frac{\sigma_s^r}{\sigma_s^r - 1}} \tag{8}$$

where $\theta_s \in (0,1)$ are Cobb–Douglas output elasticities, $\sigma_s^r > 1$ is the within-sector elasticity of substitution, and $\mathcal{R}_s$ is the set of active retail sellers in sector s. Here, $dv_s^r(j)$ denotes the equilibrium measure over active retail sellers $j \in \mathcal{R}_s$, with total mass $N_s^r \equiv v_s^r(\mathcal{R}_s)$.

Technology. Firms (buyers i) produce with Cobb–Douglas technology in labor and a Cobb–Douglas aggregator across seller sectors; we denote the buyer sector by s and the seller sector by s'

$$Q_i = z_i \, l_i^{\alpha_s^\ell} \, M_i^{1-\alpha_s^\ell}, \qquad 0 < \alpha_s^\ell < 1 \tag{9}$$

$$M_i = \prod_{s' \in \mathcal{S}} M_{is'}^{\theta_{ss'}^\ell}, \qquad \sum_{s' \in \mathcal{S}} \theta_{ss'}^\ell = 1 \quad \text{for each } (\ell, s) \tag{10}$$

where Q_i is output, z_i is firm i's productivity, l_i is firm-level labor input, α_s^ℓ is the labor output elasticity for firms of layer ℓ in sector s, and M_i is the composite materials bundle. $M_{is'}$ is the materials bundle from upstream seller sector s', and $\theta_{ss'}^\ell \geq 0$ are input elasticities for buyers in (ℓ, s) across seller sectors $s' \in \mathcal{S}$.[29] Within any seller sector s', the materials bundle is CES across firm varieties with elasticity $\sigma_{s'}^u > 1$ for each $s' \in \mathcal{S}$

$$M_{is'} = \left(\int_{j \in \mathcal{U}_{s'}} m_{ij}^{\frac{\sigma_{s'}^u - 1}{\sigma_{s'}^u}} \, dv_{s'}^u(j) \right)^{\frac{\sigma_{s'}^u}{\sigma_{s'}^u - 1}} \tag{11}$$

where m_{ij} is buyer i's input of seller variety j in seller sector s', $\sigma_{s'}^u$ is the elasticity of substitution across those varieties, and $dv_{s'}^u(j)$ is the equilibrium measure over active upstream sellers $\mathcal{U}_{s'}$, with total mass $N_{s'}^u \equiv v_{s'}^u(\mathcal{U}_{s'})$. Throughout, σ_s^ℓ denotes the elasticity of substitution across varieties sold by layer-ℓ firms in sector s.

Input price-taking. Firms are atomistic in input markets and take the wage as given. They retain market power in output markets due to product differentiation under CES demand.

[29] A zero weight $\theta_{ss'}^\ell = 0$ means sector s as a buyer does not use inputs from sector s'. Under no price discrimination and uniform prices, $\{\theta_{ss'}^\ell\}$ coincide with expenditure shares across intermediate-input sectors conditional on materials expenditure for buyer sector s, as in Acemoglu et al. (2012).

Firm entry. Firm entry follows Hopenhayn (1992) and Melitz (2003), adapted to a supply-chain environment. In each (ℓ, s), there is an unbounded pool of identical potential entrants. Entry requires paying a sunk cost $c_s^{E\,\ell} > 0$ in units of labor, after which firms draw their productivity z. Active firms exit exogenously at the end of the period with probability $(1 - \delta_s^\ell) \in (0, 1]$, which serves as the only source of time discounting.[30] Let $\pi_s^\ell(z)$ denote a potential entrant's per-period profit in numeraire units. Free entry requires that the expected discounted value of profits equals the entry cost in every (ℓ, s)

$$\frac{1}{1-\delta_s^\ell}\,\mathbb{E}_z\left[\pi_s^\ell(z)\right] \;=\; c_s^{E\,\ell}\,w \quad \forall(\ell, s)$$

where the expectation is taken over the post-entry distribution of z in (ℓ, s). Let e_s^ℓ denote the measure of new entrants in layer-sector pair (ℓ, s). In a steady state, entrant inflows replace exiting firms

$$e_s^\ell = (1 - \delta_s^\ell) N_s^\ell$$

Market clearing. All markets clear in equilibrium. Labor-market clearing requires that the total demand for labor across all active firms equals the inelastic supply of one unit

$$\sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}}\int_{i\in\mathcal{F}_{\ell s}} l_i\, dv_s^\ell(i) + \sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}} e_s^\ell c_s^{E\,\ell} = 1$$

where $\mathcal{F}_{\ell s}$ is the set of active firms of layer ℓ in sector s (so $\mathcal{F}_{us} = \mathcal{U}_s$ and $\mathcal{F}_{rs} = \mathcal{R}_s$), and v_s^ℓ is the equilibrium measure over these firms. For each upstream variety $j \in \mathcal{U}_{s'}$, market clearing requires that output equals the sum of inputs demanded by all buyers. For each retail variety $j \in \mathcal{R}_s$, market clearing requires that output equals final demand from the representative consumer

$$Q_j \;=\; \sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}}\int_{i\in\mathcal{F}_{\ell s}} m_{ij}\, dv_s^\ell(i), \quad \forall\, j \in \mathcal{U}_{s'},\ s' \in \mathcal{S}; \qquad Q_j \;=\; y_j \quad \forall s \in \mathcal{S},\ \forall\, j \in \mathcal{R}_s$$

General equilibrium under nonlinear pricing. Within a period, (i) potential entrants in each (ℓ, s) pay $c_s^{E\,\ell}$ and then draw productivity z; (ii) each upstream seller $j \in \mathcal{U}_{s'}$ observes only the buyer's pair (ℓ, s) (not z_i) and offers a pair-specific nonlinear contract menu (a quantity and a transfer) $\{m_{js}^\ell,\, T_{js}^\ell\}$; retail sellers $j \in \mathcal{R}_s$ post uniform prices to final consumers; (iii) buyers $i = (\ell, s, z_i)$ observe the offered menus and the wage w and choose labor l_i and input bundles $\{m_{ij}\}_j$ to maximize profits; (iv) production and trade occur, transfers $\{T_{ij}\}_j$ are realized, and final demand $\{y_j\}$ is met; and (v) firms exit with probability $1-\delta_s^\ell$. Contracts are enforceable, resale and arbitrage

[30] Because we focus on steady-state comparisons of macroeconomic outcomes and abstract from time discounting aside from δ, the model is isomorphic to either a constant z over time or a stochastic process for z that yields the same stationary productivity distribution.

are ruled out, and beliefs are rational.

Under the normalization $w \equiv 1$, a steady state general equilibrium consists of firm masses $\{N_s^\ell\}_{\ell,s}$, allocations $\{Q_i, l_i, \{m_{ij}\}_j\}$, retail prices $\{p_j\}_{j\in\mathcal{R}_s, s\in\mathcal{S}}$, transfers $\{T_{ij}\}_j$, consumer demands $\{y_j\}$, and the final-good price index P_Y, such that (i) technologies (9)–(11) hold for every firm; (ii) each upstream seller $j \in \mathcal{U}_{s'}$ chooses tariff schedules $T_{js}^\ell : \mathbb{R}_+ \to \mathbb{R}$ that solve its profit-maximization problem (defined below), while each retail seller $j \in \mathcal{R}_s$ sets uniform prices to the final consumer to maximize profits; (iii) each buyer chooses labor l_i and input bundles $\{m_{ij}\}_j$ to maximize profits given the normalized wage $w = 1$ and the offered contracts or prices; (iv) retail, upstream, and labor markets clear; and (v) free entry holds in each (ℓ, s).[31]

4.2 Guesses: Contracts and Revenue Shapes

We characterize the equilibrium using a guess-and-verify approach. In a supply-chain setting, firm costs and revenues are equilibrium objects shaped by price discrimination. We begin by conjecturing functional forms for firms' revenue functions and for upstream contracts, which allows us to analyze the firm's unrestricted price-discrimination problem. We then verify the conjecture by showing that the resulting equilibrium coefficients are internally consistent.

Guess 1: Two-part tariffs by buyer layer and sector (ℓ, s). Motivated by Lemma 1, for a buyer $i = (\ell, s, z_i)$, the total payment for purchasing seller variety $j \in \mathcal{U}_{s'}$ is conjectured to take the form of a two-part tariff, with the marginal price determined by an (ℓ, s, s')-specific markup $\mu_{ss'}^\ell$:

$$T_{ij} \;=\; p_{js}^\ell\, m_{ij} \;+\; F_{js}^\ell \;=\; \mu_{ss'}^\ell\, c_j\, m_{ij} \;+\; F_{js}^\ell$$

where m_{ij} is the quantity purchased by buyer i from seller j, $p_{js}^\ell = \mu_{ss'}^\ell\, c_j$ is the marginal price, ℓ and s denote the buyer's layer and sector, and c_j is the seller's marginal cost. The fixed component F_{js}^ℓ is a flat fee that varies with the seller identity j and with the buyer only through the observable pair (ℓ, s).[32]

Guess 2: Equilibrium revenue function. We conjecture that, in equilibrium, the revenue function is homogeneous of degree ψ_s^ℓ in output

$$R_i \;=\; A_s^\ell \left(Q_i\right)^{\psi_s^\ell}, \qquad A_s^\ell > 0, \quad \psi_s^\ell \in (0,1)$$

[31] Policy counterfactual equilibrium is defined analogously, except that firm maximization problems are subject to the additional constraints implied by the policy experiment (e.g., a ban on price discrimination or the introduction of output subsidies).

[32] This result applies only to triples (ℓ, s, s') for which $\theta_{ss'}^\ell > 0$. If $\theta_{ss'}^\ell = 0$, buyers in (ℓ, s) do not use inputs from sector s', so the transaction generates no surplus and both quantities and transfers are trivially zero. To simplify notation, we omit this restriction hereafter.

for unknown coefficients A_s^ℓ and ψ_s^ℓ that are constant at the buyer's layer-sector (ℓ, s) level.

Guess 3: Full Participation. We conjecture that flat fees are set so that every active buyer in the relevant layer-sector pair (ℓ, s) participates in each upstream relationship used in equilibrium. Hence, conditional on the equilibrium supplier set, flat fees are inframarginal: they redistribute surplus between buyers and sellers but do not affect marginal input demands, price indices, or variable cost functions.

4.3 Preliminaries

To proceed, it is useful to derive input demand, price indices, and cost functions under the conjectured contract structure. By Guess 3, these objects coincide with those in a uniform-pricing economy, except that marginal prices vary at the (ℓ, s) level. We then solve the unrestricted price-discrimination problem for a generic seller and verify the conjecture by matching undetermined coefficients.

4.3.1 Costs and Price Indices

CES sectoral price index. For any seller sector $s' \in \mathcal{S}$ and buyer layer–sector (ℓ, s), let p_{js}^ℓ denote the marginal price charged by seller variety $j \in \mathcal{U}_{s'}$ to buyers in (ℓ, s). The unit price of the s'-bundle faced by buyers in (ℓ, s) is

$$P_{ss'}^\ell \;=\; \left(\int_{j \in \mathcal{U}_{s'}} \left(p_{js}^\ell \right)^{1-\sigma_{s'}^u} \, dv_{s'}^u(j) \right)^{\frac{1}{1-\sigma_{s'}^u}} \tag{12}$$

Cobb–Douglas materials cost index. For firm $i = (\ell, s, z_i)$, the price aggregator of its composite materials bundle M_i in (10) is

$$P_s^{M,\ell} \;=\; \prod_{s' \in \mathcal{S}} \left(P_{ss'}^\ell \right)^{\theta_{ss'}^\ell} \tag{13}$$

Firm-level marginal cost. Define $\mathcal{S}_s^{\ell,+} \equiv \{s' \in \mathcal{S} : \theta_{ss'}^\ell > 0\}$. Given technology, wage $w > 0$, and constant returns to scale, the marginal cost of firm $i = (\ell, s, z_i)$ is

$$c_i \;=\; \frac{\Theta_s^\ell}{z_i} \, w^{\alpha_s^\ell} \left(P_s^{M,\ell} \right)^{1-\alpha_s^\ell}, \qquad \text{where} \quad \Theta_s^\ell \;\equiv\; \left(\alpha_s^\ell \right)^{-\alpha_s^\ell} \left(1 - \alpha_s^\ell \right)^{-(1-\alpha_s^\ell)} \prod_{s' \in \mathcal{S}_s^{\ell,+}} \left(\theta_{ss'}^\ell \right)^{-(1-\alpha_s^\ell)\theta_{ss'}^\ell}$$

Sectoral productivity index. We define the CES productivity index of seller firms of layer ℓ in sector s as

$$\widetilde{z}_s^{\ell} \;=\; \left(\int_{j\in\mathcal{F}_{\ell s}} z_j^{\sigma_s^{\ell}-1}\,\frac{dv_s^{\ell}(j)}{N_s^{\ell}}\right)^{\frac{1}{\sigma_s^{\ell}-1}}$$

4.3.2 Buyer Input Demand

Given the guesses in Section 4.2 and the objects defined in Section 4.3.1, each buyer $i = (\ell, s, z_i)$ chooses labor and input quantities from upstream seller varieties to maximize profits. Since flat fees are inframarginal, we formulate the maximization problem in terms of profits $\widetilde{\Pi}_i$, net of flat fees. Using Guess 2, buyer i solves

$$\widetilde{\Pi}_i = \max_{l_i,\,\{m_{ij}\}_j}\left\{A_s^{\ell}\,Q_i^{\psi_s^{\ell}} \;-\; w\,l_i \;-\; \sum_{s'\in\mathcal{S}}\int_{j\in\mathcal{U}_{s'}} p_{js}^{\ell}\,m_{ij}\,dv_{s'}^{u}(j)\right\}$$

The total variable expenditure on inputs from seller sector s' can be expressed as $P_{ss'}^{\ell}\,M_{is'}$. The first-order condition with respect to $M_{is'}$ equates the marginal revenue product of the materials bundle to its sectoral price index, similarly for labor

$$\frac{\partial R_i}{\partial M_{is'}} \;=\; P_{ss'}^{\ell}, \qquad \frac{\partial R_i}{\partial l_i} \;=\; w$$

which determines the labor-materials ratio given $\{P_{ss'}^{\ell}\}$. For a given buyer i, demand for the materials bundle from upstream sector s', $M_{is'}$, is allocated across varieties $j \in \mathcal{U}_{s'}$ according to the CES share rule. Under the conjecture that markups are (ℓ, s, s')-specific but common across sellers $j \in \mathcal{U}_{s'}$, relative input demands across varieties depend only on a seller's marginal cost relative to the sectoral price index. Buyer identity enters solely through the scale term $M_{is'}$

$$m_{ij} \;=\; M_{is'}\left(\frac{p_{js}^{\ell}}{P_{ss'}^{\ell}}\right)^{-\sigma_{s'}^{u}} \;=\; M_{is'}\left(\frac{\widetilde{z}_{s'}^{u}}{z_j\,(N_{s'}^{u})^{\frac{1}{1-\sigma_{s'}^{u}}}}\right)^{-\sigma_{s'}^{u}}$$

Using this condition, together with market clearing, we can express total production of variety j as a function of its relative productivity

$$Q_j(z_j) \;=\; \left(\frac{z_j}{\widetilde{z}_{s'}^{u}}\right)^{\sigma_{s'}^{u}}\; Q_{s'}\!\left(\widetilde{z}_{s'}^{u}\right) \tag{14}$$

where $Q_{s'}(\widetilde{z}_{s'}^{u})$ denotes the total production of the average firm with productivity $\widetilde{z}_{s'}^{u}$ in upstream sector s'.

Scaling of input demand with productivity. Input usage scales with firm productivity relative to the sectoral average.[33] For a firm in (ℓ, s) with productivity z, labor and material demand satisfy

$$l(z) \;=\; \left(\frac{z}{\widetilde{z}_s^{\ell}}\right)^{\sigma_s^{\ell}-1} l\big(\widetilde{z}_s^{\ell}\big), \qquad M(z) \;=\; \left(\frac{z}{\widetilde{z}_s^{\ell}}\right)^{\sigma_s^{\ell}-1} M\big(\widetilde{z}_s^{\ell}\big)$$

Hence, if productivities are Pareto-distributed, input demand is also Pareto-distributed, with a tail parameter that depends on the relevant elasticity of substitution[34]

$$\xi_s^{\ell} \;=\; \frac{\kappa_s^{\ell}}{\sigma_s^{\ell}-1} \;>\; 1$$

4.4 The Optimal Nonlinear Price

A seller $j \in \mathcal{U}_{s'}$ offers, for each buyer layer–sector pair (ℓ, s), an unrestricted menu $\{x_{ss'}^{\ell}(\tau), T_{ss'}^{\ell}(\tau)\}$, where $x_{ss'}^{\ell}(\tau)$ denotes the allocated quantity and $T_{ss'}^{\ell}(\tau)$ the transfer for buyer type τ (we suppress the seller index j). Denote buyer i's profit (inclusive of transfers) by Π_i. For buyer i, the total surplus from transacting with seller j of productivity z_j is defined as

$$TS_{is'}(m_{ij}) := \left.\frac{d\,\Pi_i}{d\big(\nu_{s'}^{u}(z_j)\big)}\right|_{\arg\max \Pi_i}$$

namely, the marginal value to buyer i of access to an additional infinitesimal mass of sellers with productivity z_j within sector s', evaluated at buyer i optimal input choices. The gross value of this relationship is the surplus that an infinitesimal seller j seeks to appropriate through its contract. Under CES aggregation within sector s', the extractable surplus from a purchase of a generic size m can be written as

$$TS_{is'}(m) \;=\; \frac{\sigma_{s'}^{u}}{\sigma_{s'}^{u}-1}\,\frac{\partial R_i}{\partial M_{is'}}\,M_{is'}^{\frac{1}{\sigma_{s'}^{u}}}\;m^{\frac{\sigma_{s'}^{u}-1}{\sigma_{s'}^{u}}} \;-\; T$$

where m is the quantity purchased and T the associated transfer. Since the individual seller is infinitesimal, it treats the marginal revenue product as given. Using the first-order condition, this simplifies to

$$TS_{is'}(m) \;=\; \frac{\sigma_{s'}^{u}}{\sigma_{s'}^{u}-1}\,P_{ss'}^{\ell}\,M_{is'}^{\frac{1}{\sigma_{s'}^{u}}}\;m^{\frac{\sigma_{s'}^{u}-1}{\sigma_{s'}^{u}}} \;-\; T$$

[33] This follows from the homogeneity of the Cobb–Douglas technology: $y(x_1,\ldots) = x_1 \cdot y(1, x_2/x_1, \ldots)$ implies that input ratios are pinned down by common input prices. Scaling by relative productivity yields $\frac{l(z)}{l(\widetilde{z})} = (\frac{z}{\widetilde{z}})^{\sigma-1}$, so input demand inherits a Pareto distribution with effective tail parameter $\xi = \kappa/(\sigma-1)$.

[34] Equilibrium feasibility requires that $\xi_s^{\ell} > 1$. Aggregate labor demand is $L = \int l(z)\,d\nu(z)$, which is finite only if $\xi > 1$; hence, $\kappa_s^{\ell} > \sigma_s^{\ell} - 1$.

Valuation index and its distribution. For a seller in sector s', the buyer i matters only through the one-dimensional valuation index:

$$\tau_{is'} \;\equiv\; P_{ss'}^{\ell}\, M_{is'}^{1/\sigma_{s'}^{u}}$$

The valuation index, $\tau_{is'}$, combines the sector s' price level $P_{ss'}^{\ell}$ with the buyer's scale $M_{is'}$ in the exact way that determines the buyer's gross value from a purchase: the buyer's surplus from quantity m is proportional to $\tau_{is'}\, m^{(\sigma_{s'}^{u}-1)/\sigma_{s'}^{u}}$ (up to the transfer T). Hence, buyer heterogeneity enters the seller's problem solely through $\tau_{is'}$.

When buyer productivities z_i in (ℓ, s) are Pareto, $\tau_{is'}$ is Pareto as well. Under our technology, $M_{is'}$ is strictly increasing in z_i and is distributed according to a Pareto distribution with tail parameter ξ_s^{ℓ}. Therefore, $\tau_{is'}$ inherits the Pareto law of the productivity distribution z, with

$$\tau_{is'} \;\sim\; \text{Pareto}\big(\rho_{ss'}^{\ell}\big), \qquad \rho_{ss'}^{\ell} \;=\; \sigma_{s'}^{u}\, \xi_s^{\ell}$$

Rescaling by $P_{ss'}^{\ell}$ shifts only the scale (not the tail) of the distribution. Feasibility requires $\xi_s^{\ell} > 1$, which implies $\rho_{ss'}^{\ell} > \sigma_{s'}^{u}$ for all (ℓ, s, s'). Denote $G_{ss'}^{\ell}$ and $g_{ss'}^{\ell}$ the CDF and density of $\tau_{is'}$ for buyers in partition (ℓ, s, s'). Moreover, we can rewrite buyer surplus as a (ℓ, s, s')-specific function of τ, m, and T, denoted by $TS_{ss'}^{\ell}(\tau, m, T)$.

Seller's problem. Applying the revelation principle, a seller in sector s' chooses, for each buyer-sector pair (ℓ, s), a menu of allocations $x_{ss'}^{\ell}(\tau)$ and transfers $T_{ss'}^{\ell}(\tau)$ for all buyers $i = (\ell, s, z_i)$. The total profit-maximization problem is

$$\max_{\{x_{ss'}^{\ell}(\cdot),\, T_{ss'}^{\ell}(\cdot)\}} \;\sum_{\ell \in \{u,r\}} \sum_{s \in \mathcal{S}} N_s^{\ell}\, \mathbb{E}_{\tau \sim G_{ss'}^{\ell}} \Big[\, T_{ss'}^{\ell}(\tau) - c_j\, x_{ss'}^{\ell}(\tau) \,\Big] \tag{15}$$

subject, for each (ℓ, s), to

$$\text{(LIC)} \quad TS_{ss'}^{\ell}(\tau) = TS_{ss'}^{\ell}(\underline{\tau}_{ss'}^{\ell}) + \frac{\sigma_{s'}^{u}}{\sigma_{s'}^{u} - 1} \int_{\underline{\tau}_{ss'}^{\ell}}^{\tau} x_{ss'}^{\ell}(\omega)^{\frac{\sigma_{s'}^{u}-1}{\sigma_{s'}^{u}}}\, d\omega$$

$$\text{(IR)} \quad TS_{ss'}^{\ell}(\underline{\tau}_{ss'}^{\ell}) \;\geq\; 0$$

$$\text{(Monotonicity)} \quad x_{ss'}^{\ell}(\tau') \;\geq\; x_{ss'}^{\ell}(\tau) \quad \text{for } \tau' > \tau$$

The local incentive-compatibility constraint (LIC) ensures that truth-telling is optimal for all buyer types τ locally; (IR) is the individual-rationality constraint, binding for the lowest type $\underline{\tau}$; and monotonicity requires that higher buyer types receive weakly larger allocations. Taken together, (LIC) and monotonicity guarantee that the mechanism is incentive-compatible globally. Because

the objective and constraints are additively separable across (ℓ, s, s') triples, the problem can be solved independently for each partition.

Solution concept. Each partition's problem has the structure of Lemma 1, with type τ in place of z, gross value $\frac{\sigma^u_{s'}}{\sigma^u_{s'}-1}\tau\, x^{(\sigma^u_{s'}-1)/\sigma^u_{s'}}$ in place of $zR(q)$, and Pareto tail $\rho^\ell_{ss'}$ in place of κ. Marginal cost is constant and the tail condition $\rho^\ell_{ss'} > \sigma^u_{s'}$ holds, so we solve it via the virtual-surplus approach: the problem is solved pointwise in τ, and transfers follow from the envelope formula with IR binding at $\underline{\tau}$.

Proposition 1 (Optimal Nonlinear Price for Upstream Sellers)**.** *In equilibrium, the optimal contract offered by an upstream seller $j \in \mathcal{U}_{s'}$ to any buyer $i = (\ell, s, z_i)$ is a two-part tariff*

$$T_{ij} \;=\; p^\ell_{js}\, m_{ij} \;+\; F^\ell_{js}$$

with marginal price

$$p^\ell_{js} \;=\; \mu^\ell_{ss'}\, c_j, \qquad \mu^\ell_{ss'} \;=\; \frac{\rho^\ell_{ss'}}{\rho^\ell_{ss'}-1}, \qquad \rho^\ell_{ss'} \;=\; \xi^\ell_s\, \sigma^u_{s'}$$

a constant markup over marginal cost within each buyer layer–sector (ℓ, s) for a given seller sector s'. The fixed component F^ℓ_{js} is chosen so that the lowest buyer type obtains zero surplus:

$$F^\ell_{js} = \frac{1}{\sigma^u_{s'}-1}\left(\underline{\tau}^\ell_{ss'}\right)^{\sigma^u_{s'}}\left(p^\ell_{js}\right)^{1-\sigma^u_{s'}} \;=\; \left(\frac{z_j}{\overline{z}^u_{s'}}\right)^{\sigma^u_{s'}-1}\overline{F}^\ell_{ss'}, \qquad \underline{\tau}^\ell_{ss'} = P^\ell_{ss'}\left(M^\ell_{ss'}(\underline{z}^\ell_s)\right)^{1/\sigma^u_{s'}}$$

Here, $\overline{F}^\ell_{ss'}$ denotes the average flat fee per seller in sector s':

$$\overline{F}^\ell_{ss'} \equiv \frac{1}{N^u_{s'}}\frac{1}{\sigma^u_{s'}-1}\left(\underline{\tau}^\ell_{ss'}\right)^{\sigma^u_{s'}}\left(P^\ell_{ss'}\right)^{1-\sigma^u_{s'}} = \frac{P^\ell_{ss'}M^\ell_{ss'}(\underline{z}^\ell_s)}{N^u_{s'}(\sigma^u_{s'}-1)}$$

Proof sketch and verification of the guesses. Fix an upstream seller $j \in \mathcal{U}_{s'}$. Since sellers offer separate menus to each buyer partition (ℓ, s), the mechanism-design problem is separable across partitions. Applying Lemma 1 to $\tau_{is'} \sim G^\ell_{ss'}$, with Pareto tail $\rho^\ell_{ss'}$, gives $(1 - G^\ell_{ss'}(\tau))/g^\ell_{ss'}(\tau) = \tau/\rho^\ell_{ss'}$. The virtual-surplus FOC therefore implies a constant marginal price, $p^\ell_{js} = \mu^\ell_{ss'}c_j$, with $\mu^\ell_{ss'} = \rho^\ell_{ss'}/(\rho^\ell_{ss'}-1)$. Transfers follow from the envelope theorem and lowest-type IR, yielding the flat fee F^ℓ_{js}. Since the virtual surplus is positive for all active types, the optimal mechanism serves all active buyers.

To verify the revenue conjecture, note that an upstream seller earns revenue from two sources: marginal payments and flat fees. Both components scale with the same CES demand shifter, $(p^\ell_{js}/P^\ell_{ss'})^{1-\sigma^u_{s'}}$, while output scales with $(p^\ell_{js}/P^\ell_{ss'})^{-\sigma^u_{s'}}$. Therefore upstream revenue is homogeneous in output, with $R_j \propto Q_j^{(\sigma^u_{s'}-1)/\sigma^u_{s'}}$. For retailers, CES final demand gives $R_j \propto Q_j^{(\sigma^r_s-1)/\sigma^r_s}$. Hence

$\psi_s^\ell = (\sigma_s^\ell - 1)/\sigma_s^\ell$, and the derivation verifies the two-part-tariff, homogeneous-revenue, and full-participation conjectures. See Appendix C for details and further discussion.

Existence and uniqueness. It remains to ensure that the resulting pricing structure is consistent with a well-defined general equilibrium.

Proposition 2 (Existence and Uniqueness)**.** *Suppose the sufficient regularity conditions stated in Appendix C.8 hold. Conditional on the nonlinear-pricing structure characterized in Proposition 1, the economy admits a unique steady-state general equilibrium.*

The proof is constructive and is provided in Appendix C.8.

4.5 Other Pricing Regimes for Welfare Comparisons

We compare the nonlinear-pricing benchmark to two counterfactual policies: (i) monopolistic competition with uniform prices, which corresponds to a complete ban on price discrimination; and (ii) a planner-implemented allocation in a decentralized equilibrium, which is attained with a ban on price discrimination and an output subsidy that restores marginal-cost pricing conditional on entry. Appendix C.8 also establishes existence and uniqueness of the corresponding steady-state general equilibria under the same type of regularity conditions.

Monopolistic competition (uniform pricing). Under uniform pricing, each upstream seller $j \in \mathcal{U}_{s'}$ charges the same CES markup over marginal cost to all buyers, regardless of their partition (ℓ, s)

$$p_{js}^{\ell,\text{Lin}} \;=\; \mu_{s'}^{\text{Lin}}\, c_j \qquad \mu_{s'}^{\text{Lin}} \;\equiv\; \frac{\sigma_{s'}^u}{\sigma_{s'}^u - 1}$$

where c_j is the marginal cost of seller firm j. Thus, in contrast to nonlinear pricing, marginal prices do not vary across buyer partitions. Retailers in sector s sell to final demand at the CES markup

$$\mu_s^{r,\text{Lin}} \;=\; \frac{\sigma_s^r}{\sigma_s^r - 1}$$

For buyers of layer ℓ in sector s, the CES price index for inputs from upstream sector s' is

$$P_{ss'}^{\ell,\text{Lin}} = \left(\int_{j\in\mathcal{U}_{s'}} \left(p_{js}^{\ell,\text{Lin}}\right)^{1-\sigma_{s'}^u} dv_{s'}^u(j)\right)^{\frac{1}{1-\sigma_{s'}^u}} = \mu_{s'}^{\text{Lin}} \left(\int_{j\in\mathcal{U}_{s'}} c_j^{\,1-\sigma_{s'}^u}\, dv_{s'}^u(j)\right)^{\frac{1}{1-\sigma_{s'}^u}}$$

Lemma 2 (Efficiency with uniform pricing and ad-valorem output subsidies)**.** *Consider the economy under a complete ban on price discrimination, so all sellers post uniform prices. The efficient allocation is*

decentralized if upstream sellers in sector s' and retailers in sector s receive ad-valorem output subsidies

$$\tilde{\tau}^u_{s'} = \frac{1}{\sigma^u_{s'} - 1}, \qquad \tilde{\tau}^r_s = \frac{1}{\sigma^r_s - 1}$$

Then buyer-facing prices equal marginal costs and the resulting decentralized equilibrium implements the planner allocation.

This lemma is a special case of the general result in Theorem 1 of Baqaee and Farhi (2020a) (details in Appendix C.9). Efficiency is obtained in a decentralized equilibrium when each variety charges a markup equal to its buyer-surplus ratio and receives output subsidies that exactly offset the induced within-period pricing wedge. In our CES setting, the buyer-surplus ratio for an upstream variety coincides with $\mu^{\text{Lin}}_{s'}$, and for a retail variety with $\mu^{r,\text{Lin}}_s$. Though charging the CES markup delivers the correct expected profits and thereby ensures efficient entry, it distorts input choices by acting as a tax on production. An output subsidy is therefore required to undo this distortion and restore marginal-cost pricing conditional on entry.

4.6 Theoretical Results

We now collect the main equilibrium implications of nonlinear pricing in our supply-chain model. All of them rest on two properties. First, buyers' demand is Pareto distributed, so a two-part tariff is optimal. Second, revenue functions are isoelastic in their input bundles, so buyers' inverse demands for a variety share the same curvature and differ only in level, which shifts with productivity. Both properties are passed along the supply chain. Each seller's revenue then inherits the same homogeneous, isoelastic form (Corollary 1), and because marginal prices are constant and technology has constant returns to scale, each firm's marginal cost is constant. See Appendix C.7 for derivations.

Result 1: Allocative markup. For any buyer partition (ℓ, s) and seller sector s', the allocative markup under nonlinear pricing is strictly below the uniform-CES markup

$$\mu^{\ell}_{ss'} = \frac{\rho^{\ell}_{ss'}}{\rho^{\ell}_{ss'} - 1} < \frac{\sigma^u_{s'}}{\sigma^u_{s'} - 1} \quad \text{since} \quad \rho^{\ell}_{ss'} = \xi^{\ell}_s \, \sigma^u_{s'}, \quad \xi^{\ell}_s = \frac{\kappa^{\ell}_s}{\sigma^{\ell}_s - 1} > 1$$

The allocative markup takes the form of a CES markup evaluated at an effective elasticity, $\rho^{\ell}_{ss'} = \xi^{\ell}_s \sigma^u_{s'}$, the elasticity the seller effectively faces when setting its marginal price, which is the product of two elasticities. The elasticity of substitution $\sigma^u_{s'}$ measures how easily buyers substitute across varieties of the seller's sector. Under atomistic sellers, it is the only determinant of the markup under uniform pricing, and a higher $\sigma^u_{s'}$ lowers the markup by reducing seller market power. The second term is the screening elasticity ξ^{ℓ}_s, the elasticity of the share of buyers in (ℓ, s)

above a given size with respect to that size, which is constant under a Pareto distribution.[35] It combines the dispersion of buyer productivity, κ_s^ℓ, with the buyer's elasticity of substitution, σ_s^ℓ, which maps productivity differences into differences in scale.

A higher ξ_s^ℓ lowers the allocative markup: buyer demands are less dispersed, information rents are smaller, and the seller collects more of its margin through the flat fee rather than through a distorted marginal price. Because the screening elasticity belongs to the buyer's partition, the same seller charges different allocative markups to different buyer partitions.[36]

Result 2: Margin size and margin collection. Fix a seller $j \in \mathcal{U}_{s'}$ and a buyer partition (ℓ, s), let R_{js}^ℓ denote the seller's revenue from the partition, and define its allocative and flat-fee margins as

$$\text{Margin}_{js,\text{allocative}}^\ell = \frac{1}{R_{js}^\ell} \int_{\mathcal{F}_{\ell s}} \left(p_{js}^\ell - c_j\right) m_{ij}\, dv_s^\ell(i), \qquad \text{Margin}_{js,\text{flat}}^\ell = \frac{N_s^\ell F_{js}^\ell}{R_{js}^\ell}$$

Then

$$\text{Margin}_{js,\text{allocative}}^\ell + \text{Margin}_{js,\text{flat}}^\ell = \frac{1}{\sigma_{s'}^u}, \qquad \text{Margin}_{js,\text{flat}}^\ell = \frac{\xi_s^\ell - 1}{\sigma_{s'}^u\, \xi_s^\ell - 1}$$

Because revenue functions are isoelastic, the total margin is $1/\sigma_{s'}^u$ of revenue, determined solely by the seller's elasticity of substitution, exactly as under uniform CES pricing. Price discrimination changes how a seller collects its margin in a buyer group partition, not its size, and the split depends on a single statistic, the screening elasticity ξ_s^ℓ. A higher $\sigma_{s'}^u$ makes inverse demand flatter and shrinks the total surplus that any contract can appropriate. Given this margin, the flat-fee share rises with buyer homogeneity ξ_s^ℓ, and the allocative share falls by exactly the same amount.

As $\xi_s^\ell \to 1$, flat fees vanish and the seller collects its margin entirely through the marginal price, as under uniform pricing. As $\xi_s^\ell \to \infty$, the marginal price equals marginal cost and the flat fee collects the full margin, $1/\sigma_{s'}^u$ of revenue. These shares hold for the seller's revenue from the partition as a whole. Across buyers within the partition, the flat fee is a larger share of small buyers' payments, which generates the quantity discounts characterized in Result 3.

Result 3: Average prices and quantity discounts. Fix a seller sector s' and a buyer partition (ℓ, s). For any seller $j \in \mathcal{U}_{s'}$ and buyer $i = (\ell, s, z_i)$, the average unit price is $T_{ij}/m_{ij} = p_{js}^\ell + F_{js}^\ell/m_{ij}$, and the

[35] A 1% increase in the size threshold lowers the share of buyers above it by ξ_s^ℓ%. A higher ξ_s^ℓ therefore means more homogeneous buyers.

[36] Conditioning on the buyer's sector also keeps the problem tractable: within each partition, types are Pareto and the optimal contract is a two-part tariff, whereas a mixture of Pareto distributions would make the marginal price vary with quantity. Heterogeneity in allocative markups across partitions matters little quantitatively: a common screening elasticity, under which flat fees still differ across buyer sectors, leaves welfare and optimal policy essentially unchanged (Appendix E), consistent with unit prices that differ across buyer sectors mainly at small quantities (Section 3).

resulting total markup, the average unit price over marginal cost, satisfies

$$\frac{\frac{T_{ij}}{m_{ij}}}{c_j} = \mu^{\ell}_{ss'}\left(1 + \chi^{\ell}_{ss'}(i)\right), \qquad \chi^{\ell}_{ss'}(i) = \frac{1}{\sigma^{u}_{s'} - 1}\left(\frac{z^{\ell}_{s}}{z_i}\right)^{\sigma^{\ell}_{s}-1}$$

The total markup combines the allocative markup $\mu^{\ell}_{ss'}$, common to all buyers in the partition, with the flat fee per unit, $\chi^{\ell}_{ss'}(i)$, which declines with buyer productivity because larger buyers spread the flat fee over more units. For the lowest type, $\chi^{\ell}_{ss'} = 1/(\sigma^{u}_{s'} - 1)$, while for a sufficiently large buyer $\chi^{\ell}_{ss'}(i) \to 0$ and the total markup converges to $\mu^{\ell}_{ss'}$. Hence, the limiting discount between the lowest and largest buyer types depends only on the seller's elasticity of substitution:

$$d_{s'} \equiv 1 - \frac{\mu^{\ell}_{ss'}}{\mu^{\ell}_{ss'}\left(1 + \frac{1}{\sigma^{u}_{s'}-1}\right)} = \frac{1}{\sigma^{u}_{s'}}$$

The decomposition also shows what variation in average prices drives input allocation. Within a partition, average markups vary across buyers only through $\chi^{\ell}_{ss'}(i)$, so their dispersion reflects how the seller collects its margin, not how inputs are allocated. Only the allocative markup $\mu^{\ell}_{ss'}$ distorts the quantities purchased by a given buyer, generating deadweight losses.

Result 4: Flat fees and entry incentives. By Result 2, a seller's allocative margin and flat-fee revenue sum to the CES margin. Profits as a function of revenue therefore differ from the CES benchmark only through the flat fees a firm pays to its own suppliers. For an upstream firm $j \in \mathcal{U}_{s'}$ and a retailer $i \in \mathcal{R}_s$,[37]

$$\Pi^{u}_{j} = \underbrace{\frac{1}{\sigma^{u}_{s'}} R_j}_{\text{CES margin}} \underbrace{- \sum_{t \in \mathcal{S}} \int_{h \in \mathcal{U}_t} F^{u}_{h\,s'}\, d\nu^{u}_{t}(h)}_{\text{flat-fee payments}} \qquad \Pi^{r}_{i} = \underbrace{\frac{1}{\sigma^{r}_{s}} R_i}_{\text{CES margin}} \underbrace{- \sum_{s' \in \mathcal{S}} \int_{j \in \mathcal{U}_{s'}} F^{r}_{js}\, d\nu^{u}_{s'}(j)}_{\text{flat-fee payments}}$$

Flat fees are inframarginal: they do not affect input choices or final demand, and with a representative owner (the final consumer) they net out as transfers. In general equilibrium, however, they shift free-entry conditions across (ℓ, s). Under uniform pricing, CES profits equal the surplus each variety creates, so once output subsidies remove the markup wedge on input choices, entry is efficient (Lemma 2). Under nonlinear pricing, a firm's profits fall short of its CES margin by the

[37] Appendix C.7 expands these profit functions and characterizes the determinants of flat fees, which scale with each seller's CES demand shifter and with the importance of intermediate inputs in the buyer's production, as governed by the α's and θ's. Unlike in the standard screening benchmark, the lowest type's total profits need not vanish, because participation binds within each input relationship while profits integrate across all inputs, including labor. We assume parameters that keep the lowest type's operating profits nonnegative in each partition and verify that this holds in our calibration. Aggregating across buyer-seller relationships, average firm profits are affine functions of sectoral labor expenditures; this linearity is the key step in the existence and uniqueness proof (Appendix C.8).

flat fees it pays to its own suppliers, whether it is a retailer or an upstream firm. Flat fees thus act as a tax on entry in every partition, whose size varies across layers and sectors with the importance of intermediate inputs. They distort both the level and the composition of entry, through profits rather than marginal prices, so output subsidies that correct marginal prices do not undo them. Section 5.4 assesses what this implies for optimal policy.

4.7 Welfare decomposition

We decompose welfare changes into an expenditure component, an intensive margin operating through allocative markups, and an extensive margin operating through the mass of active firms.

Let $n := |\mathcal{S}|$, and let $b \in \mathbb{R}^{1\times n}$ denote final-demand expenditure shares across retail sectors. Under Cobb–Douglas final demand, these shares are constant and equal to the preference parameters: $b_s = \theta_s$ for all $s \in \mathcal{S}$. Let $\mathcal{E}$ denote nominal household expenditure on the final consumption bundle. Welfare equals real expenditure, which in turn is equivalent to real income

$$\mathcal{W} = \frac{\mathcal{E}}{P_Y}, \qquad \Delta \log \mathcal{W} = \Delta \log \mathcal{E} - \Delta \log P_Y$$

In comparisons between the unsubsidized uniform-pricing and nonlinear-pricing equilibria, the unit labor endowment and the normalization $w = 1$ imply $\mathcal{E} = 1$ in both equilibria. Hence, $\Delta \log \mathcal{E} = 0$ for these comparisons.

Let $\tilde{\Omega}^{ru}, \tilde{\Omega}^{uu} \in \mathbb{R}^{n\times n}$ denote marginal-price input-output cost shares (excluding flat fees)

$$\tilde{\Omega}^{ru}_{ss'} = (1-\alpha^r_s)\theta^r_{ss'}, \qquad \tilde{\Omega}^{uu}_{ss'} = (1-\alpha^u_s)\theta^u_{ss'},$$

Here, $\tilde{\Omega}^{ru}_{ss'}$ is the marginal-price cost share of upstream seller sector s' in retail buyer sector s, and $\tilde{\Omega}^{uu}_{ss'}$ is the corresponding share for upstream buyer sector s. To capture direct plus indirect cost exposures of one sector to another, we define the upstream Leontief inverse and upstream cost-based Domar weights by

$$\tilde{\Psi}^{uu} := (I - \tilde{\Omega}^{uu})^{-1}, \qquad \tilde{\lambda}^{u} := b\,\tilde{\Omega}^{ru}\tilde{\Psi}^{uu}$$

Retail markups are collected in $\mu^r \in \mathbb{R}^{n\times 1}$. Upstream buyer-specific allocative markups are collected in $\mu^{ru}, \mu^{uu} \in \mathbb{R}^{n\times n}$, where $\mu^{ru}_{ss'}$ is charged by upstream seller sector s' to retail buyer sector s, and $\mu^{uu}_{ss'}$ is charged by upstream seller sector s' to upstream buyer sector s. In subsidized equilibria, these objects denote effective buyer-facing markups after accounting for the corresponding output subsidy. For each upstream seller sector s', define total exposure to retail and upstream buyers

$$\bar{\lambda}^{ru}_{s'} := \sum_{s=1}^{n} b_s \tilde{\Omega}^{ru}_{ss'}, \qquad \bar{\lambda}^{uu}_{s'} := \sum_{s=1}^{n} \tilde{\lambda}^{u}_{s} \tilde{\Omega}^{uu}_{ss'}$$

Whenever these exposures are positive, define the corresponding exposure-weighted average log-

markup changes

$$\Delta \log \bar{\mu}^{ru}_{s'} := \frac{1}{\bar{\lambda}^{ru}_{s'}} \sum_{s=1}^{n} b_s \tilde{\Omega}^{ru}_{ss'} \Delta \log \mu^{ru}_{ss'}, \qquad \Delta \log \bar{\mu}^{uu}_{s'} := \frac{1}{\bar{\lambda}^{uu}_{s'}} \sum_{s=1}^{n} \tilde{\lambda}^{u}_{s} \tilde{\Omega}^{uu}_{ss'} \Delta \log \mu^{uu}_{ss'}$$

Proposition 3 (Exact welfare decomposition)**.** *Assume CES/Cobb–Douglas aggregation, fixed marginal-price cost shares, and Pareto heterogeneity with invariant composition of active firms. Let $\sigma^r = (\sigma^r_s)_{s\in\mathcal{S}}$ and $\sigma^u = (\sigma^u_{s'})_{s'\in\mathcal{S}}$ collect the retail and upstream CES elasticities. Then welfare changes satisfy*

$$\Delta \log \mathcal{W} = \underbrace{\Delta \log \mathcal{E}}_{\text{expenditure}} \underbrace{+ - b \Delta \log \mu^r - \sum_{s'=1}^{n} \bar{\lambda}^{ru}_{s'} \Delta \log \bar{\mu}^{ru}_{s'} - \sum_{s'=1}^{n} \bar{\lambda}^{uu}_{s'} \Delta \log \bar{\mu}^{uu}_{s'}}_{\text{intensive margin: markups}} + \underbrace{b\left(\frac{\Delta \log N^r}{\sigma^r - 1}\right) + \tilde{\lambda}^u \left(\frac{\Delta \log N^u}{\sigma^u - 1}\right)}_{\text{extensive margin: firm masses}} \tag{16}$$

where divisions by $\sigma^r - 1$ and $\sigma^u - 1$ are elementwise.

The proof and further derivation details are provided in Appendix C.10. The intensive and extensive terms provide a Domar-weight decomposition of $-\Delta \log P_Y$.[38] Welfare rises when effective markups fall or when firm masses rise through CES love of variety. Flat fees do not enter the intensive margin because they are inframarginal; they affect welfare through entry and hence through the extensive margin.

Relative to uniform pricing, nonlinear pricing lowers upstream allocative markups, with gains scaled by the corresponding exposure weights. The extensive margin can amplify or offset these gains depending on how firm masses adjust in highly exposed sectors. For policy experiments and the subsidized decentralized implementation of the efficient allocation, final expenditure generally changes because subsidies are financed through the household budget, so $\Delta \log \mathcal{E}$ must also be included. We quantify these components in the next section.

5 Model Calibration and Quantification

This section takes the model to the data. Subsection 5.1 calibrates it to Chilean firm-level balance sheets and the population of firm-to-firm invoices: profit shares discipline market power, payments net of flat fees discipline the input–output structure, and employment concentration among the largest firms disciplines the screening elasticities. We then compare the calibration against the quantity discounts documented in Section 3, which we do not target. Subsection 5.2 uses the calibrated model to quantify the welfare cost of market power under nonlinear pricing,

[38] The decomposition is an accounting of the price index and does not isolate the causal effect of each distortion; Section 5.4 quantifies the value of correcting each distortion with the policy instruments.

the effects of banning price discrimination, and optimal policy, and shows how the information friction shapes each result.

5.1 Calibration

We calibrate the model to Chilean firm-to-firm transaction data and firm-level accounting data, and compare it against the quantity discounts documented in Section 3. Below we describe how each parameter is identified. Appendix D provides details.

Entry costs and survival probabilities (c^E, δ). Entry costs are sunk labor costs of creating an operating firm, and survival probabilities govern how long firms remain active. Appendix C.11 shows that, for a given entry-cost vector held fixed across regimes, the level of entry costs affects firm masses and welfare levels but not proportional welfare comparisons across pricing or policy regimes. Survival probabilities enter the equilibrium only through the effective entry cost $c_s^{E\,\ell}(1 - \delta_s^\ell)$, so the same result applies to them. Since our analysis relies only on such comparisons, we normalize $c_s^{E\,\ell}(1 - \delta_s^\ell) = 1$ for all (ℓ, s).

Final-demand elasticities (θ_s). The parameters θ_s are the Cobb–Douglas elasticities of the final-good aggregator with respect to each retail sector. Because retailers price linearly to final consumers, θ_s equals sector s's share of final expenditure. Retail and wholesale has the largest share (0.32), followed by financial services (0.24) and transport and ICT (0.12). Agriculture has the smallest (0.01).

Labor elasticities (α_s^ℓ). The model has no separate capital input, so we interpret labor as a composite primary input of labor and capital. We estimate output elasticities for each sector-layer pair following Ackerberg et al. (2015), with output and materials deflated by firm-level price indices built from invoices, set α_s^ℓ to the sum of the labor and capital elasticities, and set the materials elasticity to $1 - \alpha_s^\ell$ under constant returns to scale. The labor elasticity averages 0.41 for upstream firms and 0.42 for retailers. It is lowest in goods-producing sectors (0.22 for mining retailers) and highest in services (0.59 for business-services retailers), so the materials share ranges from 0.41 to 0.78.

Input-output elasticities ($\theta_{ss'}^\ell$). The parameters $\theta_{ss'}^\ell$ are the Cobb–Douglas elasticities of the materials bundle of firms in sector s and layer ℓ with respect to inputs from sector s'. Under cost minimization, they equal shares of materials expenditure valued at marginal prices. Observed payments also include flat fees, whose size depends on $\sigma_{s'}^u$ and ξ_s^ℓ, so we recover $\theta_{ss'}^\ell$ jointly with the elasticities of substitution (see below). Own-sector inputs account on average for 29% of up-

stream and 26% of retail materials spending. Retail and wholesale is the most important supplier sector, with 33% of upstream firms' materials on average.

Screening elasticities (ξ_s^ℓ). The screening elasticity ξ_s^ℓ governs the gap between the allocative markup and the CES markup, so it is central to the results. It is the Pareto exponent of input demand, which in the model is proportional to $z^{\sigma_s^\ell - 1}$. Under a Pareto distribution, the ratio of employment held by the top 1% of firms to that held by the top 5% equals $(0.01/0.05)^{1-1/\xi_s^\ell}$, which we invert to recover ξ_s^ℓ. Appendix D.2 shows that the estimates are stable when the lower threshold is widened to the top 10% and 25%. Given σ_s^ℓ, ξ_s^ℓ also pins down the productivity dispersion, κ_s^ℓ. Weighted by final-demand exposure, ξ_s^ℓ averages 1.29 for retailers and 1.45 for upstream firms. The largest exposure weights fall on retail and wholesale ($\xi = 1.19$ for retailers, 1.26 upstream) followed by financial services among retailers (1.14) and manufacturing among upstream firms (1.56).

Elasticities of substitution ($\sigma_{s'}^u$, σ_s^r). The elasticities of substitution govern the level of market power, which is notoriously hard to measure. Edmond et al. (2023), for instance, recalibrate their model for a range of aggregate markups rather than commit to one. We choose to discipline both elasticities with the profit decomposition in Result 4 and assess the sensitivity of our welfare results to their level. Profits equal a share $1/\sigma_{s'}^u$ of revenue for upstream firms and $1/\sigma_s^r$ for retailers, minus the flat fees the firm pays to its own suppliers. Ignoring fees, the median revenue-to-profit ratio among firms gives raw estimates $\hat{\sigma}_{s'}^u$ and $\hat{\sigma}_s^r$. Since fees depend on $\sigma_{s'}^u$, ξ_s^ℓ and $\theta_{ss'}^\ell$, we solve a fixed point: given the elasticities, we compute the fees implied by the model, net them out of observed profits and input payments, update $(\sigma^u, \sigma^r, \theta)$, and iterate to convergence. Netting out fees lowers the average upstream elasticity from 4.9 to 3.0. After the correction, $\sigma_{s'}^u$ ranges from 2.2 (financial services) to 3.9 (retail and wholesale), and σ_s^r from 2.3 (business services) to 4.2 (manufacturing). The implied uniform pricing CES markups range from 1.3 to 1.8.

Implied quantity discounts. Quantity discounts are not targeted, so they provide an untargeted check on the model's magnitude: they are pinned down by the elasticities of substitution, which we discipline with profit shares rather than prices. Result 3 implies that the limiting decline in the average unit price between the smallest and largest buyers is $d_{s'} = 1/\sigma_{s'}^u$. We compare this prediction with the discounts in the five sectors whose price schedule is decreasing and convex. These account for 93% of transactions and 76% of the upstream welfare weight, the weight with which upstream markups enter welfare. The model matches the average discount well: 27.6% against 30.8% in the data when weighting by transactions, and 30.4% against 28.1% when weighting by welfare weights. It overstates the discount in manufacturing by 6.4 percentage points and understates it in retail and wholesale by 8.4. The largest misses are in transport and ICT and in personal services, where the model overstates the discount by 23 and 18 percentage points. Together, these

two sectors account for 3% of transactions and 15% of the welfare weight. Appendix E shows how the welfare results change when we vary the level of the elasticities and when we use the discount-implied $\sigma^u_{s'}$ in the convex sectors.

5.2 Quantification

We now use the calibrated model to answer three questions. First, how does nonlinear pricing affect aggregate welfare relative to uniform pricing, through which channels, and how far does it leave the economy from the efficient allocation? Second, how should a government restricted to uniform revenue and entry subsidies set them, and how costly is it to design policy as if prices were uniform? Third, how do these answers depend on the information friction, as captured by the screening elasticity?

Nonlinear pricing raises welfare relative to uniform pricing. The economy attains 84% of efficient welfare, against 75% under uniform pricing, so nonlinear pricing closes 40% of the log welfare gap. It does so by lowering allocative markups along the supply chain, at the cost of a new distortion: flat fees reduce entry incentives. This changes optimal policy. Relative to uniform pricing, the optimal revenue subsidy falls by a third and the optimal entry subsidy rises from 2% to 17%. Applying the policy that is optimal under uniform pricing costs 4.9 log points of welfare relative to the nonlinear pricing optimal policy.[39] As buyers become more homogeneous (higher screening elasticity), sellers collect a larger share of their margin through flat fees. Welfare without policy, relative to the efficient allocation, is then hump-shaped in the screening elasticity, the optimal mix shifts toward the entry subsidy, and the cost of the uniform-pricing prescription rises, eventually leaving welfare below the no-policy level.

The first two exercises hold the calibrated primitives fixed and change only the pricing regime or the policy, so that differences across them isolate the role of nonlinear pricing. The third varies ξ and holds all other parameters fixed.[40]

5.3 The Aggregate Welfare Consequences of Nonlinear Pricing

We first ask how nonlinear pricing affects aggregate welfare. We compare the status quo with a counterfactual that bans price discrimination: upstream firms must charge every buyer the same per-unit price, with no flat fees, quantity schedules, or buyer- or sector-specific prices. We compare both regimes with the efficient allocation, which is implemented by banning price discrimination and subsidizing output to restore marginal-cost pricing (Lemma 2).

[39] Log points are 100 times log differences. We report welfare changes in log points because they add up across the components of the decomposition in Proposition 3.

[40] A different question is how large the welfare cost of market power appears when a model that ignores nonlinear pricing is calibrated to the same data. Such a model understates the cost, because it reads profits net of flat fees as weak market power. Appendix E.4 reports this exercise.

The nonlinear-pricing economy attains 83.9% of efficient welfare, against 74.5% under uniform pricing (Table 3, Panel (a)). Nonlinear pricing therefore raises welfare by 11.8 log points relative to uniform pricing. Since uniform pricing lies 29.4 log points below the efficient allocation, nonlinear pricing closes 40% of this gap.

Panel (b) decomposes each regime's welfare gap to the efficient allocation using Proposition 3, and column (3) reports the difference between nonlinear and uniform pricing. The intensive margin collects the welfare effects of markups for given firm masses, weighted by exposure to final demand. The extensive margin collects the weighted masses of retailers and upstream firms. The retail markup term does not differ across the two regimes, because retailers price linearly to final consumers in both.

The intensive margin drives the result (column (3)). Lower allocative markups along the supply chain contribute 17.8 log points: 5.5 from retail-to-upstream markups and 12.3 from upstream-to-upstream markups. Two-thirds of the gain thus comes from sales between upstream firms, where markups compound along the supply chain. The extensive margin works in the opposite direction and offsets about a third of this gain (−6.1 log points). It is driven by lower retail entry (−6.4): retail revenue is pinned down by final demand, so retailers pay flat fees out of an unchanged gross margin, and fewer of them break even. Upstream variety barely changes (+0.3).

Upstream firms also pay flat fees, but lower allocative markups raise demand for intermediate inputs, and the CES margin on their larger revenue offsets the fees they pay (Result 4). Nonlinear pricing thus trades a smaller allocative distortion for a new distortion in entry incentives. The policy analysis in Section 5.4 uses two instruments, a revenue subsidy and an entry subsidy, to quantify the welfare cost of each distortion.

The nonlinear-pricing economy remains 17.6 log points below the efficient allocation (column (1)). Almost all of this loss arises on firm-to-firm transactions: an intermediate benchmark that removes distortion between firms but keeps retail markups attains 98.9% of efficient welfare (Table 3, Panel (a)).[41] Markups raise the price of final consumption relative to the efficient allocation, but the nonlinear-pricing economy also avoids the lump-sum tax that finances the efficient subsidies (+80.9), which offsets most of the intensive loss (−82.0). Because this split depends on how the efficient allocation is implemented, we report it gross of the tax: markups account for about four-fifths of the losses, and entry for the rest.

The extensive margin shows where entry is distorted. Relative to the efficient allocation, the nonlinear-pricing economy has too few upstream firms (−28.7) and too many retailers (+12.2). In the efficient allocation, no spending is diverted to markups along the supply chain, so demand for intermediate inputs expands and upstream entry rises. With aggregate labor supply fixed, the labor these upstream firms use comes mostly from retail entry.

[41] The remaining loss is generated by heterogeneity in markups across retail sectors, which creates misallocation across sectors. If markups were homogeneous across retail sectors, correcting supply-chain distortions would restore efficiency.

Appendix E.1 breaks both decompositions down by sector. In both, retail and wholesale, transport and ICT, and financial services account for most of the effect. We also perform different robustness exercises, reported in Table A16 in Appendix E. First, we replace $\sigma^{\mu}_{s'}$ with the values implied by quantity discounts in the convex sectors. Second, we raise the labor elasticity α^{ℓ}_{s} of all firms by 10% and 20%. Third, we raise all elasticities of substitution, for both retailers and upstream firms, by 10% and 20%. Fourth, we re-estimate the screening elasticities from broader parts of the employment distribution and recalibrate the model. Fifth, we take final-demand shares from the national accounts. Sixth, we impose a common screening elasticity across buyer sectors, so that allocative markups no longer vary across them. In all exercises, the share of the gap between uniform pricing and the efficient allocation closed by nonlinear pricing is stable, ranging from 35.2% to 42.9%. The alternative elasticities, final-demand shares, and the common screening elasticity barely change welfare levels. The other exercises lower the welfare cost of market power, as expected: a higher labor elasticity weakens the accumulation of markups along the supply chain, and higher elasticities of substitution lower markups.

5.4 Optimal Policy

Banning price discrimination and subsidizing output implements the efficient allocation (Lemma 2), but a ban may not be feasible, and governments rarely have sector- or firm-specific instruments. We therefore consider a planner restricted to two uniform instruments: an ad valorem revenue subsidy τ^{y} on all sellers' receipts, which lowers every markup from μ to $\mu/(1+\tau^{y})$, and an entry subsidy τ^{e}, which lowers private entry costs to $(1-\tau^{e})c^{E}$ and leaves markups unchanged. Because only total payments are observable, the revenue subsidy also applies to flat fees, which raises the value of entry for upstream firms. Both instruments are financed by a lump-sum tax on the representative consumer, and welfare is $\mathcal{W} = \mathcal{E}/P_{Y}$, where $\mathcal{E}$ is final expenditure net of the tax. We search over a grid of (τ^{y}, τ^{e}) under each pricing regime, holding the calibrated parameters fixed (Appendix E.3). Table 4 reports the results and decomposes each policy's welfare effect into the expenditure term and the intensive and extensive margins. Because the revenue subsidy shifts every log markup by $-\log(1+\tau^{y})$, the intensive margin is mechanically positive.

Uniform pricing. Assuming uniform pricing, the optimal policy combines a revenue subsidy of 43% with an entry subsidy close to zero (2%). It raises welfare by 26.2 log points and attains 96.9% of efficient welfare. The revenue subsidy does almost all the work: markups are the only distortion, so correcting them also corrects entry incentives (Lemma 2). The residual loss reflects markup heterogeneity across sectors, which a uniform subsidy cannot undo.[42]

[42] Under uniform pricing, our model reduces to a CES economy with constant markups within sectors and abstracts from variable markups arising from demand curvature or strategic interaction. Edmond et al. (2023) find that, even with variable markups, much of the welfare cost of markups arises from an aggregate markup that acts as a uniform output tax, which a uniform subsidy can offset.

Table 3: Aggregate Welfare under Nonlinear and Uniform Pricing

Panel (a). Welfare (% of efficient allocation)	
Uniform pricing	74.5
Nonlinear pricing (status quo)	83.9
Efficient firm-to-firm transactions	98.9
Efficient allocation	100.0

Panel (b). Decomposition relative to the efficient allocation (log points)	(1) Nonlinear vs. efficient	(2) Uniform vs. efficient	(3) Nonlinear vs. uniform
Total $\Delta \log \mathcal{W}$	−17.6	−29.4	11.8
Lump-sum tax (efficient allocation)	80.9	80.9	0.0
Retail markups	−40.6	−40.6	0.0
Retail-to-upstream markups	−17.9	−23.4	5.5
Upstream-to-upstream markups	−23.6	−35.9	12.3
Total intensive margin	−82.0	−99.9	17.8
Retail variety	12.2	18.6	−6.4
Upstream variety	−28.7	−29.0	0.3
Total extensive margin	−16.5	−10.4	−6.1

Notes: Panel (a) reports steady-state welfare relative to the efficient allocation. Uniform pricing bans price discrimination (a single per-unit price, no flat fees). Efficient firm-to-firm transactions removes markups between firms but keeps retail markups. The efficient allocation bans price discrimination and adds output subsidies that restore marginal-cost pricing (Lemma 2). Panel (b) decomposes $100 \times (\log \mathcal{W} - \log \mathcal{W}^{\text{eff}})$ for each pricing regime using Equation (16). Column (3) is the difference between columns (1) and (2), the welfare change from uniform to nonlinear pricing. The first row reflects the lump-sum tax that finances the output subsidies implementing the efficient allocation; it is a transfer and does not affect allocations. Components may not add up exactly due to rounding.

Nonlinear pricing. The optimal revenue subsidy falls by about a third, to 29%, and the optimal entry subsidy rises from 2% to 17%. This mix attains 98.2% of efficient welfare, closing 90% of the gap. The revenue subsidy falls for two reasons. First, allocative markups are lower under nonlinear pricing (Result 1), so marginal prices reach marginal cost at a smaller subsidy. Second, because the subsidy also applies to flat-fee receipts, part of it raises upstream profits without lowering marginal prices, acting as a subsidy to upstream entry. The entry subsidy rises because flat fees lower profits in both layers and act as a tax on entry (Result 4). Under uniform pricing, the subsidy that corrects marginal prices also corrects entry. Under nonlinear pricing, that subsidy is smaller and leaves firm masses far from their efficient levels, and the entry subsidy offsets the fee tax. This pattern holds across all robustness exercises in Appendix E: the optimal revenue subsidy under nonlinear pricing remains well below its uniform-pricing counterpart, and the optimal

Table 4: Optimal Policy under Uniform and Nonlinear Pricing

			Welfare		Below	$\Delta \log \mathcal{W}$ vs. no policy (log points)			
	τ^y	τ^e	% eff.	Gap closed	cost (%)	Expenditure	Intensive	Extensive	Total
Panel (a). Uniform-pricing economy									
No policy	0.00	0.00	74.5	0.0	0	0.0	0.0	0.0	0.0
Optimal policy	0.43	0.02	96.9	89.3	41	−73.6	90.4	9.5	26.2
Panel (b). Nonlinear-pricing economy									
No policy	0.00	0.00	83.9	0.0	0	0.0	0.0	0.0	0.0
Optimal policy	0.29	0.17	98.2	89.7	31	−64.6	64.4	16.0	15.8
Revenue subsidy only	0.29	0.00	97.3	84.3	31	−54.7	64.4	5.2	14.8
Entry subsidy only	0.00	0.16	84.6	4.7	0	−9.3	0.0	10.1	0.8
Uniform-pricing optimum	0.43	0.02	93.5	61.8	67	−84.6	90.4	5.1	10.9

Notes: τ^y is a revenue subsidy on all sellers' receipts, including flat fees, and τ^e is a subsidy on private entry costs. Both are uniform across sectors and layers and financed by a lump-sum tax. Gap closed is the share (%) of the log welfare gap between each economy's no-policy equilibrium and the efficient allocation. "Uniform-pricing optimum" applies the optimum of Panel (a) to the nonlinear-pricing economy. "Revenue subsidy only" and "Entry subsidy only" optimize one instrument with the other set to zero. Below cost is the share (%) of markup exposure priced below marginal cost after the subsidy, weighting each retail markup by its final-demand share and each firm-to-firm link by its exposure in Equation (16). The last four columns decompose the log welfare change relative to no policy using Equation (16). The expenditure term is the change in final expenditure from the lump-sum tax that finances the subsidies; the intensive and extensive margins collect the markup and firm-mass terms.

entry subsidy well above it.

The two instruments play distinct roles. On its own, the revenue subsidy delivers 14.8 of the 15.8 log points attained by the optimal policy, as markup gains of 64.4 log points are offset by an expenditure term of -54.7. Adding the entry subsidy raises the extensive margin from 5.2 to 16.0 log points, close to the 16.5 that separate the no-policy equilibrium from the efficient allocation (Table 3, column (1)). The same holds in each layer: upstream variety rises by 26.5 log points, against 28.7, and retail variety falls by 10.5, against 12.2, for the reasons discussed in Section 5.3. Although the entry subsidy brings firm masses closer to their efficient levels, most of this gain is offset by the expenditure term, so the entry subsidy raises welfare by only 1.0 log point.

The uniform-pricing prescription. Holding the primitives fixed, we apply the policy that is optimal under uniform pricing, a 43% revenue subsidy and an almost zero entry subsidy, to the nonlinear-pricing economy.[43] It attains 93.5% of efficient welfare and closes only 62% of the gap, 4.9 log points below the optimal policy. The prescription overshoots. It prices two-thirds of markup exposure below marginal cost, against under a third at the nonlinear-pricing optimum,

[43] A different exercise calibrates a uniform-pricing model to the same data and applies its optimal policy to our economy. This mixes the effect of the pricing assumption with a calibration error, and the two partly offset. We report it in Appendix E.4.

and relative to the revenue-subsidy-only optimum, its additional markup gains (26.0 log points) are more than offset by its additional expenditure term (-29.9). Its entry subsidy is also too small to offset the fee tax. More generally, any given revenue subsidy delivers smaller welfare gains under nonlinear pricing than under uniform pricing. As the revenue subsidy increases further, welfare falls steeply: for revenue subsidies of 55% or more, welfare is below the no-policy level (Appendix E.3).

5.5 Information Frictions, Welfare, and Optimal Policy

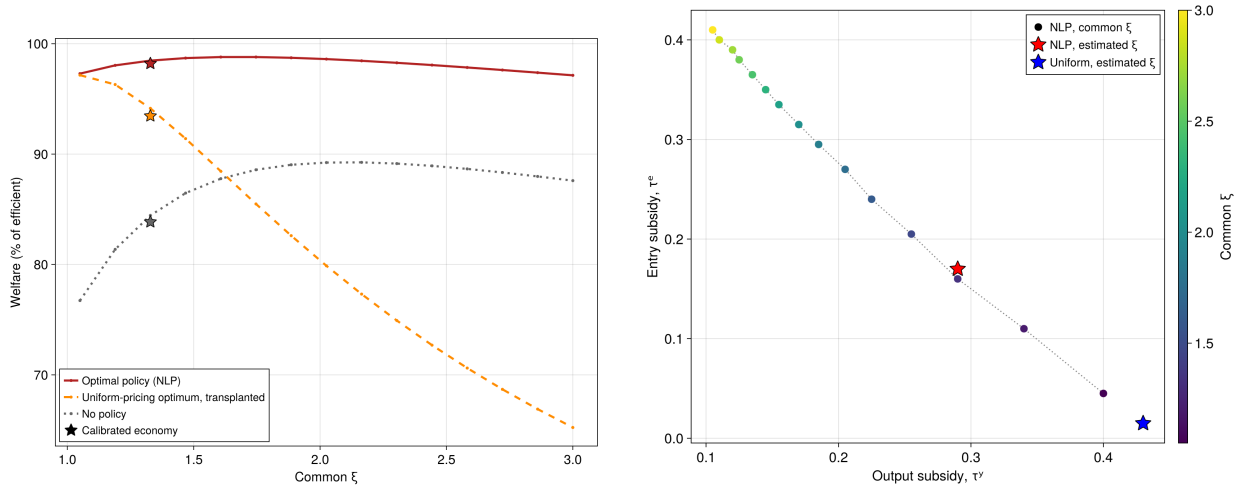
Sellers use nonlinear schedules because they do not observe their buyers' productivity: the menu screens buyers through the quantities they choose. Under Pareto distributed types, this information friction is encoded in the screening elasticity ξ. A higher ξ means more homogeneous buyers and hence weaker information frictions. The screening elasticity thus determines how sellers collect their margin: through marginal prices, which distort allocations, or through flat fees, which distort entry (Results 2 and 4).

As $\xi \to 1$, flat fees vanish and nonlinear pricing converges to uniform CES pricing. As ξ grows, marginal prices approach marginal cost and flat fees absorb a growing share of the margin. In the limit $\xi \to \infty$, marginal prices equal marginal cost and flat fees extract each buyer's surplus, a limit we refer to as full surplus extraction. Changing ξ therefore isolates the effect of the information friction.

We now ask how this trade-off shapes welfare and optimal policy, including the cost of ignoring nonlinear pricing when designing policy. We vary ξ while holding the elasticities of substitution fixed, so that the share of revenue sellers collect as margin is unchanged and only the way they collect it moves. We set a common ξ across sectors and layers, from just above one to three, and keep all other calibrated parameters fixed, so that productivity dispersion κ_s^ℓ adjusts.[44] For each value, we solve the economy without policy, under its own optimal policy, and under the policy that is optimal in the calibrated economy under uniform pricing. The calibrated economy corresponds to a policy-equivalent $\xi \approx 1.33$, the common ξ whose optimal policy is closest to the calibrated one. At this common value, allocative markups no longer vary across buyer sectors, yet welfare without policy (84.5% of the efficient level, against 83.9%) and the share of the gap closed by nonlinear pricing (42.5%, against 40.1%) are close to those in the calibrated economy. Heterogeneity in screening elasticities across buyer sectors, and hence the buyer-sector dimension of marginal prices, therefore matters little for our quantitative results; what matters is their average level.

Welfare without policy, relative to the efficient allocation, is hump-shaped in ξ (Figure 6(a)). It rises from 76.7% of the efficient level at the lowest ξ on our grid, approaching the uniform-

[44] Under uniform pricing, changing ξ affects neither welfare relative to the efficient allocation nor the optimal policy: with CES demand and atomistic firms, productivity dispersion scales welfare in the decentralized and efficient allocations by the same factor. We verify this numerically.



(a) Welfare (% of efficient)

(b) Optimal policy

Figure 6: Welfare and Optimal Policy under a Common screening elasticity ξ

Notes: All sectors and layers share a common screening elasticity ξ. All other parameters are held at their calibrated values, so productivity dispersion $\kappa_s^\ell = \xi(\sigma_s^\ell - 1)$ moves with ξ. Under uniform pricing, neither the optimal policy nor welfare relative to the efficient allocation depends on ξ, so we report uniform pricing only at the calibrated values. Panel (a): welfare relative to the efficient allocation at each ξ, without policy, under the economy's own optimal policy, and under the uniform-pricing optimum of the calibrated economy. Stars mark the calibrated economy, placed at its policy-equivalent $\xi \approx 1.33$. Panel (b): optimal revenue subsidy τ^y and entry subsidy τ^e under nonlinear pricing for each ξ (colorbar). Stars mark the optima in the calibrated economy, with estimated sector-specific ξ, under nonlinear (red) and uniform (blue) pricing.

pricing level as $\xi \to 1$, to a peak of about 89% at $\xi \approx 2$, and then declines slowly, to 88% at $\xi = 3$. A higher ξ first lowers allocative markups and the losses they compound along the supply chain, but beyond the peak, flat fees absorb so much of sellers' margins that the entry distortion dominates. This argues against the standard notion that higher surplus extraction implies higher efficiency through improved allocations. In partial equilibrium, full surplus extraction is efficient, because allocations are not distorted. Between firms with free entry, moving toward it eventually lowers welfare: marginal prices approach marginal cost, but flat fees take a growing share of buyers' surplus and tax the profits that drive entry, distorting firm participation.

As ξ rises, the optimal mix shifts monotonically from the revenue subsidy toward the entry subsidy (Figure 6(b)), from $(\tau^y, \tau^e) = (0.40, 0.045)$ near $\xi = 1$, close to the uniform-pricing optimum, to $(0.105, 0.41)$ at $\xi = 3$. Lower allocative markups call for a smaller revenue subsidy and larger fees for a larger entry subsidy, for the reasons discussed in Section 5.4. The cost of the uniform-pricing prescription rises steeply (Figure 6(a)): it is small near $\xi = 1$, reaches 4.7 percentage points of efficient welfare at the calibration (4.9 log points), and about 32 at $\xi = 3$. For ξ above about 1.7, the prescription leaves welfare below the no-policy level. As buyers become more homogeneous, distortions operate increasingly through profits rather than prices, and a policy designed for an

economy in which markups are the only distortion becomes increasingly misguided. A planner who accounts for nonlinear pricing, by contrast, attains 97–99% of efficient welfare over the whole range.

6 Conclusion

Relative to uniform pricing, price discrimination between firms changes how market power distorts the economy and how policy should correct it. Using the universe of firm-to-firm transactions in Chile, we document departures from uniform pricing: unit prices decline with quantity purchased and, for small purchases, differ across buyer sectors, consistent with both second-degree (quantity discounts) and third-degree (buyer-sector-specific schedules) price discrimination. We develop a multisector general equilibrium model, with arbitrary input-output linkages, in which firms simultaneously charge and pay nonlinear prices.

Under CES demand and Pareto-distributed firm productivities, the optimal contract is a single buyer-sector-specific two-part tariff: a flat fee and a marginal price. On one hand, flat fees let sellers extract surplus with smaller quantity distortions than uniform pricing, bringing marginal prices closer to marginal cost and mitigating deadweight losses across the supply chain. On the other hand, flat fees redistribute profits along the supply chain with no effect on quantities traded and, in general equilibrium, distort firm entry. Calibrating the model to Chilean data, we find that banning price discrimination reduces welfare from 84% to 75% of the efficient benchmark, with losses arising primarily from higher allocative markups along the supply chain. Price discrimination thus closes more than a third of the gap between uniform pricing and the efficient allocation.

Relative to the optimal policy under uniform pricing, the optimal revenue subsidy falls from 43% to 29% and the optimal entry subsidy rises from 2% to 17%. Applying the uniform-pricing prescription costs about 5 percentage points of efficient welfare at our calibration, and it does worse than no policy once buyers are homogeneous enough. Our findings carry two policy implications. First, banning price discrimination without output subsidies reduces welfare because it forces firms to extract rents through marginal prices rather than through flat fees, thereby distorting quantities. Second, a policy designed under uniform pricing over-subsidizes revenue and under-subsidizes entry, and the error grows as buyers become more homogeneous.

Our model provides a tractable framework to evaluate the aggregate welfare consequences of nonlinear pricing, decomposing welfare changes into intensive and extensive margins and mapping sector-level distortions to aggregate outcomes through input–output linkages. More broadly, the welfare cost of market power in supply chains depends not only on how large sellers' margins are, but also on how sellers collect them: margins collected through marginal prices distort quantities and call for revenue subsidies, while margins collected through flat fees distort entry

and call for entry subsidies.

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Appendix

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A Optimal Nonlinear Price Derivation

Primitives. Consider a screening problem in which a monopolist offers quantity–transfer bundles to buyers with privately observed productivity type z, drawn from distribution $\mathbb{F}(z)$ with density $f(z)$ and support $[\underline{z}, \infty)$. The seller has constant marginal cost $c > 0$. A buyer of type z who receives quantity q generates gross revenue $zR(q)$, where $R(q)$ is increasing, strictly concave, and differentiable. A direct mechanism specifies a pair $(q(z), T(z))$, so that type z obtains net surplus

$$\Pi(z) = zR(q(z)) - T(z)$$

Seller's problem. By the revelation principle, it is without loss to restrict attention to direct mechanisms. The seller therefore chooses $\{q(z), T(z)\}_{z \geq \underline{z}}$ to maximize expected profit

$$\max_{\{q(z), T(z)\}} \int_{\underline{z}}^{\infty} \Big[T(z) - cq(z)\Big] f(z)\, dz$$

subject to individual rationality (IR) and incentive compatibility (IC):

$$\Pi(z) \geq 0, \qquad \Pi(z) \geq zR(q(\tilde{z})) - T(\tilde{z}) \quad \forall z, \tilde{z} \geq \underline{z}$$

Reduced problem. Under standard single-crossing and concavity conditions, the global incentive constraints can be replaced by a local characterization together with monotonicity of the allocation. In particular, incentive compatibility implies that transfers satisfy

$$T(z) = zR(q(z)) - \Pi(\underline{z}) - \int_{\underline{z}}^{z} R(q(s))\, ds$$

and the allocation must satisfy

$$q(z) \quad \text{weakly increasing in } z$$

Moreover, since buyer surplus is increasing in type, $\Pi'(z) = R(q(z)) \geq 0$, it is sufficient to impose individual rationality only for the lowest type, $\Pi(\underline{z}) \geq 0$.

Integrating by parts, which requires expected buyer surplus to be finite so that the boundary term $(1 - \mathbb{F}(z))\Pi(z)$ vanishes as $z \to \infty$, gives

$$\begin{aligned} \Pi_{\text{seller}} &= \int_{\underline{z}}^{\infty} \left[zR(q(z)) - \frac{1 - \mathbb{F}(z)}{f(z)} R(q(z)) - cq(z) \right] f(z)\, dz - \Pi(\underline{z}) \\ &= \int_{\underline{z}}^{\infty} \Big[\phi(z, q(z)) - cq(z)\Big] f(z)\, dz - \Pi(\underline{z}) \end{aligned}$$

where $\phi(z,q) \equiv \left(z - \frac{1}{h(z)}\right) R(q)$ is the virtual surplus and $h(z) = f(z)/(1-\mathbb{F}(z))$ is the hazard rate. The seller maximizes this expression subject to $q(z)$ weakly increasing and $\Pi(\underline{z}) \geq 0$.

Since $\Pi(\underline{z})$ enters negatively and does not affect the allocation, the seller optimally sets $\Pi(\underline{z}) = 0$. The problem therefore reduces to choosing a monotone allocation $q(\cdot)$ to maximize expected virtual surplus net of production cost.

A.1 Proof of Lemma 1

Under a Pareto distribution with tail parameter κ, the hazard rate is $h(z) = \kappa/z$, so the virtual surplus in the reduced problem becomes

$$\phi(z,q) = \left(1 - \frac{1}{\kappa}\right) zR(q) = \frac{\kappa - 1}{\kappa} zR(q)$$

Hence, with $\Pi(\underline{z}) = 0$, for each type z the seller solves the pointwise problem

$$\max_{q \geq 0} \left\{ \frac{\kappa - 1}{\kappa} zR(q) - cq \right\}$$

Since $R'(0) = \infty$, the solution is interior and satisfies the first-order condition:

$$\frac{\kappa - 1}{\kappa} zR'(q(z)) = c$$

Because

$$zR'(q(z)) = \frac{\kappa}{\kappa - 1} c \equiv p^{\text{NLP}}$$

the allocation $q(z)$ is exactly the quantity that type z would choose under a constant per-unit price p^{NLP}, since a buyer facing linear price p solves $\max_q zR(q) - pq$ and therefore chooses q such that $zR'(q) = p$. Since R is strictly concave, $q(z)$ is increasing in z, so the monotonicity constraint is satisfied. It remains to characterize the transfer schedule. By incentive compatibility,

$$T'(z) = zR'(q(z))q'(z) = p^{\text{NLP}} q'(z)$$

so $T(z) - p^{\text{NLP}} q(z)$ is constant across types. Denoting this constant by F, the optimal transfer schedule is

$$T(z) = F + p^{\text{NLP}} q(z)$$

where F is pinned down by the lowest-type participation constraint, $\Pi(\underline{z}) = 0$.

A.2 Proof of Corollary 1

Under the homogeneous revenue specification

$$R(q) = q^{\frac{\sigma-1}{\sigma}}, \qquad \sigma > 1$$

and under Pareto types with tail parameter $\kappa > \sigma$, the σ-th moment of types is finite:

$$M_\sigma := \int z^\sigma \, d\mathbb{F}(z) < \infty$$

A buyer facing marginal price p^{NLP} chooses quantity $q(z)$ to solve

$$\max_{q \geq 0} \; z\, q^{\frac{\sigma-1}{\sigma}} - p^{\text{NLP}} q$$

The first-order condition is

$$z \frac{\sigma - 1}{\sigma} q(z)^{-1/\sigma} = p^{\text{NLP}}$$

which implies

$$q(z) = \left(\frac{\sigma - 1}{\sigma}\right)^\sigma \left(p^{\text{NLP}}\right)^{-\sigma} z^\sigma$$

Hence quantities are proportional to z^σ.

Total output is therefore

$$Q := \int q(z) \, d\mathbb{F}(z) = \left(\frac{\sigma - 1}{\sigma}\right)^\sigma \left(p^{\text{NLP}}\right)^{-\sigma} M_\sigma$$

so

$$p^{\text{NLP}} = \frac{\sigma - 1}{\sigma} M_\sigma^{1/\sigma} \, Q^{-1/\sigma}$$

The flat fee is pinned down by the lowest-type participation constraint

$$\Pi(\underline{z}) = 0 \quad \Longrightarrow \quad F = \underline{z}\, R(q(\underline{z})) - p^{\text{NLP}} q(\underline{z}) = \underline{z}\, q(\underline{z})^{\frac{\sigma-1}{\sigma}} - p^{\text{NLP}} q(\underline{z})$$

Using the first-order condition at $z = \underline{z}$,

$$p^{\text{NLP}} q(\underline{z}) = \frac{\sigma - 1}{\sigma} \underline{z}\, q(\underline{z})^{\frac{\sigma-1}{\sigma}}$$

so

$$F = \frac{1}{\sigma} \underline{z}\, q(\underline{z})^{\frac{\sigma-1}{\sigma}}$$

Substituting

$$q(\underline{z}) = \left(\frac{\sigma - 1}{\sigma}\right)^\sigma \left(p^{\text{NLP}}\right)^{-\sigma} \underline{z}^\sigma$$

gives

$$F = \frac{1}{\sigma}\,\underline{z}^{\sigma} M_{\sigma}^{-\frac{\sigma-1}{\sigma}}\, Q^{\frac{\sigma-1}{\sigma}}$$

Total seller revenue equals revenue from the per-unit charge plus revenue from the flat fee. Since the flat fee is constant across buyers and the type distribution has unit mass,

$$R^{\text{seller}}(Q) = \int \left[p^{\text{NLP}} q(z) + F\right] d\mathbb{F}(z) = p^{\text{NLP}} Q + F$$

Using the expression for p^{NLP} above, $p^{\text{NLP}} Q = \frac{\sigma-1}{\sigma}\, M_{\sigma}^{1/\sigma}\, Q^{\frac{\sigma-1}{\sigma}}$. Adding F gives

$$R^{\text{seller}}(Q) = A\, Q^{\frac{\sigma-1}{\sigma}}, \qquad A = \frac{\sigma-1}{\sigma}\, M_{\sigma}^{1/\sigma} + \frac{1}{\sigma}\,\underline{z}^{\sigma} M_{\sigma}^{-\frac{\sigma-1}{\sigma}} > 0$$

Thus, the seller's revenue function is homogeneous of degree $(\sigma-1)/\sigma$ in total output Q, so it inherits the same curvature as the buyers' revenue function.

A.3 Virtual Surplus and Full Participation

Under the maintained assumptions, the monopolist optimally serves all buyer types, including the lowest type $\underline{z}$. With $R(q) = q^{\frac{\sigma-1}{\sigma}}$ and Pareto types, the proof of Lemma 1 shows that, for each type z, the seller solves the pointwise problem

$$\max_{q \geq 0} \left\{ \frac{\kappa-1}{\kappa}\, z\, q^{\frac{\sigma-1}{\sigma}} - cq \right\}$$

The derivative of the objective with respect to q is

$$\frac{\partial}{\partial q}\left[\frac{\kappa-1}{\kappa}\, z\, q^{\frac{\sigma-1}{\sigma}} - cq\right] = \frac{\kappa-1}{\kappa}\, z\, \frac{\sigma-1}{\sigma}\, q^{-1/\sigma} - c$$

Since $\sigma > 1$, this derivative tends to $+\infty$ as $q \downarrow 0$. Therefore, the seller strictly prefers a positive allocation to $q = 0$ for every type $z \geq \underline{z}$. The optimal quantity is thus strictly positive for all types:

$$q(z) > 0 \qquad \forall z \geq \underline{z}$$

In particular, the lowest type is served. Equivalently, raising the exclusion cutoff above $\underline{z}$ strictly reduces profit under the tail condition $\kappa > \sigma$, as shown in Section 2.

B Additional Descriptive Evidence

B.1 Residual Price Determinants Across Frequencies and Seller Industries

We extend the residual variance-accounting exercise in two directions. First, we repeat it on the same transactions with seller-product-month rather than seller-product-day fixed effects. Monthly cells contain more repeated transactions between the same seller, product, and buyer, allowing us to include buyer fixed effects. Second, we repeat the daily and monthly exercises for Manufacturing and Retail and Wholesale, the two seller industries with the largest transaction volumes. For each frequency $f \in \{\text{day}, \text{month}\}$, we first estimate

$$\ln p_{igjt} = \beta_0 + \Psi_{jgf} + \epsilon^f_{igjt}$$

where Ψ_{jgf} are seller-product-period fixed effects. We then project the residuals on alternative fixed-effect sets S:

$$\epsilon^f_{igjt} = \beta_0 + \Psi^f_S + \nu^f_{igjt}$$

At both frequencies, S includes quantity fixed effects, buyer-group fixed effects, and quantity-by-buyer-group fixed effects. Buyer groups are defined using sector, size, and region. At the monthly frequency, we also include seller-product-month-by-buyer fixed effects.

We do not report buyer fixed effects at the daily frequency because most buyers appear only once within a seller-product-day cell. Buyer fixed effects would therefore be close to transaction-specific indicators and would not provide an informative measure of persistent buyer-specific pricing. At the monthly frequency, repeated transactions allow these fixed effects to capture average price differences across buyers purchasing the same product from the same seller within a month, potentially reflecting, for instance, differences in buyer bargaining power, or buyers who regularly place large orders and therefore pay lower unit prices.

Table A1 reports the results. Within each row, all specifications use the same sample. With monthly cells, quantity fixed effects explain 34% of residual price variation in the full sample, as with daily cells, and buyer-group fixed effects explain 38%. Buyer fixed effects explain 48%, indicating that buyer identity contains relevant information beyond observable sector, size, and region. Quantity-by-buyer-group fixed effects explain 64%, 16 percentage points more than buyer fixed effects. Thus, even after adjusting for the number of parameters, residual price variation is better described by price-quantity menus that differ across observable buyer groups than by buyer-specific price levels alone.

The same pattern holds across seller industries. For Manufacturing, quantity-by-buyer-group fixed effects explain 80% of daily residual price variation and 78% of monthly residual price variation. The corresponding values for Retail and Wholesale are 59% and 69%. In every case, the

interacted specification explains more residual variation than quantity or buyer-group fixed effects alone. At the monthly frequency, it also explains more than buyer fixed effects. The relative importance of quantity and buyer identity differs across industries, but the central result is not driven by a single seller industry.

Table A1: Price Residual Determinants Across Frequencies and Seller Industries

	Adjusted R^2 by fixed-effect set S				
	Quantity	Buyer	Buyer group	Quantity × Buyer group	N
Panel A. Full sample					
Daily	0.34	—	0.29	0.55	271,916,183
Monthly	0.34	0.48	0.38	0.64	393,136,702
Panel B. Manufacturing sellers					
Daily	0.47	—	0.46	0.80	111,286,299
Monthly	0.45	0.49	0.41	0.78	146,085,515
Panel C. Retail and Wholesale sellers					
Daily	0.46	—	0.25	0.59	129,759,973
Monthly	0.38	0.60	0.44	0.69	199,949,350

Notes: This table reports adjusted R^2 values from separate regressions of residual log prices on alternative fixed-effect sets S. Daily residuals are obtained after controlling for seller-product-day fixed effects (both use the same transactions). Monthly residuals are obtained after controlling for seller-product-month fixed effects. "Buyer" denotes seller-product-month-by-buyer fixed effects and is reported only for the monthly specifications. Buyer groups are defined using sector, size, and region. All specifications within each row use the same observations. Sample sizes differ across frequencies because singleton cells are dropped, and monthly cells contain fewer singletons. The adjusted R^2 values are not additive.

B.2 Price-Quantity Coefficients by Seller Industry

We repeat the specification in column 1 of Table 2 separately for each 1-digit seller industry. Each regression includes seller-product-day fixed effects, so the coefficient captures the conditional correlation between unit prices and quantities within seller-product-day cells.

Table A2 reports the results. The coefficient is negative in ten of the eleven industries. It is close to zero in Utilities and Financial Services, while the largest coefficients in absolute value are observed in Construction and Business Services. In Manufacturing and Retail and Wholesale, which account for most transactions in the classified sample, the coefficients are -0.026 and -0.048, respectively. These estimates imply that doubling quantity is associated with unit prices that are 1.8% lower in Manufacturing and 3.3% lower in Retail and Wholesale. The negative price-quantity relationship is therefore present across most seller industries, although its magnitude differs substantially across them.

The positive coefficient for Utilities should be interpreted separately, since pricing in electric-

ity and other network services is governed by regulated tariff schedules in Chile with fixed and usage-dependent components, rather than the market-based nonlinear pricing mechanism emphasized in the model. Similarly, the near-zero coefficient in Financial Services is not informative about pricing structure, since the main price of financial services, interest, is largely not recorded on VAT invoices.

Table A2: Price-Quantity Coefficients by Seller Industry

Seller industry	$\ln q_{igjt}$	N
Full sample	-0.0358	422,336,008
	(0.002)	
Agriculture	-0.0291	2,726,001
	(0.002)	
Mining	-0.0166	239,515
	(0.005)	
Manufacturing	-0.0257	158,307,691
	(0.003)	
Utilities	0.0055	4,603,523
	(0.024)	
Construction	-0.1572	1,070,388
	(0.007)	
Retail and Wholesale	-0.0477	219,858,685
	(0.003)	
Transport and ICTs	-0.0275	12,857,031
	(0.003)	
Financial Services	-0.0060	4,185,921
	(0.004)	
Real Estate Services	-0.0448	617,148
	(0.025)	
Business Services	-0.1143	4,576,770
	(0.006)	
Personal Services	-0.0632	728,309
	(0.009)	

Notes: This table reports coefficients from separate regressions of log unit prices on log quantities by seller industry. All regressions include seller-product-day fixed effects. The coefficients describe conditional price-quantity associations and should not be interpreted causally. Standard errors are reported in parentheses. The sector regressions include sellers with nonmissing 1-digit industry classifications.

B.3 Full Support of the Normalized-Quantity Exercise

The main text displays the portion of the normalized-quantity schedule from the lowest estimated bin through bins centered at approximately 1.5 log points above the seller-product median. This appendix reports the complete 20-bin schedule and the distribution of normalized transaction quantities. Bins not displayed in the main text remain part of the estimation; they correspond to increasingly large quantities relative to the seller-product median. Table A3 also reports the schedule estimated on transactions aggregated to the buyer-seller-product-month level. The two schedules are similar: above the median, monthly average prices are 26 to 30 log points below the lowest bin, compared with 29 to 36 log points at the transaction level.

Table A3: Full Price Schedule by Normalized Quantity

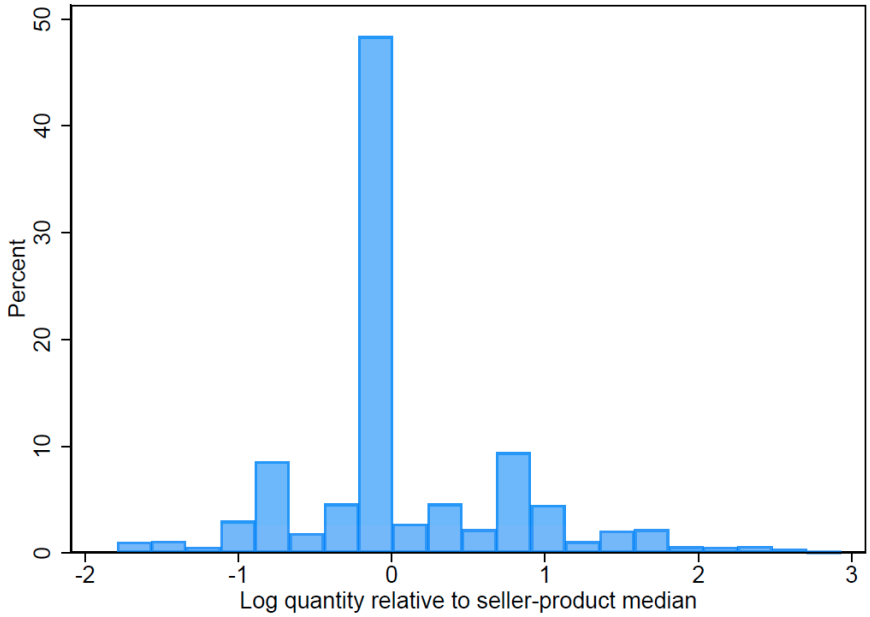
			Transaction				Monthly	
Bin	$\log(q/q_{jg}^{\text{med}})$	q/q_{jg}^{med}	Log-price difference	SE	Price difference (%)	95% CI (%)	Log-price difference	SE
1	−1.679	0.19	0.0000	–	0.00	–	0.0000	–
2	−1.454	0.23	−0.1383	0.0152	−12.92	[−15.47, −10.29]	−0.1497	0.0090
3	−1.229	0.29	−0.2646	0.0124	−23.25	[−25.08, −21.37]	−0.2360	0.0079
4	−1.004	0.37	−0.2185	0.0121	−19.62	[−21.50, −17.70]	−0.1855	0.0077
5	−0.779	0.46	−0.2506	0.0126	−22.17	[−24.07, −20.22]	−0.2196	0.0080
6	−0.554	0.57	−0.2973	0.0125	−25.72	[−27.52, −23.87]	−0.2562	0.0079
7	−0.329	0.72	−0.3089	0.0125	−26.58	[−28.36, −24.75]	−0.2532	0.0079
8	−0.104	0.90	−0.2837	0.0126	−24.70	[−26.53, −22.81]	−0.2533	0.0080
9	0.121	1.13	−0.3130	0.0129	−26.88	[−28.70, −25.01]	−0.2669	0.0083
10	0.346	1.41	−0.3383	0.0131	−28.70	[−30.50, −26.85]	−0.2739	0.0080
11	0.571	1.77	−0.3029	0.0129	−26.14	[−27.98, −24.25]	−0.2608	0.0083
12	0.796	2.22	−0.3550	0.0128	−29.88	[−31.62, −28.10]	−0.2844	0.0080
13	1.021	2.78	−0.3477	0.0129	−29.37	[−31.13, −27.56]	−0.2860	0.0080
14	1.246	3.48	−0.2911	0.0129	−25.26	[−27.13, −23.35]	−0.2614	0.0081
15	1.471	4.35	−0.3288	0.0128	−28.02	[−29.80, −26.20]	−0.2861	0.0079
16	1.696	5.45	−0.3164	0.0128	−27.12	[−28.92, −25.28]	−0.2874	0.0080
17	1.921	6.83	−0.2913	0.0129	−25.27	[−27.13, −23.36]	−0.2707	0.0080
18	2.146	8.55	−0.2992	0.0129	−25.86	[−27.71, −23.96]	−0.2791	0.0079
19	2.371	10.70	−0.2867	0.0130	−24.93	[−26.81, −22.99]	−0.2817	0.0081
20	2.596	13.40	−0.3069	0.0128	−26.43	[−28.25, −24.55]	−0.2953	0.0081

Notes: The table reports all 20 normalized-quantity-bin estimates underlying Figure 3. For each seller-product (j, g), quantity is normalized by its 2024 median, $x_{igjt} = \log(q_{igjt}/q_{jg}^{\text{med}})$. The pooled 1st–99th percentile support of x_{igjt} is divided into 20 equal-width intervals. Column 2 reports each bin midpoint in log relative quantity and column 3 reports the corresponding quantity as a multiple of the seller-product median, $\exp(x)$. Bin 1 is the omitted category. The log-price coefficients are therefore differences relative to bin 1. Percentage price differences are calculated as $100[\exp(\widehat{\beta}_b) - 1]$, and column 7 applies the same transformation to the endpoints of the 95% confidence interval. Columns 4–7 include seller-product-day fixed effects. Columns 8–9 aggregate transactions to the buyer-seller-product-month level: the dependent variable is the log of the monthly average unit price, monthly quantity is normalized by the seller-product median across buyer-months and assigned to the same bins, the specifications include seller-product-month fixed effects. Standard errors are clustered by seller-product. The main-text figure displays transaction bins 1–15; bins 16–20 remain in the estimation and are reported here in full.

The portion displayed in the main text therefore covers quantities from approximately 0.19 to 4.35 times the seller-product median. Bins 16–20 extend the estimated schedule from approximately 5.45 to 13.4 times the median. Prices in these bins lie between 1 and 4 log points above bin 15, small relative to the 33 log-point decline up to that bin, and their confidence intervals overlap with that of bin 15: unit prices are approximately 25–30% below those in the lowest normalized-

quantity bin throughout bins 9 through 20.

Figure A1: Distribution of Normalized Transaction Quantity



Notes: The figure reports the distribution of normalized transaction quantity, $x_{igjt} = \log(q_{igjt}/q_{jg}^{\text{med}})$, over the same pooled 1st–99th percentile support used to estimate Equation (6). The horizontal intervals correspond to the 20 equal-width bins underlying Table A3.

Figure A1 shows the distribution of normalized transaction quantity underlying this exercise. Transaction mass is concentrated near the seller-product median and becomes progressively sparser at larger normalized quantities, consistent with the wider confidence intervals in the upper bins of Table A3.

B.4 Alternative Normalization

Alternative rank normalization. The main text normalizes quantity by the seller-product median, preserving proportional distances in quantity across products. We assess whether the results depend on this cardinal normalization using an alternative measure that relies only on the ordering of quantities within each seller-product. For each seller-product pair (j, g), let $\mathcal{Q}_{jg}$ denote the set of distinct quantities traded in 2024 and let $K_{jg} = |\mathcal{Q}_{jg}|$. We order these quantities from smallest to largest and define

$$k_{jg}(q) \equiv \sum_{\widetilde{q} \in \mathcal{Q}_{jg}} \mathbb{1}\{\widetilde{q} \leq q\}, \qquad r_{igjt} \equiv \frac{k_{jg}(q_{igjt})}{K_{jg}}.$$

We assign each transaction to one of 50 relative-rank bins, $B^R_{igjt} = \lceil 50 r_{igjt} \rceil$. Seller-product pairs with only one distinct quantity are assigned to the first bin.[45]

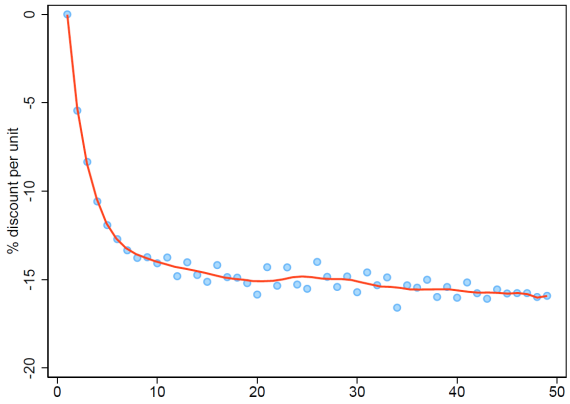
We estimate

$$\ln p_{igjt} = \alpha + \sum_{b=2}^{50} \gamma_b \mathbb{1}\{B^R_{igjt} = b\} + \Psi_{jgd} + \varepsilon_{igjt}, \tag{17}$$

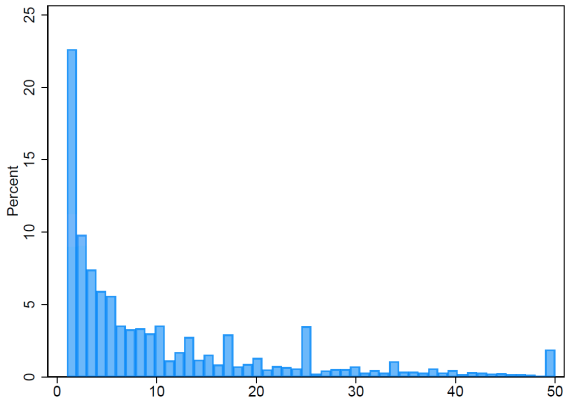
where Ψ_{jgd} are seller-product-day fixed effects and the lowest quantity-rank bin is omitted. The rank transformation places products with different units, scales, and quantity supports on a common ordinal metric. It therefore provides a useful robustness check for monotonicity and cross-industry heterogeneity, although ranks do not preserve cardinal distances in quantity and are not used to assess curvature.

Figure A2: Prices by Relative Quantity Rank

Panel A. Price Differences by Quantity Rank



Panel B. Distribution of Quantity-Rank Bins



Notes: Panel A reports percentage price differences relative to the lowest quantity-rank bin, estimated from Equation (17). The regression includes seller-product-day fixed effects. The red line is a smoothed fit through the estimated bin effects. Panel B reports the share of transactions assigned to each rank bin. Because ranks are constructed from distinct quantity levels within seller-product rather than transaction-weighted quantiles, the distribution across bins need not be uniform.

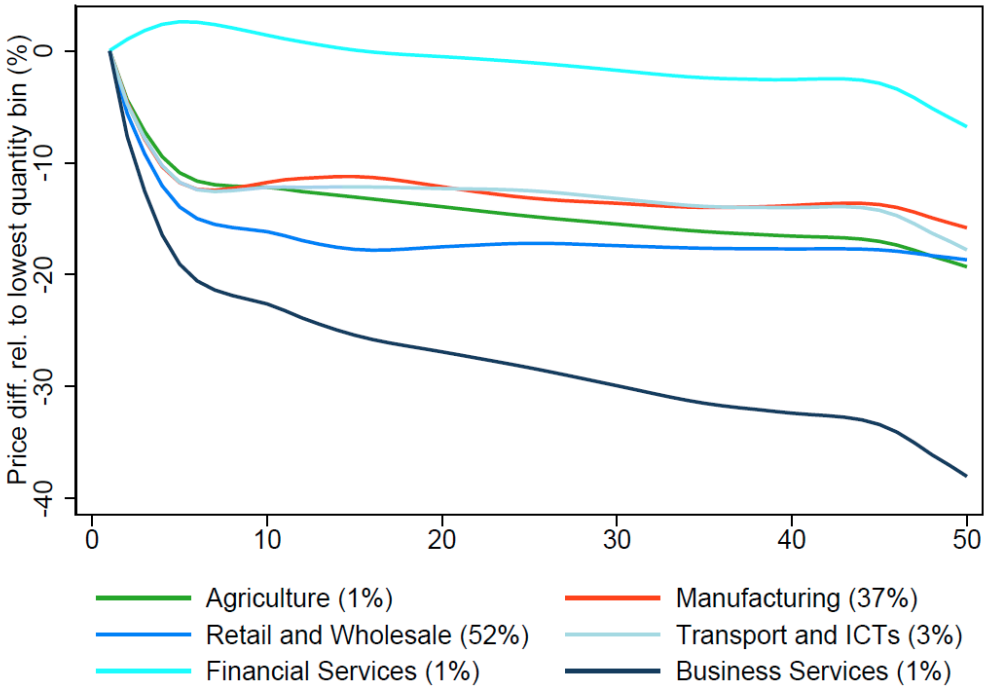
Figure A2 yields the same qualitative price–quantity relationship as the normalized-quantity exercise in the main text. Unit prices fall sharply as transactions move from the bottom of a seller-product's quantity support toward higher ranks. Most of the decline occurs over the lower part of the support, after which the schedule becomes considerably flatter. The highest-rank transactions are also substantially cheaper than transactions in the lowest rank. Thus, the negative price–

[45] All transactions with the same quantity receive the same rank. Seller-products with fewer than 50 distinct quantities therefore occupy only a subset of the bins, while seller-products with more than 50 quantities can have multiple adjacent quantities assigned to the same bin. The bins are constructed from distinct quantity levels rather than transaction counts and therefore need not contain equal numbers of transactions.

quantity relationship is not specific to measuring quantity relative to the seller-product median.

Seller-industry heterogeneity under the rank normalization. We next reestimate Equation (17) separately by 1-digit seller industry. Figure A3 shows that the substantial seller-industry heterogeneity documented in the main text is also present under the rank normalization. Industries with relatively steep normalized-quantity schedules continue to display large differences between low- and high-rank transactions, while sectors with relatively flat schedules remain comparatively flat.

Figure A3: Prices by Relative Quantity Rank, by Seller Industry



Notes: The figure reports price–quantity schedules estimated separately by 1-digit seller industry using Equation (17). Each line is a smoothed fit through the estimated rank-bin effects. The y-axis reports percentage price differences relative to the lowest quantity-rank bin. All regressions include seller-product-day fixed effects.

B.5 Convexity of the Price Schedule

The rank exercise establishes that the monotonicity and heterogeneity results are robust to an ordinal normalization of quantities, but ranks cannot be used to test convexity because they do not preserve distances between quantities. We therefore complement the nonparametric evidence with a parametric curvature test. Let

$$x_{igjt} = \log\left(\frac{q_{igjt}}{\widetilde{q}_{jgd}}\right)$$

where $\widetilde{q}_{jgd}$ is the median quantity across transactions of seller j and product g on day d, so that the comparison group is the one defined by the fixed effects. For the full economy and separately by seller industry s, we estimate a two-piece linear spline with a kink at the median

$$\ln p_{igjt} = \beta_{1s} x_{igjt} + \beta_{2s} \max\{x_{igjt}, 0\} + \Psi_{jgd} + \varepsilon_{igjt} \tag{18}$$

The coefficient of price with respect to quantity is β_{1s} below the median and $\beta_{1s} + \beta_{2s}$ above it. A positive β_{2s} implies that the schedule flattens above the median, that is, it is convex in log quantity. We classify a schedule as decreasing and convex when $\beta_{2s} > 0$ and $\beta_{1s} + \beta_{2s} < 0$, both at the 5% level.

Table A4: Price–Quantity Curvature by Seller Industry

	Full economy	Agric.	Mining	Manuf.	Utilities	Constr.	Retail & Wholesale	Transp. & ICTs	Financial Services	Real Estate	Business Services	Personal Services
Slope below median (β_1)	−0.0901	−0.0392	−0.0166	−0.0940	0.0166	−0.0951	−0.0913	−0.0637	−0.0028	−0.1398	−0.1232	−0.0679
	(0.0025)	(0.0040)	(0.0055)	(0.0055)	(0.0129)	(0.0065)	(0.0029)	(0.0044)	(0.0024)	(0.0640)	(0.0061)	(0.0093)
Change above median (β_2)	0.0782	0.0041	−0.0012	0.0847	−0.0366	−0.0123	0.0738	0.0532	−0.0151	0.1645	0.0546	0.0266
	(0.0031)	(0.0045)	(0.0067)	(0.0067)	(0.0298)	(0.0101)	(0.0034)	(0.0054)	(0.0066)	(0.0741)	(0.0071)	(0.0103)
Slope above median ($\beta_1 + \beta_2$)	−0.0119	−0.0350	−0.0177	−0.0094	−0.0201	−0.1074	−0.0176	−0.0105	−0.0179	0.0247	−0.0686	−0.0414
	(0.0013)	(0.0023)	(0.0051)	(0.0029)	(0.0336)	(0.0067)	(0.0013)	(0.0025)	(0.0067)	(0.0130)	(0.0041)	(0.0052)
Convex ($\beta_2 > 0$)	Yes	No	No	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes
Decreasing above median	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Yes
Decreasing and convex	Yes	No	No	Yes	No	No	Yes	Yes	No	No	Yes	Yes
Transaction share (%)	100.00	0.57	0.04	26.44	0.70	0.20	63.59	2.34	0.61	0.12	0.98	0.14
Observations	963,216,319	5,460,422	422,361	254,701,614	6,749,388	1,916,422	612,507,278	22,526,189	5,920,349	1,140,483	9,469,923	1,392,581
Seller-product-day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimates of Equation (18), where $x_{igjt} = \log(q_{igjt}/\widetilde{q}_{jgd})$ and $\widetilde{q}_{jgd}$ is the seller-product-day median quantity. "Slope above median" reports $\hat{\beta}_{1s} + \hat{\beta}_{2s}$. Standard errors, clustered by seller-product, are reported in parentheses. A schedule is classified as convex when $\hat{\beta}_{2s} > 0$, as decreasing above the median when $\hat{\beta}_{1s} + \hat{\beta}_{2s} < 0$, and as decreasing and convex when both hold, in each case at the 5% level. Transaction shares are computed relative to the full estimation sample; the eleven reported seller industries account for approximately 96% of observations. All specifications include seller-product-day fixed effects; singleton observations are dropped.

For the economy as a whole, the schedule is decreasing and convex: the price coefficient is −0.090 below the median and −0.012 above it. The same pattern holds in Manufacturing, Retail and Wholesale, Transport and ICTs, Business Services, and Personal Services, which account for approximately 98% of transactions in the eleven reported seller sectors and 93% of the full estimation sample. Real Estate is convex but flat above the median. In Agriculture, Mining, and Construction, prices decline at a roughly constant coefficient, and in Utilities and Financial Services the schedule is not significantly decreasing below the median. Convexity is therefore not universal across industries, but the decreasing-and-convex pattern characterizes the majority of transactions.

B.6 Alternative Explanations.

We interpret quantity discounts as evidence of quantity-based screening by buyer industry; nevertheless, the preceding results do not rule out alternative mechanisms. We next assess alternative

interpretations of the price-quantity patterns documented above, focusing on persistent buyer heterogeneity, bilateral bargaining, repeated relationships, learning, and cost-side scale economies. Table A5 reports the results. Columns 0–5 use a common sample throughout.[46] Across specifications, the coefficient on log quantity is not absorbed by these alternative mechanisms: it remains large and negative throughout, while the interactions of the additional controls with quantity are small.

Table A5: Alternative Explanations for the Average Price-Quantity Schedule

		Buyer heterogeneity	Bilateral bargaining		Relationship mechanisms		Scale economies
	(0)	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Buyer FE	Supplier options	Seller dependence	Repeated transactions	Relationship tenure	High-price products
$\log q$	-0.0415	-0.0771	-0.0524	-0.0501	-0.0496	-0.0444	-0.1730
	(0.0016)	(0.0017)	(0.0037)	(0.0017)	(0.0022)	(0.0028)	(0.04672)
log(Suppliers)			0.0431				
			(0.0015)				
$\log q \times$ log(Suppliers)			-0.00012				
			(0.0007)				
Buyer share				0.8202			
				(0.0197)			
$\log q \times$ Buyer share				-0.0501			
				(0.0064)			
$\log n_{ijg}$					0.0230		
					(0.0011)		
$\log q \times \log n_{ijg}$					0.00019		
					(0.0004)		
log(1 + Relationship age)						0.0211	
						(0.0026)	
$\log q \times$ log(1 + Relationship age)						0.00073	
						(0.0013)	
Buyer FE		✓					
Seller × Product × Day FE	✓	✓	✓	✓	✓	✓	✓
N	986M	986M	986M	986M	986M	986M	0.9M

Notes: The table reports regressions of log unit prices on log transaction quantities and variables used to distinguish alternative explanations for the average price-quantity schedule. Columns 0–5 use a common estimation sample. Column 0 includes only seller-product-day fixed effects. Column 1 adds buyer fixed effects, which absorb persistent buyer-specific price differences. In column 2, Suppliers is the number of distinct suppliers from which the buyer purchases and proxies for the buyer's outside options. In column 3, Buyer share is the buyer's share of seller-product sales and proxies for the seller's dependence on that buyer; the coefficients on $\log q$ and on its interaction with buyer share coincide to three decimals by chance. In column 4, n_{ijg} is the number of transactions observed between seller j and buyer i for product g and measures relationship intensity. In column 5, Relationship age is the number of years since the seller-buyer relationship first appears in the data; relationships first observed in 2024 have age zero. Column 6 restricts the sample to seller-products in the top 1% of average unit price in 2024. All columns include seller-product-day fixed effects, and column 1 additionally includes buyer fixed effects. Standard errors are clustered by seller-product and reported in parentheses. The coefficients describe conditional equilibrium associations.

A first possibility is that the negative price-quantity relationship reflects persistent buyer-specific pricing advantages. Column 0 reports the baseline on this sample, with a coefficient of

[46]The sample is larger relative to Table 2 as we do not interact buyer side fixed effects with seller-product-day fixed effects. This explains also the slightly different coefficient reported.

−0.041. Column 1 absorbs such differences through buyer fixed effects. The price-quantity coefficient remains negative, at −0.077, nearly twice the baseline. Persistent buyer-level price heterogeneity therefore does not seem to account for the negative price-quantity relationship.

A related alternative is bilateral bargaining. Buyers with stronger outside options, or buyers on whom sellers are more dependent, may obtain larger discounts while purchasing systematically higher quantities, thus driving the quantity discounts reported previously. Column 2 proxies for buyer outside options using the number of distinct suppliers from which the buyer purchases. The interaction between log quantity and log supplier count is negative, precisely estimated, and small in magnitude: doubling the number of suppliers changes the price-quantity relationship by only about −0.00008, compared with a relationship of approximately −0.05. Differences in buyer outside options, therefore, seem to have little quantitative relationship with the average price-quantity schedule.

Column 3 instead measures the seller's dependence on the buyer using the buyer's share of seller-product sales. The interaction is negative, so the price-quantity relationship is steeper when a seller is more dependent on a buyer. Its economic magnitude, however, is modest over most of the buyer-share distribution. Moving from the 25th to the 75th percentile of buyer share increases the magnitude of the price-quantity gradient by about 0.0014, relative to a coefficient of approximately −0.05. Seller dependence is therefore associated with heterogeneity in quantity discounts, particularly among highly dependent relationships, but does not seem to quantitatively account for the average price-quantity gradient for most transactions. The level coefficients also run against a bargaining interpretation: at typical quantities, buyers with more suppliers and buyers on whom the seller depends more pay higher, not lower, unit prices.

Columns 4 and 5 examine relationship-based explanations. If repeated interaction allows sellers to learn buyer type, the role of quantity-based screening may decline as relationships mature or intensify. Column 4 measures relationship intensity by the number of transactions observed for a seller-buyer-product relationship. Column 5 considers relationship tenure instead. Consistent with this mechanism, in both specifications the interaction between log quantity and each measure is positive (0.00019 and 0.00073 respectively). Its magnitude is, however, small, and the price-quantity relationship remains strongly negative. Repeated trading and relationship tenure may therefore attenuate the quantity gradient, but neither appears to generate the average relationship documented above.[47]

A final alternative is that the negative price-quantity relationship reflects fixed per-order costs, such as processing, customization, or delivery costs that are spread over the number of units purchased. When shipping is reported as a separate invoice line, it is excluded from the unit prices used in our analysis. As an additional check, if remaining fixed per-order costs are largely independent of the value of the underlying product, they should represent a smaller fraction of unit

[47] Why would quantity-based screening persist between buyers and sellers who trade intensively? See footnote 26.

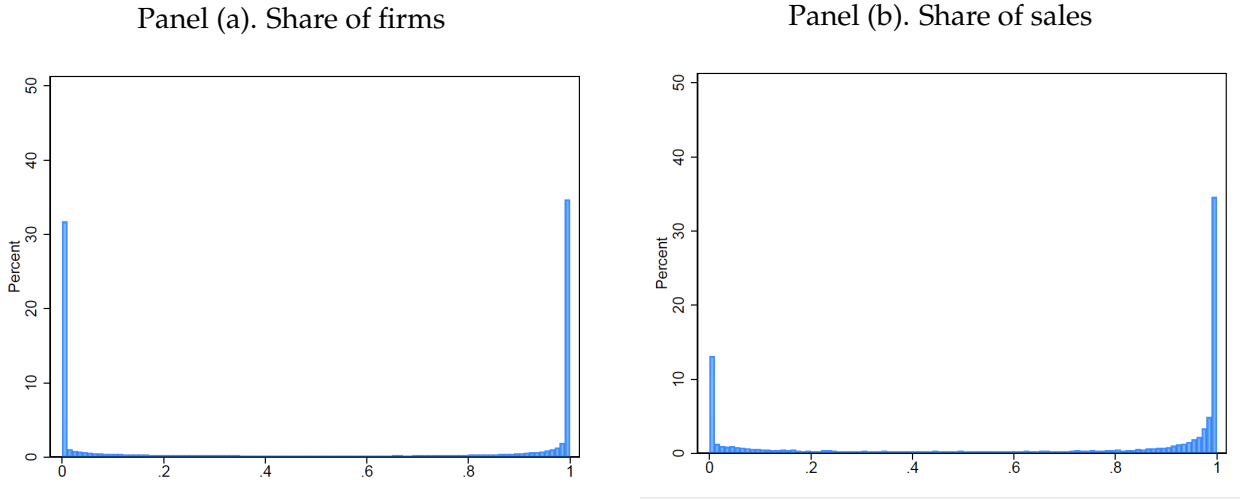
value for high-priced products. Following Bornstein and Peter (2025), column 6 restricts the sample to seller-products in the top 1% of average unit price. The price-quantity coefficient is −0.173, more than four times the baseline in column 0. Because this restriction substantially changes the size and composition of the sample, we do not interpret this difference as measuring the contribution of fixed costs. Still, a steeper rather than flatter relationship among high-priced products suggests that fixed per-order costs are unlikely to drive the negative price-quantity relationship.

B.7 Firm Sales Partition

We measure each firm's sales destinations in 2018 by combining firm-to-firm invoices with reported total sales. Sales to final demand, including exports, are the difference between the two, and firms whose invoiced sales to other firms exceed reported sales are assigned entirely to other firms. A firm is upstream if at least half of its sales are invoiced to other domestic firms and a retailer otherwise, and this assignment is held fixed in all years (Appendix D). Upstream firms account for 67% of sales.

Sales destinations are strongly bimodal (Figure A4). Across firms, 80% sell at least 90% of their sales to a single destination; weighted by sales, these firms account for 73% of sales, as large firms more often sell to both destinations. The layer assignment is therefore insensitive to the cutoff: the layer of these firms is the same for any cutoff between 0.1 and 0.9. Table A6 shows that the pattern holds within every sector.

Figure A4: Distribution of Sales Destinations, 2018



Notes: The figure shows the distribution of the share of each firm's 2018 sales invoiced to other domestic firms, in 100 bins. Panel (a) weights firms equally; panel (b) weights them by sales, winsorized at the 99th percentile. The remainder of sales, including exports, is final demand. Firms whose invoiced sales exceed reported sales are assigned a share of one.

Table A6: Firm Sales Partition, 2018

	Sales from	≥90% to one destination	
Sector (share of firm-to-firm value)	upstream firms	Firms	Sales
All firms (100%)	0.67	0.80	0.73
Agriculture (4%)	0.77	0.72	0.64
Mining (2%)	0.66	0.85	0.70
Manufacturing (21%)	0.79	0.76	0.71
Utilities (6%)	0.80	0.76	0.68
Construction (9%)	0.88	0.87	0.83
Retail and wholesale (36%)	0.61	0.76	0.68
Transport and ICT (8%)	0.71	0.86	0.76
Financial services (4%)	0.44	0.88	0.76
Real estate services (1%)	0.64	0.78	0.64
Business services (6%)	0.67	0.87	0.84
Personal services (1%)	0.16	0.90	0.83

Notes: A firm's destination share is the fraction of its 2018 sales invoiced to other domestic firms; the remainder, including exports, is final demand. Firms whose invoiced sales exceed reported sales are assigned a share of one. A firm is upstream if this share is at least 0.5. Within each sector, column 1 reports the share of sales from upstream firms, and columns 2 and 3 report the share of firms and of sales from firms selling at least 90% of their sales to a single destination. Sales weights are winsorized at the 99th percentile, so that a few very large firms do not dominate sector shares. Sector shares of firm-to-firm value are in parentheses. The all-firms row also includes public administration.

C Model Details and Derivations

This appendix provides the derivations underlying the supply-chain model in the main text. We first characterize input demand, price indices, and variable cost functions under the conjectured contract structure. We then derive the seller's nonlinear-pricing problem, solve for the optimal two-part tariff, and verify the homogeneous revenue functions used in the equilibrium characterization. Section C.7 establishes the proofs for Results 1–4. Section C.8 proves existence and uniqueness of the general equilibrium. The remaining sections derive the Kuhn–Tucker implementation of the efficient allocation (Section C.9), detail the aggregate welfare decomposition (Section C.10), and show the entry-cost invariance of relative welfare comparisons (Section C.11). Throughout, we keep the wage w explicit; under the normalization used in the main text, $w \equiv 1$. Results involving buyer–seller triples (ℓ, s, s') are understood to apply only when $\theta^{\ell}_{ss'} > 0$.

C.1 Costs and Price Indices

We derive the price indices and unit cost functions used in the main text from the firm's nested cost-minimization problem. Consider a buyer $i = (\ell, s, z_i)$. Under the conjectured two-part tariff, the payment to seller $j \in \mathcal{U}_{s'}$ is

$$T_{ij} = p^{\ell}_{js} m_{ij} + F^{\ell}_{js}$$

By the full-participation conjecture, the flat fee is inframarginal conditional on the equilibrium supplier set. Hence only the marginal prices $\{p^{\ell}_{js}\}$ enter variable input choices and variable costs. First consider the cost of producing one unit of the sector-s' materials bundle $M_{is'}$. The buyer solves

$$\min_{\{m_{ij}\}_{j\in\mathcal{U}_{s'}}} \int_{j\in\mathcal{U}_{s'}} p^{\ell}_{js} m_{ij}\, dv^{u}_{s'}(j) \qquad \text{s.t.} \qquad M_{is'} = \left(\int_{j\in\mathcal{U}_{s'}} m_{ij}^{\frac{\sigma^{u}_{s'}-1}{\sigma^{u}_{s'}}}\, dv^{u}_{s'}(j) \right)^{\frac{\sigma^{u}_{s'}}{\sigma^{u}_{s'}-1}}$$

The first-order conditions imply the standard CES demand system

$$m_{ij} = M_{is'} \left(\frac{p^{\ell}_{js}}{P^{\ell}_{ss'}} \right)^{-\sigma^{u}_{s'}}$$

where the unit cost of the sector-s' bundle is

$$P^{\ell}_{ss'} = \left(\int_{j\in\mathcal{U}_{s'}} \left(p^{\ell}_{js}\right)^{1-\sigma^{u}_{s'}}\, dv^{u}_{s'}(j) \right)^{\frac{1}{1-\sigma^{u}_{s'}}}$$

Therefore variable expenditure on inputs from seller sector s' is

$$\int_{j\in\mathcal{U}_{s'}} p^{\ell}_{js} m_{ij}\, dv^{u}_{s'}(j) = P^{\ell}_{ss'} M_{is'}$$

Next, consider the cost of producing one unit of the composite materials bundle

$$M_i = \prod_{s'\in\mathcal{S}} M_{is'}^{\theta^{\ell}_{ss'}}, \qquad \sum_{s'\in\mathcal{S}} \theta^{\ell}_{ss'} = 1$$

The buyer solves

$$\min_{\{M_{is'}\}_{s'}} \sum_{s'\in\mathcal{S}} P^{\ell}_{ss'} M_{is'} \quad \text{subject to} \quad \prod_{s'\in\mathcal{S}} M_{is'}^{\theta^{\ell}_{ss'}} = M_i$$

The Cobb–Douglas cost function is

$$C_i^M(M_i) = \left[\prod_{\substack{s' \in \mathcal{S}: \\ \theta_{ss'}^\ell > 0}} \left(\frac{P_{ss'}^\ell}{\theta_{ss'}^\ell} \right)^{\theta_{ss'}^\ell} \right] M_i$$

Equivalently, defining

$$P_s^{M,\ell} \equiv \prod_{s' \in \mathcal{S}} \left(P_{ss'}^\ell \right)^{\theta_{ss'}^\ell}$$

the unit cost of the materials composite is

$$\prod_{\substack{s' \in \mathcal{S}: \\ \theta_{ss'}^\ell > 0}} \left(\frac{P_{ss'}^\ell}{\theta_{ss'}^\ell} \right)^{\theta_{ss'}^\ell} = P_s^{M,\ell} \prod_{\substack{s' \in \mathcal{S}: \\ \theta_{ss'}^\ell > 0}} \left(\theta_{ss'}^\ell \right)^{-\theta_{ss'}^\ell}$$

Finally, the firm combines labor and materials according to

$$Q_i = z_i l_i^{\alpha_s^\ell} M_i^{1-\alpha_s^\ell}$$

The variable cost of producing Q_i units is obtained from

$$\min_{l_i, M_i} w l_i + C_i^M(M_i) \quad \text{subject to} \quad z_i l_i^{\alpha_s^\ell} M_i^{1-\alpha_s^\ell} \geq Q_i$$

The resulting unit cost is

$$c_i = \frac{\Theta_s^\ell}{z_i} w^{\alpha_s^\ell} \left(P_s^{M,\ell} \right)^{1-\alpha_s^\ell}$$

where

$$\Theta_s^\ell \equiv \left(\alpha_s^\ell \right)^{-\alpha_s^\ell} \left(1 - \alpha_s^\ell \right)^{-(1-\alpha_s^\ell)} \prod_{\substack{s' \in \mathcal{S}: \\ \theta_{ss'}^\ell > 0}} \left(\theta_{ss'}^\ell \right)^{-(1-\alpha_s^\ell)\theta_{ss'}^\ell}$$

Thus marginal cost is constant in output and inversely proportional to firm productivity z_i.

C.2 Buyer Input Demand

We now derive the input demand conditions used in the main text. Consider a buyer $i = (\ell, s, z_i)$. Since flat fees are inframarginal conditional on full participation, the buyer's marginal input choices solve the variable-profit problem

$$\widetilde{\Pi}_i = \max_{l_i, \{M_{is'}\}_{s'}} \left\{ A_s^\ell Q_i^{\psi_s^\ell} - w l_i - \sum_{s' \in \mathcal{S}} P_{ss'}^\ell M_{is'} \right\} \qquad \text{s.t.} \qquad Q_i = z_i l_i^{\alpha_s^\ell} \prod_{s' \in \mathcal{S}} M_{is'}^{(1-\alpha_s^\ell)\theta_{ss'}^\ell}$$

Equivalently, $Q_i = z_i l_i^{\alpha_s^\ell} M_i^{1-\alpha_s^\ell}$, where $M_i = \prod_{s'} M_{is'}^{\theta_{ss'}^\ell}$.

Let $R_i = A_s^\ell Q_i^{\psi_s^\ell}$. The first-order condition for labor is

$$\frac{\partial R_i}{\partial l_i} = \psi_s^\ell R_i \frac{\alpha_s^\ell}{l_i} = w$$

For each materials sector s' with $\theta_{ss'}^\ell > 0$, the first-order condition is

$$\frac{\partial R_i}{\partial M_{is'}} = \psi_s^\ell R_i \frac{(1-\alpha_s^\ell)\theta_{ss'}^\ell}{M_{is'}} = P_{ss'}^\ell$$

Taking ratios gives

$$\frac{l_i}{M_{is'}} = \frac{\alpha_s^\ell}{(1-\alpha_s^\ell)\theta_{ss'}^\ell} \frac{P_{ss'}^\ell}{w}$$

Thus, within a buyer layer-sector pair (ℓ, s), input ratios are pinned down by common input prices and technology parameters. Buyer productivity affects the scale of production and input use, but not input ratios.

We next derive demand for individual seller varieties within a seller sector s'. Conditional on total demand $M_{is'}$, the CES cost-minimization problem derived above implies

$$m_{ij} = M_{is'} \left(\frac{p_{js}^\ell}{P_{ss'}^\ell} \right)^{-\sigma_{s'}^u}, \qquad j \in \mathcal{U}_{s'}$$

Under the conjectured pricing rule $p_{js}^\ell = \mu_{ss'}^\ell c_j$, and using the unit-cost expression $c_j = C_{s'}^u / z_j$ for upstream sellers in sector s', where $C_{s'}^u$ is common within the sector, we have

$$P_{ss'}^\ell = \mu_{ss'}^\ell C_{s'}^u \left(\int_{j \in \mathcal{U}_{s'}} z_j^{\sigma_{s'}^u - 1} \, dv_{s'}^u(j) \right)^{\frac{1}{1-\sigma_{s'}^u}}$$

Using

$$\widetilde{z}_{s'}^u = \left(\int_{j \in \mathcal{U}_{s'}} z_j^{\sigma_{s'}^u - 1} \frac{dv_{s'}^u(j)}{N_{s'}^u} \right)^{\frac{1}{\sigma_{s'}^u - 1}}$$

this implies

$$\frac{p_{js}^\ell}{P_{ss'}^\ell} = \frac{\widetilde{z}_{s'}^u}{z_j \left(N_{s'}^u \right)^{\frac{1}{1-\sigma_{s'}^u}}}$$

Therefore

$$m_{ij} = M_{is'} \left(\frac{\widetilde{z}^u_{s'}}{z_j \left(N^u_{s'} \right)^{\frac{1}{1-\sigma^u_{s'}}}} \right)^{-\sigma^u_{s'}}$$

Taking the ratio of demand for seller j to demand for a seller with productivity $\widetilde{z}^u_{s'}$ gives

$$\frac{m_{ij}(z_j)}{m_{ij}(\widetilde{z}^u_{s'})} = \left(\frac{z_j}{\widetilde{z}^u_{s'}} \right)^{\sigma^u_{s'}}$$

This relative demand is independent of buyer identity. Aggregating across buyers and imposing market clearing for upstream variety j, total production of seller $j \in \mathcal{U}_{s'}$ satisfies

$$Q_j = \left(\frac{z_j}{\widetilde{z}^u_{s'}} \right)^{\sigma^u_{s'}} Q_j(\widetilde{z}^u_{s'})$$

Because production is constant returns to scale, input use scales with output divided by productivity. Hence, for an upstream seller $j \in \mathcal{U}_{s'}$

$$l_j(z_j) = \left(\frac{z_j}{\widetilde{z}^u_{s'}} \right)^{\sigma^u_{s'}-1} l_j(\widetilde{z}^u_{s'}), \qquad M_j(z_j) = \left(\frac{z_j}{\widetilde{z}^u_{s'}} \right)^{\sigma^u_{s'}-1} M_j(\widetilde{z}^u_{s'})$$

The same argument applies to retail firms, with the final-demand elasticity σ^r_s in place of $\sigma^u_{s'}$. Thus, if productivity in (ℓ, s) is Pareto with tail parameter κ^ℓ_s, input demand is Pareto with effective tail parameter

$$\xi^\ell_s = \frac{\kappa^\ell_s}{\sigma^\ell_s - 1}$$

Feasibility of aggregate input demand requires $\xi^\ell_s > 1$, or equivalently $\kappa^\ell_s > \sigma^\ell_s - 1$.

C.3 Retailer Revenue Function

Within retail sector s, CES demand over differentiated varieties implies

$$y_j \;=\; Y_s \left(\frac{p_j}{P_s} \right)^{-\sigma^r_s}, \qquad P_s Y_s = \theta_s \mathcal{E} = \theta_s P_Y Y$$

where $\mathcal{E}$ denotes nominal final expenditure. Under the normalization $w \equiv 1$ and household income equal to labor income, $\mathcal{E} = 1$, so welfare is $Y = 1/P_Y$. Hence the inverse demand is $p_j = P_s \left(y_j / Y_s \right)^{-1/\sigma^r_s}$. Revenue as a function of own output Q_j is

$$R_j \;=\; p_j Q_j \;=\; P_s^{1-\frac{1}{\sigma^r_s}} \left(\theta_s \mathcal{E} \right)^{1/\sigma^r_s} Q_j^{(\sigma^r_s - 1)/\sigma^r_s}$$

Therefore, the revenue guess holds for retailers ($\ell = r$) with

$$\psi_s^r = \frac{\sigma_s^r - 1}{\sigma_s^r}, \qquad A_s^r = P_s^{1-\frac{1}{\sigma_s^r}} (\theta_s \mathcal{E})^{1/\sigma_s^r}$$

Moreover, retailer profit maximization implies the usual CES markup over marginal cost:

$$p_j = \frac{\sigma_s^r}{\sigma_s^r - 1} c_j, \qquad c_j = \frac{\Theta_s^r}{z_j} w^{\alpha_s^r} \left(P_s^{M,r}\right)^{1-\alpha_s^r}$$

C.4 Seller's Nonlinear-Pricing Problem

We now derive the unrestricted nonlinear-pricing problem faced by an upstream seller. Fix a seller sector s', a seller $j \in \mathcal{U}_{s'}$, and a buyer layer-sector pair (ℓ, s). The seller offers buyers $i = (\ell, s, z_i)$ an unrestricted menu of quantity-transfer pairs. We first rewrite the buyer's surplus from transacting with seller j in a form that depends on a one-dimensional buyer type.

Consider a buyer $i = (\ell, s, z_i)$ purchasing quantity m from seller $j \in \mathcal{U}_{s'}$. Since seller j is atomistic, it takes as given the buyer's marginal value of the sector-s' input bundle and the aggregate bundle $M_{is'}$. The gross value generated by quantity m is obtained by integrating the marginal contribution of m to the CES bundle

$$\frac{\partial R_i}{\partial M_{is'}} M_{is'}^{1/\sigma_{s'}^u} m^{-1/\sigma_{s'}^u}$$

Thus the buyer surplus from a quantity-transfer pair (m, T) is

$$TS_{is'}(m, T) = \frac{\sigma_{s'}^u}{\sigma_{s'}^u - 1} \frac{\partial R_i}{\partial M_{is'}} M_{is'}^{1/\sigma_{s'}^u} m^{\frac{\sigma_{s'}^u - 1}{\sigma_{s'}^u}} - T$$

Using the buyer's first-order condition

$$\frac{\partial R_i}{\partial M_{is'}} = P_{ss'}^{\ell},$$

this becomes

$$TS_{is'}(m, T) = \frac{\sigma_{s'}^u}{\sigma_{s'}^u - 1} P_{ss'}^{\ell} M_{is'}^{1/\sigma_{s'}^u} m^{\frac{\sigma_{s'}^u - 1}{\sigma_{s'}^u}} - T$$

Define the buyer valuation index

$$\tau_{is'} \equiv P_{ss'}^{\ell} M_{is'}^{1/\sigma_{s'}^u}$$

Then buyer surplus can be written as

$$TS^{\ell}_{ss'}(\tau, m, T) = \frac{\sigma^u_{s'}}{\sigma^u_{s'} - 1} \tau m^{\frac{\sigma^u_{s'} - 1}{\sigma^u_{s'}}} - T$$

Hence, for seller j, all buyer heterogeneity within the partition (ℓ, s, s') is summarized by the one-dimensional type τ.

We now characterize the distribution of τ. From the buyer input-demand results, input demand $M_{is'}$ is Pareto distributed within each buyer layer-sector pair (ℓ, s), with effective tail parameter ξ^{ℓ}_{s}. Since

$$\tau_{is'} = P^{\ell}_{ss'} M^{1/\sigma^u_{s'}}_{is'}$$

multiplication by $P^{\ell}_{ss'}$ changes only the scale of the distribution, while raising $M_{is'}$ to the power $1/\sigma^u_{s'}$ multiplies the Pareto tail parameter by $\sigma^u_{s'}$. Therefore

$$\tau_{is'} \sim \text{Pareto}(\rho^{\ell}_{ss'}), \qquad \rho^{\ell}_{ss'} = \sigma^u_{s'} \xi^{\ell}_{s}$$

The lower bound of the type distribution is denoted by

$$\underline{\tau}^{\ell}_{ss'} = P^{\ell}_{ss'} \left(M^{\ell}_{ss'}(\underline{z}^{\ell}_{s}) \right)^{1/\sigma^u_{s'}}$$

Feasibility of aggregate input demand requires $\xi^{\ell}_{s} > 1$, which implies $\rho^{\ell}_{ss'} > \sigma^u_{s'}$.

Applying the revelation principle, seller $j \in \mathcal{U}_{s'}$ can equivalently choose, for each buyer layer-sector pair (ℓ, s), a direct mechanism

$$\left\{ x^{\ell}_{ss'}(\tau), T^{\ell}_{ss'}(\tau) \right\}_{\tau \geq \underline{\tau}^{\ell}_{ss'}}$$

where $x^{\ell}_{ss'}(\tau)$ is the quantity allocated to a buyer of type τ and $T^{\ell}_{ss'}(\tau)$ is the associated transfer. The seller's problem is

$$\max_{\{x^{\ell}_{ss'}(\cdot), T^{\ell}_{ss'}(\cdot)\}_{\ell,s}} \sum_{\ell \in \{u,r\}} \sum_{s \in \mathcal{S}} N^{\ell}_{s} \mathbb{E}_{\tau \sim G^{\ell}_{ss'}} \left[T^{\ell}_{ss'}(\tau) - c_j x^{\ell}_{ss'}(\tau) \right]$$

subject, for each (ℓ, s), to incentive compatibility, individual rationality, and monotonicity. Given

$$TS^{\ell}_{ss'}(\tau) = \frac{\sigma^u_{s'}}{\sigma^u_{s'} - 1} \tau x^{\ell}_{ss'}(\tau)^{\frac{\sigma^u_{s'} - 1}{\sigma^u_{s'}}} - T^{\ell}_{ss'}(\tau)$$

local incentive compatibility implies the envelope condition

$$TS^{\ell}_{ss'}(\tau) = TS^{\ell}_{ss'}(\underline{\tau}^{\ell}_{ss'}) + \frac{\sigma^u_{s'}}{\sigma^u_{s'} - 1} \int_{\underline{\tau}^{\ell}_{ss'}}^{\tau} x^{\ell}_{ss'}(\omega)^{\frac{\sigma^u_{s'} - 1}{\sigma^u_{s'}}} \, d\omega$$

The individual-rationality constraint is

$$TS^{\ell}_{ss'}(\underline{\tau}^{\ell}_{ss'}) \geq 0$$

and monotonicity requires

$$x^{\ell}_{ss'}(\tau') \geq x^{\ell}_{ss'}(\tau) \qquad \text{for } \tau' > \tau$$

We solve the relaxed problem and verify monotonicity ex post. Substituting the envelope representation into the seller's objective and integrating by parts yields the virtual-surplus problem

$$\max_{\{x^{\ell}_{ss'}(\cdot)\}_{\ell,s}} \sum_{\ell\in\{u,r\}} \sum_{s\in\mathcal{S}} N^{\ell}_{s} \mathbb{E}_{\tau\sim G^{\ell}_{ss'}} \left[\frac{\sigma^{u}_{s'}}{\sigma^{u}_{s'}-1} \left(\tau - \frac{1-G^{\ell}_{ss'}(\tau)}{g^{\ell}_{ss'}(\tau)} \right) x^{\ell}_{ss'}(\tau)^{\frac{\sigma^{u}_{s'}-1}{\sigma^{u}_{s'}}} - c_j x^{\ell}_{ss'}(\tau) \right]$$

where $G^{\ell}_{ss'}$ and $g^{\ell}_{ss'}$ denote the CDF and density of τ in buyer partition (ℓ, s, s'). The lowest-type surplus is set to zero

$$TS^{\ell}_{ss'}(\underline{\tau}^{\ell}_{ss'}) = 0$$

Since τ is Pareto with tail parameter $\rho^{\ell}_{ss'}$,

$$\frac{1-G^{\ell}_{ss'}(\tau)}{g^{\ell}_{ss'}(\tau)} = \frac{\tau}{\rho^{\ell}_{ss'}}$$

Hence the virtual valuation is

$$\tau - \frac{1-G^{\ell}_{ss'}(\tau)}{g^{\ell}_{ss'}(\tau)} = \tau \frac{\rho^{\ell}_{ss'}-1}{\rho^{\ell}_{ss'}}$$

The first-order condition for $x^{\ell}_{ss'}(\tau)$ is therefore

$$\tau \frac{\rho^{\ell}_{ss'}-1}{\rho^{\ell}_{ss'}} x^{\ell}_{ss'}(\tau)^{-1/\sigma^{u}_{s'}} = c_j$$

or equivalently

$$\tau x^{\ell}_{ss'}(\tau)^{-1/\sigma^{u}_{s'}} = \frac{\rho^{\ell}_{ss'}}{\rho^{\ell}_{ss'}-1} c_j \equiv \mu^{\ell}_{ss'} c_j$$

This is exactly the buyer first-order condition under a constant marginal price

$$p^{\ell}_{js} = \mu^{\ell}_{ss'} c_j$$

Thus the optimal allocation can be implemented by a two-part tariff

$$T_{ij} = p^{\ell}_{js} m_{ij} + F^{\ell}_{js}, \qquad p^{\ell}_{js} = \frac{\rho^{\ell}_{ss'}}{\rho^{\ell}_{ss'}-1} c_j$$

Because $x^{\ell}_{ss'}(\tau) = (\tau/p^{\ell}_{js})^{\sigma^u_{s'}}$, the allocation is increasing in τ, so the monotonicity constraint is satisfied.

It remains to pin down the flat fee. Under the two-part tariff, a buyer of type τ chooses

$$m(\tau) = \left(\frac{\tau}{p^{\ell}_{js}}\right)^{\sigma^u_{s'}}$$

The buyer's surplus before paying the flat fee is

$$\frac{\sigma^u_{s'}}{\sigma^u_{s'}-1}\tau m(\tau)^{\frac{\sigma^u_{s'}-1}{\sigma^u_{s'}}} - p^{\ell}_{js}m(\tau) = \frac{1}{\sigma^u_{s'}-1}\tau^{\sigma^u_{s'}}\left(p^{\ell}_{js}\right)^{1-\sigma^u_{s'}}$$

Imposing zero surplus for the lowest type gives

$$F^{\ell}_{js} = \frac{1}{\sigma^u_{s'}-1}\left(\underline{\tau}^{\ell}_{ss'}\right)^{\sigma^u_{s'}}\left(p^{\ell}_{js}\right)^{1-\sigma^u_{s'}} = \frac{1}{\sigma^u_{s'}-1}P^{\ell}_{ss'}M^{\ell}_{ss'}(\underline{z}^{\ell}_{s})\left(\frac{p^{\ell}_{js}}{P^{\ell}_{ss'}}\right)^{1-\sigma^u_{s'}}$$

This expression shows that the flat fee varies across sellers only through the CES demand shifter $(p^{\ell}_{js}/P^{\ell}_{ss'})^{1-\sigma^u_{s'}}$, and varies across buyers only through the observable layer-sector pair (ℓ, s).

C.5 Verification of Upstream Revenue Homogeneity

We now verify that the optimal nonlinear tariff implies a homogeneous revenue function for upstream sellers. Fix an upstream seller $j \in \mathcal{U}_{s'}$. From the previous subsection, the optimal marginal price charged to buyers in layer-sector pair (ℓ, s) is $p^{\ell}_{js} = \mu^{\ell}_{ss'}c_j$. Since $c_j = C^u_{s'}/z_j$ for all sellers in sector s', the relative price of seller j's variety within any buyer partition (ℓ, s) is

$$r_j \equiv \frac{p^{\ell}_{js}}{P^{\ell}_{ss'}} = \frac{\widetilde{z}^u_{s'}}{z_j\left(N^u_{s'}\right)^{\frac{1}{1-\sigma^u_{s'}}}}$$

which is independent of the buyer layer-sector pair (ℓ, s).

Define average sector-s' materials demand among buyers in (ℓ, s) by

$$M^{\ell}_{ss'} \equiv \frac{1}{N^{\ell}_{s}}\int_{i\in\mathcal{F}_{\ell s}} M_{is'}\, dv^{\ell}_{s}(i)$$

Then aggregate demand for seller j's variety from buyer group (ℓ, s) is $N^{\ell}_{s}M^{\ell}_{ss'}r_j^{-\sigma^u_{s'}}$. Aggregating

over all buyer groups, total output of seller j is

$$Q_j = r_j^{-\sigma^u_{s'}} D_{s'}, \qquad D_{s'} \equiv \sum_{\ell\in\{u,r\}} \sum_{s\in\mathcal{S}} N_s^\ell M_{ss'}^\ell$$

Variable revenue from the marginal-price component is

$$R_j^V = \sum_{\ell\in\{u,r\}} \sum_{s\in\mathcal{S}} p_{js}^\ell N_s^\ell M_{ss'}^\ell r_j^{-\sigma^u_{s'}} = r_j^{1-\sigma^u_{s'}} \sum_{\ell\in\{u,r\}} \sum_{s\in\mathcal{S}} P_{ss'}^\ell N_s^\ell M_{ss'}^\ell$$

where the second equality uses $p_{js}^\ell = r_j P_{ss'}^\ell$. Using the flat fee derived above, total flat-fee revenue is

$$R_j^F = r_j^{1-\sigma^u_{s'}} \sum_{\ell\in\{u,r\}} \sum_{s\in\mathcal{S}} N_s^\ell \frac{1}{\sigma^u_{s'}-1} P_{ss'}^\ell M_{ss'}^\ell(\underline{z}_s^\ell)$$

where $M_{ss'}^\ell(\underline{z}_s^\ell)$ denotes the sector-s' materials demand of the lowest active buyer type in (ℓ, s). Combining variable and flat-fee revenue,

$$R_j = R_j^V + R_j^F = r_j^{1-\sigma^u_{s'}} B_{s'}, \qquad B_{s'} \equiv \sum_{\ell\in\{u,r\}} \sum_{s\in\mathcal{S}} N_s^\ell P_{ss'}^\ell \left[M_{ss'}^\ell + \frac{1}{\sigma^u_{s'}-1} M_{ss'}^\ell(\underline{z}_s^\ell) \right]$$

Since $Q_j = r_j^{-\sigma^u_{s'}} D_{s'}$, we have $r_j^{1-\sigma^u_{s'}} = (Q_j/D_{s'})^{(\sigma^u_{s'}-1)/\sigma^u_{s'}}$. Therefore

$$R_j = A_{s'}^u Q_j^{\psi^u_{s'}}, \qquad \psi_{s'}^u = \frac{\sigma_{s'}^u - 1}{\sigma_{s'}^u}, \qquad A_{s'}^u = B_{s'} D_{s'}^{-\frac{\sigma^u_{s'}-1}{\sigma^u_{s'}}}$$

This verifies the homogeneous-revenue conjecture for upstream sellers. Moreover, because both marginal payments and flat fees scale with the same seller-specific CES demand shifter, the revenue shares coming from marginal payments and flat fees are common across sellers within sector s'.

C.6 Recap: Verified Coefficients

The derivations above verify the conjectured contract and revenue forms used in the main text. For upstream sellers $j \in \mathcal{U}_{s'}$, the optimal nonlinear contract is a two-part tariff with marginal price $p_{js}^\ell = \mu_{ss'}^\ell c_j$, where $\mu_{ss'}^\ell = \rho_{ss'}^\ell/(\rho_{ss'}^\ell - 1)$ and $\rho_{ss'}^\ell = \sigma_{s'}^u \xi_s^\ell$, and a flat fee pinned down by lowest-type individual rationality. Upstream revenue is homogeneous of degree $\psi_{s'}^u = (\sigma_{s'}^u - 1)/\sigma_{s'}^u$, with shifter $A_{s'}^u = B_{s'} D_{s'}^{-(\sigma^u_{s'}-1)/\sigma^u_{s'}}$. For retailers, CES final demand implies $\psi_s^r = (\sigma_s^r - 1)/\sigma_s^r$ and $A_s^r = P_s^{1-1/\sigma_s^r} (\theta_s \mathcal{E})^{1/\sigma_s^r}$. Thus, all conjectured coefficients in Guess 1 and Guess 2 are pinned down by equilibrium objects, with $\psi_s^\ell = (\sigma_s^\ell - 1)/\sigma_s^\ell$ for both layers. Since the flat fee leaves the lowest buyer type with zero surplus from each bilateral input relationship, every buyer weakly participates in

each transaction, conditional on operating.

C.7 Derivations for the Theoretical Results

C.7.1 Result 1: Allocative markup

Under nonlinear pricing, the allocative markup charged by an upstream seller in sector s' to buyers in partition (ℓ, s) is

$$\mu^{\ell}_{ss'} = \frac{\rho^{\ell}_{ss'}}{\rho^{\ell}_{ss'} - 1}, \qquad \rho^{\ell}_{ss'} = \sigma^{u}_{s'} \xi^{\ell}_{s}$$

The uniform-pricing CES markup is

$$\mu^{\text{Lin}}_{s'} = \frac{\sigma^{u}_{s'}}{\sigma^{u}_{s'} - 1}$$

Since $x/(x-1)$ is strictly decreasing for $x > 1$, and since $\xi^{\ell}_{s} > 1$ implies $\rho^{\ell}_{ss'} > \sigma^{u}_{s'}$, we have

$$\mu^{\ell}_{ss'} = \frac{\rho^{\ell}_{ss'}}{\rho^{\ell}_{ss'} - 1} < \frac{\sigma^{u}_{s'}}{\sigma^{u}_{s'} - 1} = \mu^{\text{Lin}}_{s'}$$

Thus nonlinear pricing lowers the allocative markup relative to uniform CES pricing.

C.7.2 Result 2: Margin size and margin collection

Margin invariance. For an upstream seller, the allocative margin and flat-fee revenue sum to the same share of revenue as under uniform CES pricing.

Lemma 3 (Margin invariance)**.** *For any upstream seller $j \in \mathcal{U}_{s'}$, the allocative margin and flat-fee revenue satisfy*

$$\sum_{\ell,s} \int_{i \in \mathcal{F}_{\ell s}} (p^{\ell}_{js} - c_j) m_{ij} \, dv^{\ell}_{s}(i) + \sum_{\ell,s} N^{\ell}_{s} F^{\ell}_{js} = \frac{1}{\sigma^{u}_{s'}} R_j, \qquad \textit{equivalently} \qquad R_j = \frac{\sigma^{u}_{s'}}{\sigma^{u}_{s'} - 1} c_j Q_j$$

Proof. Fix a buyer partition (ℓ, s) and write $\sigma \equiv \sigma^{u}_{s'}$, $G \equiv G^{\ell}_{ss'}$, $g \equiv g^{\ell}_{ss'}$, $c \equiv c_j$. A buyer of type τ receiving quantity x obtains gross value

$$v(x, \tau) = \frac{\sigma}{\sigma - 1} \tau \, x^{\frac{\sigma - 1}{\sigma}}$$

which is linear in τ, so $\partial v / \partial \tau = v/\tau$. Let $U(\tau) = v(x(\tau), \tau) - T(\tau)$ denote the buyer's surplus under the seller's mechanism. Local incentive compatibility gives the envelope condition $U'(\tau) = \frac{\sigma}{\sigma - 1} x(\tau)^{\frac{\sigma - 1}{\sigma}}$. Integrating by parts and using that individual rationality binds at the bottom, $U(\underline{\tau}) = 0$,

$$\mathbb{E}[U] = \int_{\underline{\tau}}^{\infty} U'(\tau) \, [1 - G(\tau)] \, d\tau = \mathbb{E}\left[\frac{1 - G(\tau)}{g(\tau)} \, U'(\tau) \right]$$

Since $T = v - U$, expected transfers are

$$\mathbb{E}[T] = \mathbb{E}\left[\frac{\sigma}{\sigma-1}\,\hat{\tau}(\tau)\,x(\tau)^{\frac{\sigma-1}{\sigma}}\right], \qquad \hat{\tau}(\tau) \equiv \tau - \frac{1-G(\tau)}{g(\tau)}$$

The seller maximizes $\mathbb{E}[T] - c\,\mathbb{E}[x]$ pointwise, with first-order condition $\hat{\tau}(\tau)\,x(\tau)^{-1/\sigma} = c$ for every type with $x(\tau) > 0$. Multiplying by $x(\tau)$,

$$\hat{\tau}(\tau)\,x(\tau)^{\frac{\sigma-1}{\sigma}} = c\,x(\tau)$$

which also holds trivially for excluded types with $x(\tau) = 0$. Substituting into $\mathbb{E}[T]$,

$$\mathbb{E}[T] = \frac{\sigma}{\sigma-1}\,c\,\mathbb{E}[x]$$

Multiplying by N_s^ℓ and summing over buyer partitions, total revenue equals $\frac{\sigma}{\sigma-1}$ times total variable cost, $R_j = \frac{\sigma}{\sigma-1}c_jQ_j$. Therefore $R_j - c_jQ_j = R_j/\sigma^u_{s'}$, and the allocative margin plus flat-fee revenue equals $R_j - c_jQ_j$. □

The lemma uses neither the Pareto assumption nor the two-part tariff implementation; it requires only that buyer value be multiplicatively separable in type and isoelastic in quantity, that marginal cost be constant, and that individual rationality binds at the bottom. As a check, under Pareto types with the two-part tariff of Proposition 1, flat fees relative to variable receipts from partition (ℓ, s) equal $\frac{1}{\sigma^u_{s'}-1}\frac{M^\ell_{ss'}(\underline{z}^\ell_s)}{M^\ell_{ss'}} = \frac{1-1/\xi^\ell_s}{\sigma^u_{s'}-1}$, using $M^\ell_{ss'}/M^\ell_{ss'}(\underline{z}^\ell_s) = \xi^\ell_s/(\xi^\ell_s - 1)$, so revenue relative to variable receipts is $(\sigma^u_{s'} - 1/\xi^\ell_s)/(\sigma^u_{s'} - 1)$; variable cost relative to variable receipts is $1/\mu^\ell_{ss'} = (\sigma^u_{s'} - 1/\xi^\ell_s)/\sigma^u_{s'}$. The ratio of the two is $(\sigma^u_{s'} - 1)/\sigma^u_{s'}$ for every ξ^ℓ_s, as the lemma requires.

Composition of the margin. The same calculation gives the share of revenue from partition (ℓ, s) collected through flat fees,

$$\frac{R^{\ell,F}_{j,s}}{R^\ell_{j,s}} = \frac{1 - 1/\xi^\ell_s}{\sigma^u_{s'} - 1/\xi^\ell_s}$$

which is strictly increasing in ξ^ℓ_s since $\sigma^u_{s'} > 1$. It converges to 0 as $\xi^\ell_s \to 1$, where the allocative markup converges to the CES markup $\sigma^u_{s'}/(\sigma^u_{s'} - 1)$, and to $1/\sigma^u_{s'}$ as $\xi^\ell_s \to \infty$, where the allocative markup converges to one and the entire margin is collected through the fee. With heterogeneous ξ^ℓ_s across partitions, the seller's total fee share is the revenue-weighted average of these partition-level shares.

C.7.3 Result 3: Average prices and quantity discounts

Fix a seller sector s', a buyer partition (ℓ, s), and a seller $j \in \mathcal{U}_{s'}$. The optimal two-part tariff is

$$T_{ij} = p^{\ell}_{js} m_{ij} + F^{\ell}_{js}, \qquad p^{\ell}_{js} = \mu^{\ell}_{ss'} c_j$$

with flat fee

$$F^{\ell}_{js} = \frac{1}{\sigma^u_{s'} - 1} P^{\ell}_{ss'} M^{\ell}_{ss'}(\underline{z}^{\ell}_s) \left(\frac{p^{\ell}_{js}}{P^{\ell}_{ss'}} \right)^{1-\sigma^u_{s'}}$$

Buyer i's CES demand for seller j's variety is

$$m_{ij} = M_{is'} \left(\frac{p^{\ell}_{js}}{P^{\ell}_{ss'}} \right)^{-\sigma^u_{s'}}$$

Therefore,

$$\frac{F^{\ell}_{js}}{m_{ij}} = \frac{1}{\sigma^u_{s'} - 1} \frac{M^{\ell}_{ss'}(\underline{z}^{\ell}_s)}{M_{is'}} p^{\ell}_{js}$$

Hence the total markup satisfies

$$\frac{T_{ij}/m_{ij}}{c_j} = \frac{p^{\ell}_{js}}{c_j} \left[1 + \frac{1}{\sigma^u_{s'} - 1} \frac{M^{\ell}_{ss'}(\underline{z}^{\ell}_s)}{M_{is'}} \right] = \mu^{\ell}_{ss'} \left[1 + \chi^{\ell}_{ss'}(i) \right], \qquad \chi^{\ell}_{ss'}(i) \equiv \frac{1}{\sigma^u_{s'} - 1} \frac{M^{\ell}_{ss'}(\underline{z}^{\ell}_s)}{M_{is'}}$$

The term $\chi^{\ell}_{ss'}(i)$ depends on the buyer's scale relative to the lowest type in the same partition, but not on seller identity j. Therefore, within a buyer partition (ℓ, s) and seller sector s', the total markup is invariant to the seller's identity.

By the scaling of input demand derived above, the sector-s' materials bundle scales with buyer productivity according to

$$M_{is'} = M^{\ell}_{ss'}(\overline{z}^{\ell}_s) \left(\frac{z_i}{\overline{z}^{\ell}_s} \right)^{\sigma^{\ell}_s - 1}$$

Hence

$$\frac{M^{\ell}_{ss'}(\underline{z}^{\ell}_s)}{M_{is'}} = \left(\frac{\underline{z}^{\ell}_s}{z_i} \right)^{\sigma^{\ell}_s - 1}, \qquad \chi^{\ell}_{ss'}(i) = \frac{1}{\sigma^u_{s'} - 1} \left(\frac{\underline{z}^{\ell}_s}{z_i} \right)^{\sigma^{\ell}_s - 1}$$

Thus, the total markup depends on the buyer only through its productivity relative to the lowest active type in the same partition, and remains independent of seller identity.

C.7.4 Result 4: Flat fees and entry incentives

Throughout, recall that $\psi_s^\ell = (\sigma_s^\ell - 1)/\sigma_s^\ell$, where σ_s^ℓ is the demand elasticity faced by a firm in (ℓ, s) as a seller, and define the fee intensity of buyers in (ℓ, s)

$$\mathcal{H}_s^\ell \equiv \sum_{s' \in \mathcal{S}} \frac{\theta_{ss'}^\ell}{\sigma_{s'}^u - 1}$$

Decomposition. For an upstream seller $j \in \mathcal{U}_{s'}$, total revenue consists of marginal payments and flat fees

$$R_j = \sum_{\ell \in \{u,r\}} \sum_{s \in \mathcal{S}} \left[\int_{i \in \mathcal{F}_{\ell s}} p_{js}^\ell m_{ij} \, d\nu_s^\ell(i) + N_s^\ell F_{js}^\ell \right]$$

Variable production costs are $c_j Q_j$, and upstream market clearing implies $Q_j = \sum_{\ell,s} \int_{\mathcal{F}_{\ell s}} m_{ij} \, d\nu_s^\ell(i)$. Hence

$$c_j Q_j = \sum_{\ell \in \{u,r\}} \sum_{s \in \mathcal{S}} \int_{i \in \mathcal{F}_{\ell s}} c_j m_{ij} \, d\nu_s^\ell(i)$$

Subtracting variable costs and the flat fees paid by j to its own upstream suppliers gives

$$\Pi_j^u = \sum_{\ell \in \{u,r\}} \sum_{s \in \mathcal{S}} \int_{i \in \mathcal{F}_{\ell s}} (p_{js}^\ell - c_j) m_{ij} \, d\nu_s^\ell(i) + \sum_{\ell \in \{u,r\}} \sum_{s \in \mathcal{S}} N_s^\ell F_{js}^\ell - \sum_{t \in \mathcal{S}} \int_{h \in \mathcal{U}_t} F_{hs'}^u \, d\nu_t^u(h)$$

By Lemma 3, the first two terms sum to $R_j/\sigma_{s'}^u$, so

$$\Pi_j^u = \frac{1}{\sigma_{s'}^u} R_j - \sum_{t \in \mathcal{S}} \int_{h \in \mathcal{U}_t} F_{hs'}^u \, d\nu_t^u(h)$$

This is the decomposition in the main text.

For a retailer $i \in \mathcal{R}_s$, revenue is R_i. Since retail demand is CES, $\psi_s^r = (\sigma_s^r - 1)/\sigma_s^r$, and constant returns imply variable costs equal $\psi_s^r R_i$. Thus variable profits before flat fees are $R_i - \psi_s^r R_i = R_i/\sigma_s^r$, and

$$\Pi_i^r = \frac{1}{\sigma_s^r} R_i - \sum_{s' \in \mathcal{S}} \int_{j \in \mathcal{U}_{s'}} F_{js}^r \, d\nu_{s'}^u(j)$$

Average flat fee. For buyer partition (ℓ, s) and seller sector s', define the average flat fee paid to sellers in s' by

$$\overline{F}_{ss'}^\ell \equiv \frac{1}{N_{s'}^u} \int_{j \in \mathcal{U}_{s'}} F_{js}^\ell \, d\nu_{s'}^u(j) = \frac{1}{N_{s'}^u} \frac{1}{\sigma_{s'}^u - 1} P_{ss'}^\ell M_{ss'}^\ell(\underline{z}_s^\ell) \int_{j \in \mathcal{U}_{s'}} \left(\frac{p_{js}^\ell}{P_{ss'}^\ell} \right)^{1 - \sigma_{s'}^u} d\nu_{s'}^u(j)$$

By definition of the CES price index, the integral equals one. Thus

$$\overline{F}^{\ell}_{ss'} = \frac{P^{\ell}_{ss'} M^{\ell}_{ss'}(\underline{z}^{\ell}_{s})}{N^{u}_{s'}(\sigma^{u}_{s'} - 1)}$$

The buyer first-order condition for materials from sector s', evaluated at the lowest buyer type, gives

$$P^{\ell}_{ss'} M^{\ell}_{ss'}(\underline{z}^{\ell}_{s}) = \psi^{\ell}_{s} R^{\ell}_{s}(\underline{z}^{\ell}_{s})(1 - \alpha^{\ell}_{s})\theta^{\ell}_{ss'},$$

Therefore,

$$\overline{F}^{\ell}_{ss'} = \frac{\psi^{\ell}_{s}}{\sigma^{u}_{s'} - 1}(1 - \alpha^{\ell}_{s})\theta^{\ell}_{ss'} \frac{R^{\ell}_{s}(\underline{z}^{\ell}_{s})}{N^{u}_{s'}}$$

Fees paid and net profit shares. By the expression for the average flat fee above, every buyer in (ℓ, s) pays the same total flat fee to sellers in s', $N^{u}_{s'} \overline{F}^{\ell}_{ss'} = P^{\ell}_{ss'} M^{\ell}_{ss'}(\underline{z}^{\ell}_{s})/(\sigma^{u}_{s'} - 1)$. Using the materials first-order condition at the lowest type, $P^{\ell}_{ss'} M^{\ell}_{ss'}(\underline{z}^{\ell}_{s}) = \psi^{\ell}_{s}(1 - \alpha^{\ell}_{s})\theta^{\ell}_{ss'} R^{\ell}_{s}(\underline{z}^{\ell}_{s})$, and summing over supplier sectors, total fees paid by any firm in (ℓ, s) are

$$\Phi^{\ell}_{s} = \psi^{\ell}_{s}\,(1 - \alpha^{\ell}_{s})\,\mathcal{H}^{\ell}_{s}\,R^{\ell}_{s}(\underline{z}^{\ell}_{s})$$

They depend on the network and on the lowest type, but not on the paying firm's own size. Combining with Lemma 3 for upstream firms and with the CES margin for retailers, the profit share of a firm i in (ℓ, s) is

$$\frac{\Pi_i}{R_i} = \frac{1}{\sigma^{\ell}_{s}} - \frac{\sigma^{\ell}_{s} - 1}{\sigma^{\ell}_{s}}\,(1 - \alpha^{\ell}_{s})\,\mathcal{H}^{\ell}_{s}\,\frac{R^{\ell}_{s}(\underline{z}^{\ell}_{s})}{R_i}$$

Profit shares therefore rise with firm size and converge to the CES margin $1/\sigma^{\ell}_{s}$ for the largest firms, whose fees are negligible relative to revenue.

Lowest-type profits and the hold-up condition. Evaluating at $R_i = R^{\ell}_{s}(\underline{z}^{\ell}_{s})$,

$$\Pi^{\ell}_{s}(\underline{z}^{\ell}_{s}) = \frac{R^{\ell}_{s}(\underline{z}^{\ell}_{s})}{\sigma^{\ell}_{s}}\left[1 - (\sigma^{\ell}_{s} - 1)(1 - \alpha^{\ell}_{s})\mathcal{H}^{\ell}_{s}\right]$$

Thus lowest-type profits are nonnegative if and only if

$$(\sigma^{\ell}_{s} - 1)(1 - \alpha^{\ell}_{s}) \sum_{s' \in \mathcal{S}} \frac{\theta^{\ell}_{ss'}}{\sigma^{u}_{s'} - 1} \leq 1$$

The condition summarizes the hold-up force generated by multiple suppliers: flat fees are larger when firms are more materials intensive (low α^{ℓ}_{s}) and when supplier varieties are harder to substitute (low $\sigma^{u}_{s'}$). The lowest type retains positive profits because labor, priced competitively, is not

subject to flat fees: a larger labor share α_s^ℓ relaxes the condition.

Affine representation of average profits. Finally, sector-average upstream profits are affine in sectoral labor expenditures. Let L_s^ℓ denote aggregate production labor in (ℓ, s). Fix an upstream sector m, and consider the average upstream seller with productivity $\widetilde{z}_m^u$. From the revenue decomposition above, the average seller's revenue from buyers in partition (ℓ, s) is

$$\frac{1}{N_m^u} N_s^\ell P_{sm}^\ell \left[M_{sm}^\ell + \frac{1}{\sigma_m^u - 1} M_{sm}^\ell(\underline{z}_s^\ell) \right]$$

where M_{sm}^ℓ denotes average sector-m materials demand among buyers in (ℓ, s). The factor $1/N_m^u$ follows from evaluating the CES demand shifter at the average upstream seller

$$\left(\frac{Q_j}{D_m} \right)^{\frac{\sigma_m^u - 1}{\sigma_m^u}} = \frac{1}{N_m^u} \qquad \text{when } z_j = \widetilde{z}_m^u$$

Using the input-demand first-order condition

$$N_s^\ell P_{sm}^\ell M_{sm}^\ell = w L_s^\ell \frac{(1 - \alpha_s^\ell)\theta_{sm}^\ell}{\alpha_s^\ell}$$

and the scaling of lowest-type materials demand

$$M_{sm}^\ell(\underline{z}_s^\ell) = M_{sm}^\ell \left(\frac{\underline{z}_s^\ell}{\widetilde{z}_s^\ell} \right)^{\sigma_s^\ell - 1}$$

we can write revenue received by the average upstream seller in sector m as

$$R_m^u(\widetilde{z}_m^u) = \frac{w}{N_m^u} \sum_{\ell \in \{u,r\}} \sum_{s \in \mathcal{S}} L_s^\ell v_{sm}^\ell, \qquad v_{sm}^\ell \equiv \frac{(1 - \alpha_s^\ell)\theta_{sm}^\ell}{\alpha_s^\ell} \left[1 + \frac{1}{\sigma_m^u - 1} \left(\frac{\underline{z}_s^\ell}{\widetilde{z}_s^\ell} \right)^{\sigma_s^\ell - 1} \right]$$

The average upstream firm in sector m also pays labor costs, variable materials payments, and flat fees to its own suppliers. Its direct labor cost is $w L_m^u / N_m^u$. For inputs purchased from upstream supplier sector q, its variable materials payments and flat-fee payments are

$$\frac{w L_m^u}{N_m^u} \frac{(1 - \alpha_m^u)\theta_{mq}^u}{\alpha_m^u} \left[1 + \frac{1}{\sigma_q^u - 1} \left(\frac{\underline{z}_m^u}{\widetilde{z}_m^u} \right)^{\sigma_m^u - 1} \right]$$

Therefore define

$$\bar{v}_m^u \equiv 1 + \sum_{q \in \mathcal{S}} \frac{(1 - \alpha_m^u)\theta_{mq}^u}{\alpha_m^u} \left[1 + \frac{1}{\sigma_q^u - 1} \left(\frac{\underline{z}_m^u}{\widetilde{z}_m^u} \right)^{\sigma_m^u - 1} \right]$$

where the leading 1 is the direct labor-cost term. Hence average profit of an upstream firm in sector m is

$$\mathbb{E}\Pi_m^u = \frac{w}{N_m^u}\left[\sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}} L_s^\ell v_{sm}^\ell - L_m^u \bar{v}_m^u\right]$$

This affine representation is the object used in the equilibrium-existence argument. It is consistent with Lemma 3: the labor and variable-materials terms in $\bar{v}_m^u$ equal the average firm's variable cost, which the lemma pins at a share $(\sigma_m^u - 1)/\sigma_m^u$ of its revenue.

C.8 Proof: Existence and Uniqueness

This section establishes existence and uniqueness of the steady-state equilibrium under the nonlinear-pricing benchmark. The proof proceeds constructively. We first show that final demand pins down the relative allocation of production labor across retail sectors. We then use input-output demand to map retail labor into upstream labor. Free-entry conditions determine entry flows as linear functions of sectoral production labor. Aggregate labor clearing then pins down the unique scale of the economy. Finally, conditional on firm masses, the price and cost indices solve a log-linear system.

Let L_s^ℓ denote total production labor used by active firms of layer $\ell \in \{u, r\}$ in sector s, and let N_s^ℓ denote the corresponding mass of active firms. Active firms survive to the next period with probability δ_s^ℓ and exit with probability $1 - \delta_s^\ell$. Hence the steady-state entry flow is

$$e_s^\ell = (1 - \delta_s^\ell) N_s^\ell$$

Entry requires $c_s^{E\,\ell}$ units of labor. Therefore aggregate labor clearing is

$$1 = \sum_{s\in\mathcal{S}} L_s^r + \sum_{s\in\mathcal{S}} L_s^u + \sum_{s\in\mathcal{S}} c_s^{E\,r} e_s^r + \sum_{s\in\mathcal{S}} c_s^{E\,u} e_s^u$$

Throughout this subsection, $M_{ss'}^\ell$ denotes average demand for inputs from upstream sector s' among active buyers in layer-sector pair (ℓ, s). Thus aggregate spending by buyers in (ℓ, s) on inputs from sector s' is $N_s^\ell P_{ss'}^\ell M_{ss'}^\ell$.

Step 1. Retail labor shares

Final demand is Cobb–Douglas across retail sectors, so aggregate revenue of retailers in sector s is

$$\mathcal{R}_s^r = \theta_s \mathcal{E}$$

For a retailer in sector s, the revenue function is homogeneous of degree $\psi_s^r = (\sigma_s^r - 1)/\sigma_s^r$. The labor first-order condition implies

$$wL_s^r = \alpha_s^r \psi_s^r \mathcal{R}_s^r = \alpha_s^r \psi_s^r \theta_s \mathcal{E}$$

Therefore, for any two retail sectors s and k

$$\frac{L_s^r}{L_k^r} = \frac{\alpha_s^r \psi_s^r \theta_s}{\alpha_k^r \psi_k^r \theta_k}$$

Taking sector 1 as the reference sector, define

$$\omega_s^r \equiv \frac{\alpha_s^r \psi_s^r \theta_s}{\alpha_1^r \psi_1^r \theta_1}, \qquad \omega_1^r = 1$$

Then retail production labor lies on a one-dimensional ray:

$$\mathbf{L}_r = \boldsymbol{\omega}^r L_1^r, \qquad \boldsymbol{\omega}^r = (\omega_1^r, \ldots, \omega_S^r)'$$

Step 2. Upstream labor requirements

We next express upstream production labor as a linear function of retail production labor. Aggregating the buyer first-order conditions over active firms in (ℓ, s), we have

$$N_s^\ell P_{ss'}^\ell M_{ss'}^\ell = wL_s^\ell \frac{(1-\alpha_s^\ell)\theta_{ss'}^\ell}{\alpha_s^\ell}$$

Under the optimal nonlinear tariff, the marginal price charged by upstream sellers in sector s' to buyers in (ℓ, s) is $p_{js}^\ell = \mu_{ss'}^\ell c_j$. Hence variable payments to upstream sector s' are a markup $\mu_{ss'}^\ell$ over production costs of that seller sector. It follows that the labor used in upstream sector s' to serve buyers in (ℓ, s) is

$$L_{s' \leftarrow (\ell,s)}^u = \frac{\alpha_{s'}^u}{\mu_{ss'}^\ell} L_s^\ell \frac{(1-\alpha_s^\ell)\theta_{ss'}^\ell}{\alpha_s^\ell}$$

Aggregating over all upstream and retail buyers gives, for each upstream sector s'

$$L_{s'}^u = \sum_{s \in \mathcal{S}} \frac{\alpha_{s'}^u}{\mu_{ss'}^u} L_s^u \frac{(1-\alpha_s^u)\theta_{ss'}^u}{\alpha_s^u} + \sum_{s \in \mathcal{S}} \frac{\alpha_{s'}^u}{\mu_{ss'}^r} L_s^r \frac{(1-\alpha_s^r)\theta_{ss'}^r}{\alpha_s^r}$$

In vector form,

$$\mathbf{L}_u = \mathbf{A}\mathbf{L}_u + \mathbf{B}\mathbf{L}_r$$

where

$$\mathbf{A}_{s's} = \frac{\alpha_{s'}^u}{\mu_{ss'}^u} \frac{(1-\alpha_s^u)\theta_{ss'}^u}{\alpha_s^u}, \qquad \mathbf{B}_{s's} = \frac{\alpha_{s'}^u}{\mu_{ss'}^r} \frac{(1-\alpha_s^r)\theta_{ss'}^r}{\alpha_s^r}$$

The upstream feedback matrix is nonnegative. We impose the regularity condition that its spectral radius satisfies $\text{spr}(\mathbf{A}) < 1$, which holds at our calibrated parameter values. A simple sufficient condition is

$$\max_{s' \in \mathcal{S}} \sum_{s \in \mathcal{S}} \frac{\alpha^u_{s'}}{\mu^u_{ss'}} \frac{(1-\alpha^u_s)\theta^u_{ss'}}{\alpha^u_s} < 1$$

Define $\mathbf{\Psi} \equiv (I - \mathbf{A})^{-1}\mathbf{B}$. Then, using Step 1,

$$\mathbf{L}_u = \mathbf{\Psi}\mathbf{L}_r = \mathbf{\Psi}\boldsymbol{\omega}^r L^r_1$$

Step 3. Retail entry flows

Retail free entry pins down average production labor per active retailer. Let $\bar{R}^r_s \equiv R^r_s(\bar{z}^r_s)$ denote revenue of the average-productivity retailer in sector s. From the retail profit expression

$$\mathbb{E}\Pi^r_s = \frac{\bar{R}^r_s}{\sigma^r_s}\left[1 - \left(\frac{z^r_s}{\bar{z}^r_s}\right)^{\sigma^r_s - 1}(1-\alpha^r_s)(\sigma^r_s - 1)\mathcal{H}^r_s\right]$$

where $\mathcal{H}^r_s$ is the fee intensity defined above. Using

$$\frac{\bar{R}^r_s}{\sigma^r_s} = \frac{1}{\sigma^r_s - 1}\frac{w\bar{l}^r_s}{\alpha^r_s}$$

and imposing free entry

$$\mathbb{E}\Pi^r_s = w c^{E\,r}_s (1 - \delta^r_s)$$

we obtain

$$\bar{l}^r_s = \frac{c^{E\,r}_s(1-\delta^r_s)(\sigma^r_s - 1)\alpha^r_s}{1 - \left(\frac{z^r_s}{\bar{z}^r_s}\right)^{\sigma^r_s - 1}(1-\alpha^r_s)(\sigma^r_s - 1)\mathcal{H}^r_s}$$

We assume the denominator is strictly positive, so that $\bar{l}^r_s > 0$.

Since $L^r_s = N^r_s \bar{l}^r_s$, we have

$$N^r_s = \frac{L^r_s}{\bar{l}^r_s}, \qquad e^r_s = (1-\delta^r_s)\frac{L^r_s}{\bar{l}^r_s}$$

Define

$$\Gamma_r \equiv \text{diag}\left(\frac{1-\delta^r_s}{\bar{l}^r_s}\right)_{s \in \mathcal{S}}$$

Then

$$\mathbf{e}_r = \Gamma_r \mathbf{L}_r$$

Step 4. Upstream profits and entry flows

Recall from Appendix C.7 that average profits of an upstream firm in sector m can be written as

$$\mathbb{E}\Pi_m^u = \frac{w}{N_m^u}\left[\sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}} L_s^\ell v_{sm}^\ell - L_m^u \bar{v}_m^u\right]$$

Free entry requires

$$\mathbb{E}\Pi_m^u = w c_m^{Eu}(1-\delta_m^u)$$

Multiplying by N_m^u/w and using $e_m^u = (1-\delta_m^u)N_m^u$, we obtain

$$\sum_{s\in\mathcal{S}} L_s^r v_{sm}^r + \sum_{s\in\mathcal{S}} L_s^u v_{sm}^u - L_m^u \bar{v}_m^u = c_m^{Eu} e_m^u$$

Stacking across upstream sectors gives

$$(U_u - \bar{U}_u)\mathbf{L}_u + U_r \mathbf{L}_r = C_u \mathbf{e}_u$$

where

$$[U_r]_{ms} = v_{sm}^r, \qquad [U_u]_{ms} = v_{sm}^u, \qquad \bar{U}_u = \operatorname{diag}(\bar{v}_1^u,\ldots,\bar{v}_S^u), \qquad C_u = \operatorname{diag}(c_1^{Eu},\ldots,c_S^{Eu})$$

Substituting $\mathbf{L}_u = \mathbf{\Psi}\mathbf{L}_r$ yields

$$\mathbf{e}_u = C_u^{-1}\left[(U_u - \bar{U}_u)\mathbf{\Psi} + U_r\right]\mathbf{L}_r \equiv \mathcal{X}_u \mathbf{L}_r$$

For every upstream sector that is used in equilibrium, we require $\mathcal{X}_u \boldsymbol{\omega}^r > 0$, so that implied upstream entry flows are positive along the retail labor ray.

Step 5. Labor clearing and uniqueness of the quantity block

Steps 1–4 express all production labor and entry flows as linear functions of retail labor:

$$\mathbf{L}_r = \boldsymbol{\omega}^r L_1^r, \qquad \mathbf{L}_u = \mathbf{\Psi}\boldsymbol{\omega}^r L_1^r, \qquad \mathbf{e}_r = \Gamma_r \boldsymbol{\omega}^r L_1^r, \qquad \mathbf{e}_u = \mathcal{X}_u \boldsymbol{\omega}^r L_1^r$$

Aggregate labor clearing requires

$$1 = \mathbf{1}^\top \mathbf{L}_r + \mathbf{1}^\top \mathbf{L}_u + (c^{Er})^\top \mathbf{e}_r + (c^{Eu})^\top \mathbf{e}_u$$

where $c^{E\ell} \equiv (c_s^{E\ell})_{s\in\mathcal{S}}$. Substituting the linear representations above gives

$$L_1^r = \left\{\left[\mathbf{1}^\top(I+\mathbf{\Psi}) + (c^{Er})^\top\Gamma_r + (c^{Eu})^\top\mathcal{X}_u\right]\boldsymbol{\omega}^r\right\}^{-1}$$

Under the maintained regularity conditions $\mathrm{spr}(\mathbf{A}) < 1$, $\bar{l}_s^r > 0$, and $\mathcal{X}_u\boldsymbol{\omega}^r \geq 0$, the denominator is strictly positive. Therefore L_1^r exists and is unique. It follows that $\mathbf{L}_r$, $\mathbf{L}_u$, $\mathbf{e}_r$, and $\mathbf{e}_u$ are also uniquely pinned down. Firm masses then follow from $N_s^\ell = e_s^\ell/(1-\delta_s^\ell)$.

Step 6. Price and cost indices

It remains to recover prices, costs, and the final-good price index. Unlike the quantity block, this step is purely nominal: firm masses and production labor have already been pinned down by Steps 1–5. We take labor as the numeraire, $w \equiv 1$, and solve for all sectoral price indices and for P_Y.

Let lowercase letters denote logs. Since $w \equiv 1$, the log wage is zero. For buyers in layer $\ell \in \{u, r\}$ in sector s purchasing from upstream sector s', the CES input price index satisfies

$$p_{ss'}^\ell = \ln\mu_{ss'}^\ell + \ln\Theta_{s'}^u - \ln\widetilde{z}_{s'}^u + (1-\alpha_{s'}^u)\sum_{q\in\mathcal{S}}\theta_{s'q}^u p_{s'q}^u + \frac{1}{1-\sigma_{s'}^u}n_{s'}^u$$

The corresponding retail price index for sector s sold to final demand is

$$p_s^c = \ln\mu_s^r + \ln\Theta_s^r - \ln\widetilde{z}_s^r + (1-\alpha_s^r)\sum_{q\in\mathcal{S}}\theta_{sq}^r p_{sq}^r + \frac{1}{1-\sigma_s^r}n_s^r$$

The final-good price index is

$$p_Y = \ln\Theta_c + \sum_{s\in\mathcal{S}}\theta_s p_s^c, \qquad \Theta_c \equiv \prod_{s\in\mathcal{S}}\theta_s^{-\theta_s}$$

Collect the unknowns in

$$\mathbf{p} = \left(\{p_{ss'}^u\}_{s,s'}, \{p_{ss'}^r\}_{s,s'}, \{p_s^c\}_s\right)'$$

Given the masses $\{N_s^u, N_s^r\}$, the terms $n_s^\ell = \ln N_s^\ell$ are known, so the equations above form a square linear system in $\mathbf{p}$. Because intermediate-input shares sum to $1-\alpha_s^\ell < 1$ within each producing sector, the price-index feedback matrix has spectral radius below one. Hence the log-linear system has a unique solution. The final-good price index P_Y is then uniquely recovered from the final equation.

Other pricing regimes.

The same constructive argument applies under uniform pricing. In that case there are no flat fees and upstream marginal prices are seller-sector specific: $\mu^{\ell}_{ss'}$ is replaced by $\mu^{\text{Lin}}_{s'} = \sigma^{u}_{s'}/(\sigma^{u}_{s'} - 1)$. The retail entry term simplifies to

$$\bar{l}^{r,\text{Lin}}_{s} = c^{E\,r}_{s}(1 - \delta^{r}_{s})(\sigma^{r}_{s} - 1)\alpha^{r}_{s}$$

and the upstream profit coefficients become

$$v^{\ell,\text{Lin}}_{sm} = \frac{(1 - \alpha^{\ell}_{s})\theta^{\ell}_{sm}}{\alpha^{\ell}_{s}}, \qquad \bar{v}^{u,\text{Lin}}_{m} = 1 + \sum_{q \in \mathcal{S}} \frac{(1 - \alpha^{u}_{m})\theta^{u}_{mq}}{\alpha^{u}_{m}}$$

Thus Steps 1–5 go through with the corresponding uniform-pricing matrices $\mathbf{A}^{\text{Lin}}$, $\mathbf{B}^{\text{Lin}}$, U^{Lin}_{u}, U^{Lin}_{r}, and labor clearing again pins down a unique scale whenever the analogue of $\text{spr}(\mathbf{A}) < 1$ holds. The log-price block is also unchanged except that $\ln \mu^{\ell}_{ss'}$ is replaced by $\ln \mu^{\text{Lin}}_{s'}$.

The planner-implemented decentralized equilibrium is similar. Firms post CES markups, but output subsidies exactly undo the within-period pricing wedge, so buyers face marginal-cost prices in their input and final-demand choices. Equivalently, the allocative blocks are obtained from the uniform-pricing system by replacing the effective markups with one. The same linear quantity system and log-price system therefore apply with $\mu^{\text{eff}} = 1$, while entry is supported by the CES markup/subsidy scheme. Hence existence and uniqueness follow from the same spectral and positivity conditions as above.

C.9 Kuhn–Tucker Implementation of the Efficient Allocation

This appendix proves the implementation result used in the main text. The proof follows the logic in Baqaee and Farhi (2020a), specialized to the functional forms of our model. The planner chooses both quantities and active firm masses. We write the planner problem in real units, so shadow values are measured in units of the final good. In the decentralized implementation, we maintain the main-text normalization $w \equiv 1$; nominal prices are therefore converted into real units by dividing by the final-good price index P_Y.

Let $F_i(l_i, \{M_{is'}\}_{s' \in \mathcal{S}})$ denote the production technology of firm i, and let $\mathcal{Y}(\{y_j\}_{j \in \cup_s \mathcal{R}_s})$ denote the final-good aggregator defined in the main text. Write the equilibrium measure in layer-sector pair (ℓ, s) as

$$dv^{\ell}_{s} = N^{\ell}_{s} dH^{\ell}_{s}$$

where H^{ℓ}_{s} is the (exogenous) normalized distribution of active producers. Since active firms exit with probability $1 - \delta^{\ell}_{s}$, maintaining mass N^{ℓ}_{s} requires entry flow $(1 - \delta^{\ell}_{s})N^{\ell}_{s}$, which uses $c^{E\,\ell}_{s}(1 - \delta^{\ell}_{s})N^{\ell}_{s}$ units of labor.

Ignoring nonnegativity constraints for notational simplicity, the planner's Lagrangian is

$$\begin{aligned}\mathcal{L} = Y &+ \Lambda\left(\mathcal{Y}(\{y_j\}_{j\in\cup_s\mathcal{R}_s}) - Y\right)\\ &+ \sum_{s'\in\mathcal{S}}\int_{j\in\mathcal{U}_{s'}}\lambda_j\left[Q_j - \sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}}\int_{i\in\mathcal{F}_{\ell s}} m_{ij}\,dv_s^\ell(i)\right]dv_{s'}^u(j)\\ &+ \sum_{s\in\mathcal{S}}\int_{j\in\mathcal{R}_s}\lambda_j(Q_j - y_j)\,dv_s^r(j)\\ &+ \sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}}\int_{i\in\mathcal{F}_{\ell s}}\eta_i\left[F_i(l_i,\{M_{is'}\}_{s'\in\mathcal{S}}) - Q_i\right]dv_s^\ell(i)\\ &+ \varpi\left[1 - \sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}}\int_{i\in\mathcal{F}_{\ell s}} l_i\,dv_s^\ell(i) - \sum_{\ell\in\{u,r\}}\sum_{s\in\mathcal{S}} c_s^{E\,\ell}(1-\delta_s^\ell)N_s^\ell\right]\end{aligned}$$

Here λ_j is the shadow value of variety j, η_i is the multiplier on producer i's technology constraint, Λ is the multiplier on the final-good aggregator, and ϖ is the shadow value of labor.

The first-order condition with respect to final output gives $\Lambda = 1$. For each retail variety $j \in \mathcal{R}_s$, the first-order condition with respect to y_j gives

$$\lambda_j = \frac{\partial \mathcal{Y}}{\partial y_j}$$

For each producer i, the first-order condition with respect to Q_i gives $\lambda_i = \eta_i$. Hence the shadow value of producer i's output equals the shadow cost of producing it.

For labor, the planner's first-order condition is

$$\varpi = \lambda_i \frac{\partial F_i}{\partial l_i}$$

For materials, m_{ij} affects the planner's objective through two channels. It raises buyer i's sector-s' materials bundle, and hence relaxes buyer i's production constraint, but it also uses one unit of upstream variety j. The first-order condition with respect to m_{ij} is therefore

$$\eta_i \frac{\partial F_i}{\partial M_{is'}}\frac{\partial M_{is'}}{\partial m_{ij}} = \lambda_j$$

Using the output FOC $\eta_i = \lambda_i$, define the shadow value of buyer i's sector-s' materials bundle as

$$\mathcal{P}_{is'} \equiv \lambda_i \frac{\partial F_i}{\partial M_{is'}}$$

Since the sector-s' materials bundle is CES, $\partial M_{is'}/\partial m_{ij} = M_{is'}^{1/\sigma_{s'}^u} m_{ij}^{-1/\sigma_{s'}^u}$, and the FOC for an individ-

ual input variety $j \in \mathcal{U}_{s'}$ becomes

$$\mathcal{P}_{is'} M_{is'}^{1/\sigma^u_{s'}} m_{ij}^{-1/\sigma^u_{s'}} = \lambda_j$$

We now derive the planner's entry condition. Consider an additional infinitesimal mass of an upstream variety $j \in \mathcal{U}_{s'}$. Holding buyer choices fixed and applying the envelope theorem, the gross marginal value to buyer i of adding variety j to the sector-s' CES materials bundle is

$$\frac{\sigma^u_{s'}}{\sigma^u_{s'}-1} \mathcal{P}_{is'} M_{is'}^{1/\sigma^u_{s'}} m_{ij}^{\frac{\sigma^u_{s'}-1}{\sigma^u_{s'}}} = \frac{\sigma^u_{s'}}{\sigma^u_{s'}-1} \lambda_j m_{ij}$$

where the equality uses the planner's first-order condition for m_{ij}. Producing m_{ij} units of variety j has social cost $\lambda_j m_{ij}$. Therefore the net social surplus generated by the buyer-seller relationship (i, j) is $\lambda_j m_{ij}/(\sigma^u_{s'} - 1)$. Aggregating over all buyers, the social surplus generated by an additional active upstream variety $j \in \mathcal{U}_{s'}$ is

$$\mathcal{V}^u_j = \sum_{\ell \in \{u,r\}} \sum_{s \in \mathcal{S}} \int_{i \in \mathcal{F}_{\ell s}} \frac{1}{\sigma^u_{s'}-1} \lambda_j m_{ij} \, dv^\ell_s(i)$$

For a retail variety $j \in \mathcal{R}_s$, the same argument applied to the within-sector CES final-demand aggregator gives gross marginal value $\frac{\sigma^r_s}{\sigma^r_s - 1} \lambda_j y_j$. Net of the resource cost $\lambda_j y_j$, the social surplus generated by an additional active retail variety is

$$\mathcal{V}^r_j = \frac{1}{\sigma^r_s - 1} \lambda_j y_j$$

Since $dv^\ell_s = N^\ell_s dH^\ell_s$, the derivative of the optimized planner objective with respect to N^ℓ_s is the H^ℓ_s-average surplus of an additional active variety, net of the labor cost of replacing exiting firms. The planner's KKT condition for active masses is therefore

$$\mathbb{E}_{j \sim H^\ell_s}[\mathcal{V}^\ell_j] \leq \omega c^{E\,\ell}_s (1 - \delta^\ell_s)$$

with complementary slackness

$$N^\ell_s \left(\mathbb{E}_{j \sim H^\ell_s}[\mathcal{V}^\ell_j] - \omega c^{E\,\ell}_s (1 - \delta^\ell_s) \right) = 0$$

For every active layer-sector pair (ℓ, s), this condition holds with equality.

We now show that the planner allocation can be decentralized with uniform pricing and ad-valorem output subsidies. Consider an upstream seller $j \in \mathcal{U}_{s'}$. Under uniform pricing, demand for its variety has constant elasticity $\sigma^u_{s'}$. Suppose the government pays an ad-valorem output

subsidy at rate $\tilde{\tau}^u_{s'} = 1/(\sigma^u_{s'} - 1)$. The seller chooses its buyer-facing price p_j to maximize

$$(1 + \tilde{\tau}^u_{s'})p_j Q_j(p_j) - C_j(Q_j(p_j))$$

The first-order condition is $(1 + \tilde{\tau}^u_{s'})MR_j = C'_j(Q_j)$. Since CES demand implies $MR_j = p_j(1 - 1/\sigma^u_{s'})$ and $1 + \tilde{\tau}^u_{s'} = \sigma^u_{s'}/(\sigma^u_{s'} - 1)$, the seller's first-order condition implies

$$p_j = C'_j(Q_j)$$

Thus buyers face the social marginal cost of every upstream variety. The same logic applies to retailers: a retailer $j \in \mathcal{R}_s$ faces final demand elasticity σ^r_s, and with ad-valorem output subsidy $\tilde{\tau}^r_s = 1/(\sigma^r_s - 1)$ its first-order condition implies $p_j = C'_j(Q_j)$. Thus final consumers also face marginal-cost prices.

Given these subsidies, the decentralized intratemporal first-order conditions coincide with the planner's. Since every buyer-facing price equals marginal cost, cost minimization by buyer i implies

$$p_j = p_i \frac{\partial F_i}{\partial M_{is'}} \frac{\partial M_{is'}}{\partial m_{ij}}, \qquad j \in \mathcal{U}_{s'}, \qquad\qquad w = p_i \frac{\partial F_i}{\partial l_i}$$

These conditions coincide with the planner's first-order conditions under the identification

$$\lambda_k = \frac{p_k}{P_Y} \qquad \text{for every producer } k, \qquad \varpi = \frac{w}{P_Y}$$

For retail varieties, final demand gives $p_j/P_Y = \partial \mathcal{Y}/\partial y_j = \lambda_j$, which coincides with the planner's retail first-order condition.

It remains to verify the entry margin. Under the upstream subsidy, buyers pay the marginal-cost price $p_j = C'_j(Q_j)$. The subsidy revenue from sales to buyer i, expressed in units of the final good, is

$$\frac{\tilde{\tau}^u_{s'} p_j m_{ij}}{P_Y} = \frac{1}{\sigma^u_{s'} - 1} \frac{p_j}{P_Y} m_{ij} = \frac{1}{\sigma^u_{s'} - 1} \lambda_j m_{ij}$$

where the last equality uses $p_j/P_Y = \lambda_j$ at the implemented allocation. This is exactly the social surplus generated by the buyer-seller relationship. Therefore the real operating profits of upstream seller j satisfy $\Pi^{u,\mathrm{op}}_j/P_Y = \mathcal{V}^u_j$. For a retail seller $j \in \mathcal{R}_s$, real subsidy revenue is

$$\frac{\tilde{\tau}^r_s p_j y_j}{P_Y} = \frac{1}{\sigma^r_s - 1} \frac{p_j}{P_Y} y_j = \frac{1}{\sigma^r_s - 1} \lambda_j y_j = \mathcal{V}^r_j$$

Thus $\Pi^{\ell,\mathrm{op}}_j/P_Y = \mathcal{V}^\ell_j$ for every active producer j in every layer-sector pair (ℓ, s).

The decentralized free-entry condition is

$$\frac{1}{1-\delta_s^\ell}\mathbb{E}_{j\sim H_s^\ell}\left[\Pi_j^{\ell,\mathrm{op}}\right] = w c_s^{E\,\ell}$$

Dividing by P_Y and using $\Pi_j^{\ell,\mathrm{op}}/P_Y = \mathcal{V}_j^\ell$ and $\omega = w/P_Y$, it becomes

$$\mathbb{E}_{j\sim H_s^\ell}[\mathcal{V}_j^\ell] = \omega c_s^{E\,\ell}(1-\delta_s^\ell)$$

which is exactly the planner's KKT condition for active masses.

Hence the decentralized equilibrium with uniform pricing and ad-valorem output subsidies implements the planner allocation. The subsidies make buyer-facing prices equal marginal costs, so marginal input choices and final demand satisfy the planner's intratemporal first-order conditions. At the same time, subsidy payments generate operating profits that, when expressed in units of the final good, exactly equal the CES surplus created by an additional active variety, so decentralized free entry coincides with the planner's entry condition.

C.10 Welfare Decomposition

This appendix first derives a decomposition of the final-good price index and then maps it into welfare. The price-index decomposition is valid across all pricing and policy regimes. The mapping from prices to welfare additionally requires accounting for equilibrium final expenditure. Throughout, changes are computed across two equilibria, lower-case letters denote logarithms, and Δx denotes the change in x.

Let $n = |\mathcal{S}|$. Let $b \in \mathbb{R}^{1\times n}$ denote the vector of Cobb–Douglas final-demand shares, so that

$$\Delta \log P_Y = b\,\Delta p^r$$

where $p^r \in \mathbb{R}^{n\times 1}$ is the vector of retail-sector price indices.

Price-index decomposition. The relevant input-output weights are cost shares with respect to marginal prices, not total payments. Under nonlinear pricing, total payments include flat fees, but flat fees are inframarginal and do not enter marginal costs or input price indices. Define

$$\tilde{\Omega}^{ru}_{ss'} \equiv (1-\alpha_s^r)\theta^r_{ss'}, \qquad \tilde{\Omega}^{uu}_{ss'} \equiv (1-\alpha_s^u)\theta^u_{ss'}$$

where $\tilde{\Omega}^{ru}_{ss'}$ is the marginal-price cost share of upstream seller sector s' in retail buyer sector s, and $\tilde{\Omega}^{uu}_{ss'}$ is the marginal-price cost share of upstream seller sector s' in upstream buyer sector s. These shares are the weights that enter the log marginal-cost equations.

Let $\mu^{ru}_{ss'}$ denote the allocative markup charged by upstream sellers in sector s' to retail buyers

in sector s, and let $\mu^{uu}_{ss'}$ denote the allocative markup charged by upstream sellers in sector s' to upstream buyers in sector s. Let μ^{r}_{s} denote the retail markup faced by final consumers in retail sector s. In subsidized equilibria, these objects are understood as effective buyer-facing markups, after accounting for the corresponding output subsidy.

Let $\mathbf{1} \in \mathbb{R}^{n\times 1}$ denote a vector of ones. Let $\Delta \log \mu^{ru} \in \mathbb{R}^{n\times n}$ and $\Delta \log \mu^{uu} \in \mathbb{R}^{n\times n}$ denote the matrices with entries $\Delta \log \mu^{ru}_{ss'}$ and $\Delta \log \mu^{uu}_{ss'}$, respectively. Define the buyer-level marginal-cost shocks induced by changes in buyer-specific upstream markups

$$g^{ru} := (\tilde{\Omega}^{ru} \odot \Delta \log \mu^{ru})\mathbf{1} \in \mathbb{R}^{n\times 1}, \qquad g^{uu} := (\tilde{\Omega}^{uu} \odot \Delta \log \mu^{uu})\mathbf{1} \in \mathbb{R}^{n\times 1}$$

where $\odot$ denotes elementwise multiplication, so that

$$g^{ru}_{s} = \sum_{s'=1}^{n} \tilde{\Omega}^{ru}_{ss'} \Delta \log \mu^{ru}_{ss'}, \qquad g^{uu}_{s} = \sum_{s'=1}^{n} \tilde{\Omega}^{uu}_{ss'} \Delta \log \mu^{uu}_{ss'}$$

Let $\Delta \log N^{u}, \sigma^{u} \in \mathbb{R}^{n\times 1}$ denote the vectors with entries $\Delta \log N^{u}_{s'}$ and $\sigma^{u}_{s'}$, and let $\Delta \log N^{r}, \sigma^{r} \in \mathbb{R}^{n\times 1}$ denote the vectors with entries $\Delta \log N^{r}_{s}$ and σ^{r}_{s}. Define the variety terms

$$h^{u} \equiv \frac{\Delta \log N^{u}}{\sigma^{u} - 1}, \qquad h^{r} \equiv \frac{\Delta \log N^{r}}{\sigma^{r} - 1}$$

where division is elementwise.

Let $c^{u} \in \mathbb{R}^{n\times 1}$ denote the vector of upstream seller-sector marginal-cost components, excluding the buyer-specific markup charged on the current sale. For a buyer in sector s purchasing from upstream seller sector s', the log input price index changes according to

$$\Delta p^{\ell u}_{ss'} = \Delta \log \mu^{\ell u}_{ss'} + \Delta c^{u}_{s'} - h^{u}_{s'}, \qquad \ell \in \{r, u\}$$

The last term is the CES variety effect from changes in the mass of upstream sellers in sector s'. Selection terms are omitted here and discussed below.

For upstream buyer sector s, the change in its marginal-cost component is the marginal-price cost-share weighted average of its upstream input price changes

$$\Delta c^{u}_{s} = \sum_{s'=1}^{n} \tilde{\Omega}^{uu}_{ss'} \Delta p^{uu}_{ss'}$$

Substituting the previous equation gives

$$\Delta c^{u} = g^{uu} + \tilde{\Omega}^{uu} \Delta c^{u} - \tilde{\Omega}^{uu} h^{u}$$

Assuming the upstream Leontief inverse $\tilde{\Psi}^{uu} := (I - \tilde{\Omega}^{uu})^{-1}$ exists,

$$\Delta c^u = \tilde{\Psi}^{uu}\left(g^{uu} - \tilde{\Omega}^{uu}h^u\right) \tag{19}$$

Retail-sector prices combine retail markups, upstream input costs, and retail variety effects. For retail sector s

$$\Delta p_s^r = \Delta \log \mu_s^r + \sum_{s'=1}^{n} \tilde{\Omega}_{ss'}^{ru} \Delta p_{ss'}^{ru} - h_s^r$$

Equivalently, in vector form,

$$\Delta p^r = \Delta \log \mu^r + g^{ru} + \tilde{\Omega}^{ru}\Delta c^u - \tilde{\Omega}^{ru}h^u - h^r \tag{20}$$

Substituting (19) into (20) and using the identity $\tilde{\Psi}^{uu}\tilde{\Omega}^{uu} + I = \tilde{\Psi}^{uu}$ gives

$$\Delta p^r = \Delta \log \mu^r + g^{ru} + \tilde{\Omega}^{ru}\tilde{\Psi}^{uu}g^{uu} - \tilde{\Omega}^{ru}\tilde{\Psi}^{uu}h^u - h^r$$

Since $\Delta \log P_Y = b\,\Delta p^r$, and defining the upstream cost-based Domar weights $\tilde{\lambda}^u := b\,\tilde{\Omega}^{ru}\tilde{\Psi}^{uu} \in \mathbb{R}^{1\times n}$, the buyer-level price-index decomposition is

$$-\Delta \log P_Y = -\,b\,\Delta \log \mu^r - \,b\,g^{ru} - \,\tilde{\lambda}^u g^{uu} + \,b\left(\frac{\Delta \log N^r}{\sigma^r - 1}\right) + \,\tilde{\lambda}^u\left(\frac{\Delta \log N^u}{\sigma^u - 1}\right) \tag{21}$$

This expression is the collapsed form of the full log price-index system. The first three terms are intensive markup wedges. The last two terms are extensive variety effects from changes in the mass of active retail and upstream firms.

The buyer-level shocks g^{ru} and g^{uu} can be unpacked into bilateral markup wedges. Define the link-level marginal-cost exposure matrices

$$\tilde{\Lambda}^{ru} := \mathrm{Diag}(b^\top)\tilde{\Omega}^{ru}, \qquad \tilde{\Lambda}^{uu} := \mathrm{Diag}\left((\tilde{\lambda}^u)^\top\right)\tilde{\Omega}^{uu}$$

with entries $\tilde{\Lambda}_{ss'}^{ru} = b_s\tilde{\Omega}_{ss'}^{ru}$ and $\tilde{\Lambda}_{ss'}^{uu} = \tilde{\lambda}_s^u\tilde{\Omega}_{ss'}^{uu}$. Hence

$$b\,g^{ru} = \sum_{s=1}^{n}\sum_{s'=1}^{n}\tilde{\Lambda}_{ss'}^{ru}\Delta \log \mu_{ss'}^{ru}, \qquad \tilde{\lambda}^u g^{uu} = \sum_{s=1}^{n}\sum_{s'=1}^{n}\tilde{\Lambda}_{ss'}^{uu}\Delta \log \mu_{ss'}^{uu}$$

Thus a bilateral upstream–upstream markup $\mu_{ss'}^{uu}$ is weighted by the product of buyer sector s's aggregate upstream Domar exposure and seller sector s''s direct marginal-price cost share in buyer s's input bundle. Define seller-level exposures

$$\bar{\lambda}_{s'}^{ru} := \sum_{s=1}^{n}\tilde{\Lambda}_{ss'}^{ru}, \qquad \bar{\lambda}_{s'}^{uu} := \sum_{s=1}^{n}\tilde{\Lambda}_{ss'}^{uu}$$

Whenever the denominator is positive, define normalized exposure weights

$$\omega^{ru}_{s|s'} := \frac{\tilde{\Lambda}^{ru}_{ss'}}{\bar{\lambda}^{ru}_{s'}}, \qquad \omega^{uu}_{s|s'} := \frac{\tilde{\Lambda}^{uu}_{ss'}}{\bar{\lambda}^{uu}_{s'}}$$

The corresponding exposure-weighted seller-level average log-markup changes are

$$\Delta \log \bar{\mu}^{ru}_{s'} := \sum_{s=1}^{n} \omega^{ru}_{s|s'} \Delta \log \mu^{ru}_{ss'}, \qquad \Delta \log \bar{\mu}^{uu}_{s'} := \sum_{s=1}^{n} \omega^{uu}_{s|s'} \Delta \log \mu^{uu}_{ss'}$$

By construction, $\sum_{s,s'} \tilde{\Lambda}^{ru}_{ss'} \Delta \log \mu^{ru}_{ss'} = \sum_{s'} \bar{\lambda}^{ru}_{s'} \Delta \log \bar{\mu}^{ru}_{s'}$, and analogously for the upstream–upstream terms. The seller-level price-index decomposition is therefore

$$\begin{aligned} -\Delta \log P_Y = &- b\, \Delta \log \mu^r - \sum_{s'=1}^{n} \bar{\lambda}^{ru}_{s'} \Delta \log \bar{\mu}^{ru}_{s'} - \sum_{s'=1}^{n} \bar{\lambda}^{uu}_{s'} \Delta \log \bar{\mu}^{uu}_{s'} \\ &+ b \left(\frac{\Delta \log N^r}{\sigma^r - 1} \right) + \tilde{\lambda}^u \left(\frac{\Delta \log N^u}{\sigma^u - 1} \right) \end{aligned} \tag{22}$$

From the price-index decomposition to welfare. Let $\mathcal{E}$ denote nominal household expenditure on the final consumption bundle, measured in units of the numeraire. Welfare is $\mathcal{W} = \mathcal{E}/P_Y$, so that $\Delta \log \mathcal{W} = \Delta \log \mathcal{E} - \Delta \log P_Y$. Combining this identity with (22) gives

$$\begin{aligned} \Delta \log \mathcal{W} = \Delta \log \mathcal{E} - &\ b\, \Delta \log \mu^r - \sum_{s'=1}^{n} \bar{\lambda}^{ru}_{s'} \Delta \log \bar{\mu}^{ru}_{s'} - \sum_{s'=1}^{n} \bar{\lambda}^{uu}_{s'} \Delta \log \bar{\mu}^{uu}_{s'} \\ &+ b \left(\frac{\Delta \log N^r}{\sigma^r - 1} \right) + \tilde{\lambda}^u \left(\frac{\Delta \log N^u}{\sigma^u - 1} \right) \end{aligned} \tag{23}$$

which is the decomposition stated in Proposition 3.

For comparisons between the unsubsidized uniform-pricing and nonlinear-pricing equilibria, there are no taxes or subsidies. With one unit of labor and the wage normalized to one, $\mathcal{E} = 1$ in both equilibria, so $\Delta \log \mathcal{E} = 0$ and the price-index decomposition is also a welfare decomposition. For comparisons involving policy experiments or the decentralized implementation of the efficient allocation, subsidies are financed through the household budget, so final expenditure may differ across equilibria and the expenditure term in (23) must also be included.

Selection. The decomposition above abstracts from changes in the composition of active firms. If selection changes the CES productivity indices $\widetilde{z}^u$ or $\widetilde{z}^r$, additional terms enter the upstream cost recursion and retail price recursion additively. Under Pareto heterogeneity with invariant truncation, these selection terms are unchanged across the equilibria compared here and therefore difference out. They are omitted from (23).

C.11 Entry-Cost Invariance of Welfare Comparisons

This subsection shows that the levels of sectoral entry costs affect equilibrium firm masses and absolute welfare levels, but not welfare comparisons across pricing or policy regimes. The result holds when the same vector of entry-cost primitives is maintained across the equilibria being compared.

Let

$$c^{E\,r} \equiv (c_s^{E\,r})_{s\in\mathcal{S}}, \qquad c^{E\,u} \equiv (c_s^{E\,u})_{s\in\mathcal{S}}, \qquad C_r := \mathrm{Diag}(c^{E\,r}), \qquad C_u := \mathrm{Diag}(c^{E\,u})$$

denote the vectors and diagonal matrices of retail and upstream entry costs. For any pricing or policy regime g, the entry equations can be written as

$$e_r^g = C_r^{-1}\widehat{\Gamma}_r^g L_r^g, \qquad e_u^g = C_u^{-1}\widehat{\mathcal{X}}_u^g L_r^g$$

where $\widehat{\Gamma}_r^g$ and $\widehat{\mathcal{X}}_u^g$ depend on technology, demand elasticities, markups, and the policy regime, but not on the levels of $c^{E\,r}$ or $c^{E\,u}$. Proportional entry subsidies, when present, are included in these regime-specific matrices.

The production-labor allocations satisfy $L_u^g = \mathbf{\Psi}^g L_r^g$ and $L_r^g = \boldsymbol{\omega}^{r,g} L_1^{r,g}$. Labor-market clearing is

$$1 = \mathbf{1}^\top L_r^g + \mathbf{1}^\top L_u^g + (c^{E\,r})^\top e_r^g + (c^{E\,u})^\top e_u^g$$

Because $(c^{E\,r})^\top C_r^{-1} = \mathbf{1}^\top$ and $(c^{E\,u})^\top C_u^{-1} = \mathbf{1}^\top$, substituting the entry equations into labor-market clearing gives

$$L_1^{r,g} = \left\{\left[\mathbf{1}^\top(I + \mathbf{\Psi}^g) + \mathbf{1}^\top\widehat{\Gamma}_r^g + \mathbf{1}^\top\widehat{\mathcal{X}}_u^g\right]\boldsymbol{\omega}^{r,g}\right\}^{-1}$$

which is independent of the entry-cost vectors. It follows that the production labor allocations L_r^g and L_u^g are also independent of entry-cost levels.

Entry costs instead determine the number and average size of firms. Using the stationarity conditions $e_s^{\ell,g} = (1-\delta_s^\ell)N_s^{\ell,g}$, the equilibrium masses can be written as

$$N_s^{r,g} = \frac{\widehat{N}_s^{r,g}}{c_s^{E\,r}}, \qquad N_{s'}^{u,g} = \frac{\widehat{N}_{s'}^{u,g}}{c_{s'}^{E\,u}}$$

where $\widehat{N}^{r,g}$ and $\widehat{N}^{u,g}$ do not depend on entry-cost levels. Consequently, for two regimes g and g' with the same entry-cost vectors,

$$\Delta\log N^r = \log\widehat{N}^{r,g} - \log\widehat{N}^{r,g'}, \qquad \Delta\log N^u = \log\widehat{N}^{u,g} - \log\widehat{N}^{u,g'}$$

where logarithms are taken elementwise. The entry-cost terms therefore cancel from the extensive-margin components of the price-index decomposition in (22). Because the markup terms and the

production-labor allocations are also independent of entry-cost levels, $\Delta \log P_Y$ is independent of $c^{E\,r}$ and $c^{E\,u}$.

The same conclusion applies to equilibrium final expenditure. The retail labor first-order condition implies, for every retail sector s,

$$\mathcal{E}^g = \frac{w L_s^{r,g} \mu_s^{r,g}}{\alpha_s^r b_s}$$

where $\mu_s^{r,g}$ is the effective buyer-facing retail markup. Since $L_s^{r,g}$ and $\mu_s^{r,g}$ do not depend on entry-cost levels, neither does $\mathcal{E}^g$. This includes subsidized equilibria, in which $\mathcal{E}^g$ may differ from one but remains independent of the baseline entry-cost vectors.

To make the effect on welfare levels explicit, define

$$D^r := \operatorname{Diag}\left(\frac{1}{\sigma^r - \mathbf{1}}\right), \qquad D^u := \operatorname{Diag}\left(\frac{1}{\sigma^u - \mathbf{1}}\right)$$

where division is elementwise. The final-good price index in regime g can be written as

$$\log P_Y^g = \widehat{p}_Y^g + b D^r \log c^{E\,r} + \tilde{\lambda}^u D^u \log c^{E\,u}$$

where $\widehat{p}_Y^g$ is independent of entry-cost levels. It follows that

$$\log \mathcal{W}^g = \log \widehat{\mathcal{W}}^g - b D^r \log c^{E\,r} - \tilde{\lambda}^u D^u \log c^{E\,u}, \qquad \log \widehat{\mathcal{W}}^g := \log \mathcal{E}^g - \widehat{p}_Y^g$$

and therefore, for any two regimes g and g',

$$\frac{\mathcal{W}^g}{\mathcal{W}^{g'}} = \frac{\widehat{\mathcal{W}}^g}{\widehat{\mathcal{W}}^{g'}}$$

Welfare ratios, log-welfare differences, and percentage welfare gains across regimes are thus invariant to the levels of the entry-cost primitives.

Entry costs nevertheless affect absolute welfare levels: changing entry costs multiplies welfare in every regime by the same common factor $\exp\{-b D^r \log c^{E\,r} - \tilde{\lambda}^u D^u \log c^{E\,u}\}$. Raw welfare differences in levels are correspondingly rescaled, even though proportional comparisons are unchanged.

Finally, suppose welfare is evaluated as a function of the policy vector $\tilde{\boldsymbol{\tau}}$. Since the entry-cost component of $\log \mathcal{W}(\tilde{\boldsymbol{\tau}}; c^{E\,r}, c^{E\,u})$ is constant with respect to policy, provided that entry costs do not alter the admissible policy set,

$$\arg\max_{\tilde{\boldsymbol{\tau}}} \mathcal{W}(\tilde{\boldsymbol{\tau}}; c^{E\,r}, c^{E\,u}) = \arg\max_{\tilde{\boldsymbol{\tau}}} \widehat{\mathcal{W}}(\tilde{\boldsymbol{\tau}})$$

The calibration of entry-cost levels therefore matters for firm counts, average firm size, and absolute welfare levels, but not for welfare comparisons across regimes or for the optimal policy.

Survival probabilities enter the equilibrium only through the effective entry cost $c_s^{E\,\ell}(1-\delta_s^\ell)$: retail average labor $\bar{l}_s^r$ and upstream entry flows scale with it, and firm masses equal $N_s^\ell = e_s^\ell/(1-\delta_s^\ell)$. Replacing c^E by $c^E(1-\delta)$ throughout, the same argument shows that welfare comparisons and the optimal policy do not depend on δ.

D Parameter Calibration and Estimation

Each parameter is measured according to the following rules. Parameters that describe technology and market power, which the model treats as stable, pool all available years, and we use medians for profit shares so that the estimates are robust to unusual years. Parameters that describe the cross-sectional structure of the supply chain are measured in a single benchmark year, 2018, the first year in which invoices cover all firms and the last before the aggregate shocks of 2019–2023. The quantity discounts we compare the model with come from 2024, the latest year available, so the comparison is also out of sample in time.[48]

A firm is upstream if at least half of its domestic sales are invoiced to other domestic firms, and a retailer otherwise. We apply this rule once, to 2018 sales, when sales to other firms are fully observed, and hold each firm's layer fixed in every year we use.[49] The model treats the layer as a permanent attribute, and a fixed assignment ensures that the same firms define each layer for every parameter, so that no estimate is driven by firms switching layers. For most activity, the assignment does not depend on the cutoff: firms that sell at least 90% of their sales exclusively to other firms or exclusively to final consumers account for 73% of sales (Appendix B.7). Table A7 summarizes the sources and periods.

[48] Estimating the quantity-discount schedule on 2018 invoices yields a very similar schedule.

[49] Accordingly, in every year we use, we restrict attention to firms active in 2018. The calibration describes the 2018 economy, so its parameters are estimated on the firms that compose it. In the model, exit is independent of productivity, so firms active in 2018 that are observed in other years are a random sample of the stationary distribution. Selection on survival arises only to the extent that exit in the data depends on profitability. It is weakest in the years around 2018, and aggregating profit shares by the median across years limits the influence of the years furthest from 2018, where it is strongest.

Table A7: Calibration Summary

Parameter	Source	Period	Moment
Stable structural parameters			
Elasticities of substitution $\sigma_{s'}^{u}, \sigma_{s}^{r}$	Balance sheets	2005–2023	Median profit shares, net of flat fees
Primary-input elasticities α_s^ℓ	Balance sheets, invoice deflators	2014–2023	Production functions
Supply-chain structure (benchmark year)			
Input–output elasticities $\theta_{ss'}^{\ell}$	Invoices	2018	Payment shares, net of flat fees
Final-demand elasticities θ_s	Balance sheets, invoices	2018-2023	Retailers' domestic sales to final consumers
Screening elasticities ξ_s^ℓ	Balance sheets	2018	Employment concentration
Normalization			
Effective entry costs $c^E(1-\delta)$	–	–	Normalized to one
Untargeted comparison			
Quantity discounts $d_{s'}$	Invoices	2024	Largest decline in unit-price schedules

D.1 Primary-Input Output Elasticity α

In the Chilean IRS firm balance-sheet data, we observe labor (L), measured in worker-months, expenditures on intermediate inputs (E^M), and the capital stock (K). Since the model abstracts from capital as a separate factor, we interpret its labor input in the quantitative analysis as a composite primary input comprising labor and capital.

We estimate output elasticities using a standard production function methodology, following Ackerberg et al. (2015). For each sector-layer pair, we estimate a Cobb–Douglas production function in labor, capital, and intermediate inputs

$$q_{it} = \omega_{it} + \beta_l \, l_{it} + \beta_k \, k_{it} + \beta_m \, m_{it} + \varepsilon_{it}$$

where lowercase letters denote logs, ω_{it} is firm i's productivity, and ε_{it} is measurement error or an unanticipated shock to output. We deflate sales and intermediate-input expenditures with firm-level price indices built from invoice-level prices and quantities, so that q_{it} and m_{it} measure quantities rather than revenue. This addresses a well-known limitation of production function estimation: output and inputs are usually observed only as revenue and expenditure, so that output elasticities are confounded with prices and markups (Klette and Griliches, 1996; De Loecker et al., 2016; Bond et al., 2021). We therefore apply a standard estimator to data that observe quantities directly, on both the output and the input side, for the firms whose transactions are invoiced.[50]

[50]Because the deflators remove price changes over time but not price-level differences across firms, large buyers, who pay lower average prices because of quantity discounts, have their materials quantities understated, so measured materials grow less than one-for-one with true materials. This mainly biases the materials coefficient. Output deflators likewise retain cross-firm price levels, which could bias all coefficients if output prices vary with scale; the unconstrained elasticities sum to close to one (Table A8), as expected if measured output tracks quantities rather than revenue. This motivates constructing α_s^ℓ from $\hat{\beta}_l + \hat{\beta}_k$ rather than from $1 - \hat{\beta}_m$, and we show that our results are robust to raising all α_s^ℓ by 10% and 20% (Appendix E).

Because the model has constant returns to scale in the primary-input composite and materials, we set the primary-input elasticity to $\alpha_s^\ell = \hat{\beta}_l + \hat{\beta}_k$ and impose constant returns by setting the materials elasticity to $1 - \alpha_s^\ell$. The data support this restriction: without imposing it, the estimated elasticities sum to 0.99 for retailers and 1.00 for upstream firms on average, and to within 10% of one in 20 of the 22 sector-layer pairs (Table A8).

Table A8 reports the estimates, which are held fixed across all pricing regimes and counterfactuals. The estimation sample covers 2014–2023. It starts in 2014, the first year in which invoice-level prices and quantities are available to construct firm-level deflators. Firm-level price indices are available only for firms that issue or receive invoices with prices and quantities, which excludes sales to final consumers recorded without invoices and most small firms before electronic invoicing became universal in 2018. Firm-years without an output or input price index are excluded from the estimation.

Table A8: Primary-Input Elasticities and Returns to Scale by Sector and Firm Layer

	$\alpha_s^\ell = \hat{\beta}_l + \hat{\beta}_k$		$\hat{\beta}_l + \hat{\beta}_k + \hat{\beta}_m$	
Sector	Retailers	Upstream	Retailers	Upstream
Agriculture	0.28	0.32	0.97	0.96
Mining	0.22	0.46	1.00	1.15
Manufacturing	0.31	0.36	1.02	1.00
Utilities	0.51	0.35	1.05	0.98
Construction	0.51	0.45	0.99	1.02
Retail and wholesale	0.29	0.32	1.03	1.03
Transport and ICT	0.51	0.46	1.01	1.02
Financial services	0.52	0.51	1.00	0.99
Real estate services	0.49	0.44	0.88	0.94
Business services	0.59	0.48	1.00	0.97
Personal services	0.41	0.37	0.93	0.94
Sector mean	0.42	0.41	0.99	1.00

Notes: $\hat{\beta}_l$, $\hat{\beta}_k$, and $\hat{\beta}_m$ are the estimated output elasticities from a Cobb–Douglas production function in labor, capital, and intermediate inputs, estimated separately for each sector-layer pair following Ackerberg et al. (2015), without imposing constant returns to scale. The last two columns report their sum. Sector means are unweighted means across the 11 sectors.

D.2 Productivity Dispersion

We recover the Pareto exponent of the productivity distribution, κ_s^ℓ, from the Pareto exponent of the employment distribution, ξ_s^ℓ, which we estimate from the upper end of the firm employment distribution. Heavy-tailed firm-size distributions are well documented (Axtell, 2001; Gabaix, 2009; Luttmer, 2007; Rossi-Hansberg and Wright, 2007; Carvalho and Grassi, 2019). We estimate ξ_s^ℓ directly from employment concentration among the largest firms, which avoids rank regressions and extrapolating the Pareto approximation to small firms (Gabaix and Ibragimov, 2011).

We use the 2018 cross-section of firms with positive employment, measured in worker-years, separately for each of the eleven sectors and the retail and upstream layers. We use 2018, the year of the input–output shares, so that the distribution of buyers matches the supply-chain data used in the correction fixed point. Suppose that, in the upper tail, employment in sector s and layer ℓ satisfies $\Pr(L > x) = (L_{\min}/x)^{\xi_s^\ell}$. Then the share of total employment held by the largest fraction p of firms is p^{1-1/ξ_s^ℓ}. Let $E_{q,s}^\ell$ denote total employment among firms above the qth employment percentile within cell (s, ℓ). The ratio of employment held by the top 1 percent to that held by the top $(100 - q)$ percent satisfies

$$\frac{E_{99,s}^\ell}{E_{q,s}^\ell} = \left(\frac{0.01}{1 - q/100}\right)^{1-1/\xi_s^\ell} \qquad \Longrightarrow \qquad \widehat{\xi}_{q,s}^\ell = \left[1 - \frac{\log\left(E_{99,s}^\ell / E_{q,s}^\ell\right)}{\log\left(0.01/(1 - q/100)\right)}\right]^{-1}$$

Our baseline uses $q = 95$, comparing employment held by the largest 1 percent of firms with that held by the largest 5 percent. This moment is scale invariant and uses only the part of the distribution that identifies ξ_s^ℓ. As robustness, we keep the top 1 percent fixed and lower the threshold to the 90th and 75th percentiles. Table A9 shows that the estimates are generally stable, particularly in the larger sector-layer cells. We use $\widehat{\xi}_{95,s}^\ell$ throughout. The estimates matter most where the welfare exposure weights are largest (Section 4.7), and these are also the best-estimated cells. Retail and wholesale, financial services, and transport and ICT account for 68% of final demand, and retail and wholesale, manufacturing, and transport and ICT for 68% of the upstream weight. Each of these cells has at least 177 firms in the top 5%, and moving the lower threshold from the 95th to the 75th percentile changes the estimate by at most 0.14. The noisiest cells, those with fewer than 50 firms in the top 5% (mining and utilities retailers; mining, utilities and real-estate upstream), carry about 10% of the final-demand weight and under 5% of the upstream weight; Appendix E shows that recalibrating with the alternative thresholds changes the results little. All estimates exceed one, as required for employment to have a finite mean.

Table A9: Employment Pareto Exponents: Baseline and Threshold Robustness

	Retailers				Upstream			
Sector	n_{95}	$\widehat{\xi}_{95}$	$\widehat{\xi}_{90}$	$\widehat{\xi}_{75}$	n_{95}	$\widehat{\xi}_{95}$	$\widehat{\xi}_{90}$	$\widehat{\xi}_{75}$
Agriculture	159	1.81	1.67	1.53	519	1.76	1.68	1.59
Mining	5	1.37	1.40	1.41	40	1.47	1.44	1.39
Manufacturing	318	1.34	1.28	1.23	673	1.56	1.50	1.42
Utilities	20	2.12	1.77	1.50	29	1.76	1.65	1.49
Construction	126	1.27	1.24	1.22	773	1.52	1.45	1.38
Retail and wholesale	2016	1.19	1.19	1.18	960	1.26	1.25	1.24
Transport and ICT	234	1.26	1.22	1.19	687	1.52	1.46	1.39
Financial services	177	1.14	1.12	1.10	56	1.65	1.57	1.47
Real estate services	118	1.60	1.57	1.51	48	1.81	1.69	1.55
Business services	506	1.44	1.39	1.34	518	1.44	1.38	1.31
Personal services	600	1.28	1.25	1.21	76	1.30	1.28	1.26

Notes: n_{95} is the number of firms above the 95th employment percentile within each sector-layer cell. $\widehat{\xi}_q$ compares total employment above the 99th percentile with total employment above the qth percentile; $\widehat{\xi}_{95}$ is the baseline used in the calibration.

In the model, labor demand satisfies $l(z) \propto z^{\sigma_s^\ell - 1}$ (Section 4.3.2), and a power of a Pareto random variable is Pareto, so $\xi_s^\ell = \kappa_s^\ell/(\sigma_s^\ell - 1)$ and

$$\widehat{\kappa}_s^\ell = (\sigma_s^\ell - 1)\,\widehat{\xi}_{95,s}^\ell$$

where σ_s^ℓ are the corrected elasticities in Table A14. Table A10 reports the implied values.

Table A10: Implied Productivity Tail Exponents

	Retailers			Upstream		
Sector	σ_s^r	$\widehat{\xi}^r_{95,s}$	$\widehat{\kappa}^r_s$	σ_s^u	$\widehat{\xi}^u_{95,s}$	$\widehat{\kappa}^u_s$
Agriculture	4.03	1.81	5.48	3.65	1.76	4.66
Mining	3.15	1.37	2.96	3.22	1.47	3.27
Manufacturing	4.20	1.34	4.29	3.28	1.56	3.55
Utilities	3.12	2.12	4.50	2.99	1.76	3.50
Construction	2.82	1.27	2.30	3.00	1.52	3.04
Retail and wholesale	4.19	1.19	3.79	3.88	1.26	3.62
Transport and ICT	2.50	1.26	1.89	2.57	1.52	2.39
Financial services	2.51	1.14	1.73	2.23	1.65	2.03
Real estate services	2.34	1.60	2.14	2.58	1.81	2.85
Business services	2.29	1.44	1.86	2.66	1.44	2.39
Personal services	2.46	1.28	1.86	2.67	1.30	2.17

Notes: $\widehat{\kappa}^\ell_s = (\sigma^\ell_s - 1)\,\widehat{\xi}^\ell_{95,s}$, using the corrected elasticities in Table A14 and the baseline employment tails in Table A9. All estimates are computed within each sector-layer cell, without imputation or pooling across sectors.

D.3 Input–Output Expenditure Shares

For firms in buyer sector s and layer $\ell \in \{r, u\}$, the parameters $\theta^\ell_{ss'}$ are the Cobb–Douglas elasticities of the materials aggregator with respect to inputs from upstream sector s'. Under cost minimization, they equal expenditure shares across supplier sectors when inputs are valued at marginal prices. Under nonlinear pricing, however, a payment from buyer i to supplier j is $T_{ij} = p^\ell_{js}\, m_{ij} + F^\ell_{js}$, so observed payment shares

$$\widehat{\theta}^\ell_{ss'} = \frac{\sum_{i \in \mathcal{F}_{\ell s}} \sum_{j \in \mathcal{U}_{s'}} T_{ij}}{\sum_{i \in \mathcal{F}_{\ell s}} \sum_{k \in \mathcal{S}} \sum_{j \in \mathcal{U}_k} T_{ij}}$$

include flat fees and differ from the shares required by the model. Under nonlinear pricing, we net out the model-implied flat fees jointly with the elasticities of substitution, as described in Appendix D.5.1. Under uniform pricing calibration used in Appendix E.4, the raw shares are used directly.

We construct $\widehat{\theta}^\ell_{ss'}$ from Chilean firm-to-firm transactions in 2018 by aggregating payments by buyer sector, buyer layer, and supplier sector, and dividing by total intermediate payments of buyers in the corresponding sector and layer. The public sector is excluded from both buyers and suppliers, non-positive bilateral flows are dropped, and transaction values are winsorized at the 99.99th percentile before aggregation.

We use 2018 for two reasons. First, it is the first year in which firm-to-firm transactions are covered universally: electronic invoicing was phased in by firm size from November 2014 and became mandatory for all firms in February 2018.[51] Earlier years under-represent small firms, which would bias input–output shares toward the purchasing patterns of large buyers. Second, 2018 is the last year before a sequence of large aggregate shocks, the social unrest of late 2019, the COVID-19 pandemic, and high inflation in 2021–2023, and is therefore the year closest to the steady state the model describes.

Tables A11 and A12 report the raw shares.

Table A11: Observed Input–Output Payment Shares $\widehat{\theta}^{u}_{ss'}$: Upstream Buyers

Buyer sector s \ Supplier sector s'	1	2	3	4	5	6	7	8	9	10	11
1	0.2570	0.0034	0.1178	0.0194	0.0420	0.3837	0.0730	0.0609	0.0025	0.0369	0.0035
2	0.0080	0.2227	0.0975	0.0471	0.0900	0.2962	0.1021	0.0322	0.0023	0.0997	0.0021
3	0.0778	0.0100	0.2876	0.0250	0.0343	0.3254	0.1474	0.0268	0.0067	0.0551	0.0038
4	0.0074	0.0175	0.0703	0.4546	0.1291	0.1344	0.0838	0.0229	0.0013	0.0759	0.0029
5	0.0069	0.0081	0.1340	0.0081	0.2583	0.4254	0.0459	0.0247	0.0051	0.0801	0.0034
6	0.0360	0.0076	0.2030	0.0045	0.0193	0.4756	0.1679	0.0220	0.0102	0.0487	0.0052
7	0.0107	0.0026	0.0514	0.0120	0.0583	0.3385	0.3650	0.0550	0.0112	0.0858	0.0094
8	0.0074	0.0030	0.0235	0.0019	0.0388	0.2202	0.0712	0.5714	0.0100	0.0456	0.0068
9	0.0210	0.0010	0.0640	0.0240	0.2975	0.1113	0.0457	0.2235	0.0950	0.1088	0.0081
10	0.0117	0.0082	0.0860	0.0069	0.0812	0.3787	0.1630	0.0961	0.0115	0.1460	0.0108
11	0.0054	0.0004	0.0847	0.0122	0.0528	0.5291	0.0722	0.0392	0.0412	0.0920	0.0709

Notes: Each row reports observed payment shares $\widehat{\theta}^{u}_{ss'}$ across supplier sectors s' for upstream buyers in sector s. Sector codes: 1 agriculture; 2 mining; 3 manufacturing; 4 utilities; 5 construction; 6 retail and wholesale; 7 transport and ICT; 8 financial services; 9 real estate services; 10 business services; 11 personal services.

[51] Law 20.727 of 2014; Servicio de Impuestos Internos, announcement of February 1, 2018, https://www.sii.cl/noticias/2018/010218noti01er.htm.

Table A12: Observed Input–Output Payment Shares $\widehat{\theta}^{r}_{ss'}$: Retail Buyers

Buyer sector s \ Supplier sector s'	1	2	3	4	5	6	7	8	9	10	11
1	0.2520	0.0017	0.2111	0.0209	0.0318	0.3146	0.0520	0.0731	0.0016	0.0385	0.0028
2	0.0017	0.0406	0.1876	0.0634	0.1477	0.2978	0.0663	0.0167	0.0011	0.1728	0.0045
3	0.1297	0.0185	0.3487	0.0227	0.0347	0.2446	0.1086	0.0275	0.0071	0.0548	0.0030
4	0.0041	0.0205	0.0653	0.3383	0.2203	0.1047	0.0942	0.0326	0.0009	0.1153	0.0037
5	0.0068	0.0063	0.0926	0.0153	0.4939	0.1853	0.0412	0.0834	0.0055	0.0666	0.0032
6	0.0631	0.0048	0.2839	0.0056	0.0185	0.4710	0.0741	0.0158	0.0268	0.0340	0.0026
7	0.0034	0.0011	0.0405	0.0423	0.0672	0.3327	0.3123	0.0382	0.0208	0.1335	0.0080
8	0.0040	0.0011	0.0281	0.0021	0.0564	0.1945	0.0789	0.4535	0.0185	0.0664	0.0965
9	0.0068	0.0005	0.0414	0.0271	0.4245	0.1203	0.0440	0.2285	0.0184	0.0811	0.0073
10	0.0105	0.0004	0.0831	0.0078	0.0543	0.3444	0.1289	0.2070	0.0162	0.1253	0.0223
11	0.0057	0.0002	0.0753	0.0166	0.1055	0.4022	0.0917	0.0851	0.0531	0.1161	0.0486

Notes: Each row reports observed payment shares $\widehat{\theta}^{r}_{ss'}$ across supplier sectors s' for retail buyers in sector s. Sector codes as in Table A11.

D.4 Final-Demand Elasticities

We calibrate the Cobb–Douglas elasticities θ_s of the final-good aggregator from retailers' sales to final users. Because retailers sell to final consumers at uniform prices, sectoral expenditure shares identify θ_s directly. Two adjustments bring observed revenue closer to the model's object. First, since the model is a closed economy, we exclude export sales. Second, since retailers in the data may also sell part of their output to other firms, we weight each retailer's domestic revenue by $1 - \varsigma_i$, where ς_i is the share of its sales that go to other firms. For each year t, we compute

$$\widehat{\theta}_{st} = \frac{\sum_{i \in \mathcal{R}_s} \left(R_{it} - R^x_{it}\right)(1 - \varsigma_i)}{\sum_{s' \in \mathcal{S}} \sum_{i \in \mathcal{R}_{s'}} \left(R_{it} - R^x_{it}\right)(1 - \varsigma_i)}$$

where R^x_{it} denotes export sales, using the full population of retailers outside the public sector, and average $\widehat{\theta}_{st}$ over 2018–2023. Since nonlinear pricing in upstream transactions does not affect this mapping, we use the same θ_s under both the nonlinear- and uniform-pricing calibrations (Appendix E.4).

As a robustness check, we also use household final consumption shares by sector from the 2018 input–output matrix of the national accounts (Banco Central de Chile, 2023). The two measures differ in some sectors partly because the national accounts classify consumption by product rather than by the sector of the selling firm. In particular, financial services account for a smaller share in the national accounts, consistent with part of financial-sector revenue (such as interest) being outside the VAT invoices from which we infer sales to final users. Using the national-accounts shares barely changes welfare and policy results (Table A16).

Table A13: Final-Demand Elasticities by Retail Sector

Sector	Baseline	National accounts
Agriculture	0.0082	0.0228
Mining	0.0479	0.0003
Manufacturing	0.0689	0.1679
Utilities	0.0562	0.0306
Construction	0.0183	0.0001
Retail and wholesale	0.3178	0.2684
Transport and ICT	0.1236	0.1154
Financial services	0.2360	0.0827
Real estate services	0.0101	0.1461
Business services	0.0455	0.0120
Personal services	0.0674	0.1538

Notes: Baseline: θ_s is set equal to retail sector s's share of retailers' domestic sales to final users. Each retailer's revenue net of exports is weighted by the share of its sales that do not go to other firms. Shares are computed each year over the full population of retailers and averaged over 2018–2023. National accounts: household final consumption shares by sector from the 2018 input–output matrix (12 sectors) of the Central Bank of Chile, excluding public administration and renormalized to sum to one; used in the robustness exercise in Table A16.

D.5 Elasticities of Substitution

Raw moments. Upstream and retail elasticities are recovered with the same method. For each firm, we measure operating profit as sales revenue net of labor expenditures, the user cost of capital, and observed intermediate-input payments

$$\widetilde{\Pi}_{j,t} = R_{j,t} - w_{j,t}L_{j,t} - r_t K_{j,t} - X^M_{j,t}$$

Within each sector, layer, and year, we take the median operating-profit share $\widetilde{\Pi}_{j,t}/R_{j,t}$, invert it, and take the median across years. This gives raw estimates $\hat{\sigma}^u_s$ for upstream sellers and $\hat{\sigma}^r_s$ for retailers. We keep firms with positive sales, labor expenditures, materials, and capital, winsorize each at the 1st and 99th percentiles within sector and year.

Under nonlinear pricing, observed operating profits are net of the flat fees each firm pays to its suppliers, and observed payments include them. By Result 4, a firm in (ℓ, s) with revenue R has

$$\frac{\widetilde{\Pi}}{R} = \frac{1}{\sigma^\ell_s} - \frac{\sigma^\ell_s - 1}{\sigma^\ell_s}(1 - \alpha^\ell_s)\,\mathcal{H}^\ell_s\,\frac{R^\ell_s}{R}, \qquad \mathcal{H}^\ell_s = \sum_{s' \in \mathcal{S}} \frac{\theta^\ell_{ss'}}{\sigma^u_{s'} - 1}$$

so the raw estimate overstates σ_s^ℓ. Similarly, payments from buyers in (ℓ, s) to sector s' equal variable expenditure times $1 + \frac{1}{\sigma_{s'}^u - 1} \underline{M}_{ss'}^\ell / M_{is'}$, so raw payment shares overweight supplier sectors with low elasticities. Appendix D.5.1 describes the fixed point we use to correct both, and reports the resulting elasticities.

Uniform-pricing lens. For the exercise in Appendix E.4, we recalibrate the model under uniform pricing. In that case, observed payments are variable expenditures and the CES markup implies $\widetilde{\Pi}_{j,t}/R_{j,t} = 1/\sigma_s^\ell$, so the uniform-pricing calibration uses $\hat{\sigma}_s^u$, $\hat{\sigma}_s^r$, and $\widehat{\theta}_{ss'}^\ell$ directly.

D.5.1 Correction Fixed Point

We correct the raw estimates of θ and σ for flat fees as follows. The two corrections use different scale ratios because they correct different moments. Profit shares are medians across firms, so we evaluate the profit correction at the median firm, for which the Pareto distribution of buyer scale implies $\underline{R}_s^\ell / R = 2^{-1/\xi_s^\ell}$. Payment shares aggregate payments across buyers, so the fee correction depends on minimum relative to mean demand, $\underline{M}_{ss'}^\ell / \mathbb{E}[M_{is'}] = 1 - 1/\xi_s^\ell$, which implies a ratio of fees to variable payments of $(1 - 1/\xi_s^\ell)/(\sigma_{s'}^u - 1)$ (Appendix C.7). Starting from the raw moments, we iterate on

$$\frac{1}{\sigma_s^\ell} = \frac{1}{\hat{\sigma}_s^\ell} + \frac{\sigma_s^\ell - 1}{\sigma_s^\ell}(1 - \alpha_s^\ell)\,\mathcal{H}_s^\ell\, 2^{-1/\xi_s^\ell}, \qquad \theta_{ss'}^\ell \propto \frac{\widehat{\theta}_{ss'}^\ell}{1 + (1 - 1/\xi_s^\ell)/(\sigma_{s'}^u - 1)},$$

renormalizing the input shares to sum to one within each (ℓ, s), until the elasticities and input shares converge. The labor elasticities α_s^ℓ and the screening elasticities are held fixed. At convergence, the largest change in an input share relative to its raw value is 0.028 for upstream buyers and 0.018 for retail buyers.

Table A14 reports the raw and corrected elasticities. The raw moments use 2005-2023. Netting out fees lowers the average upstream elasticity from 4.86 to 2.98 and the average retail elasticity from 5.60 to 3.06. The correction is largest in agriculture, manufacturing, and retail and wholesale. The table also reports an alternative calibration implied by the observed quantity discount, $1/\widehat{d}_s$, estimated from firm-to-firm invoices for the five sectors with decreasing and convex price schedules. These values are not used in the baseline calibration; Appendix E uses them as a robustness check.

Table A14: Elasticities of Substitution

	Upstream σ_s^u			Retail σ_s^r	
Sector	Raw	Corrected	Discount-implied	Raw	Corrected
Agriculture	7.61	3.65		11.63	4.03
Mining	4.98	3.22		6.02	3.15
Manufacturing	5.65	3.28	4.15	10.36	4.20
Utilities	5.27	2.99		5.04	3.12
Construction	4.33	3.00		3.80	2.82
Retail and wholesale	7.76	3.88	2.93	9.04	4.19
Transport and ICT	3.56	2.57	6.49	3.20	2.50
Financial services	3.01	2.23		3.36	2.51
Real estate services	3.94	2.58		3.13	2.34
Business services	3.62	2.66	3.14	2.77	2.29
Personal services	3.71	2.67	5.18	3.26	2.46
Sector mean	4.86	2.98		5.60	3.06

Notes: Raw elasticities are the inverse of the median operating-profit share, aggregated across years by the median. Corrected elasticities net out the model-implied flat fees through the fixed point in Appendix D.5.1 and are used under the nonlinear-pricing lens; raw elasticities are used under the uniform-pricing lens. The discount-implied elasticity is $1/\widehat{d_s}$, where $\widehat{d_s}$ is the largest estimated proportional decline in the sector's unit-price schedule relative to the smallest-quantity bin. It is reported for sectors with decreasing and convex schedules and, as a seller-side measure, shown next to the upstream elasticities. Sector means are unweighted.

E Quantification Material

E.1 Sectoral Decomposition of Welfare

Table A15 breaks down the price-index terms of the decomposition in Table 3, Panel (b), by sector. Panel (a) reports the change from nonlinear pricing to uniform pricing, and Panel (b) the change from nonlinear pricing to the efficient allocation. Markup terms are assigned to the sector that charges them, and variety terms to the sector whose firms enter. The change in final expenditure is not sector-specific and is excluded. It is zero in Panel (a) and −80.9 log points in Panel (b).

Uniform pricing. A ban on price discrimination lowers the price-index component of welfare by 11.8 log points. The loss comes from higher markups on firm-to-firm sales, mostly upstream-to-upstream (−12.3), and is partly offset by higher retail entry (+6.4), which rises in every sector. Three sectors account for 74% of the total loss: transport and ICT (−3.8), retail and whole-

sale (−2.5), and financial services (−2.3). Retail and wholesale has the largest markup loss, since markups on its sales enter the costs of almost every other sector. In transport and ICT and in financial services, the ban also lowers upstream entry (−1.3 and −0.8). These sectors have the lowest elasticities of substitution, so fees are a large share of sellers' margins, and the value of each variety is high. In four small sectors (agriculture, mining, utilities, and personal services), higher retail entry more than offsets the markup losses.

Efficient allocation. Moving to the efficient allocation raises the price-index component by 98.5 log points. The gains are concentrated. Retail and wholesale alone accounts for 23.6 log points, 24% of the total, and together with financial services, transport and ICT, and manufacturing, it accounts for nearly three quarters. Its upstream-to-upstream term (7.7) is a third of the economy-wide total. The extensive margin follows a different pattern. Upstream variety gains are largest in transport and ICT (6.3), retail and wholesale (6.1), financial services (5.5), and manufacturing (4.1), which together account for three quarters of the total. As in Panel (a), low elasticities in transport and ICT and in financial services make each additional variety more valuable, so the under-provision of upstream firms is most costly there. Retail variety falls in every sector except agriculture, most in financial services (−3.8) and retail and wholesale (−3.7). Mining and personal services are the only sectors where the extensive margin is negative overall.

Table A15: Welfare Decomposition by Sector

	Markups			Variety		
Sector	Retail	Retail-to-up.	Up-to-up.	Retail	Upstream	Total
Panel (a). Uniform vs. nonlinear pricing						
Agriculture	0.0	−0.2	−0.5	0.2	0.6	0.0
Mining	0.0	−0.1	−0.1	0.5	0.0	0.4
Manufacturing	0.0	−0.8	−1.8	0.6	1.1	−0.9
Utilities	0.0	−0.3	−0.4	0.9	0.1	0.3
Construction	0.0	−0.5	−0.8	0.1	−0.2	−1.3
Retail and wholesale	0.0	−1.3	−3.5	1.6	0.6	−2.5
Transport and ICT	0.0	−0.7	−2.5	0.7	−1.3	−3.8
Financial services	0.0	−0.8	−1.7	0.9	−0.8	−2.3
Real estate services	0.0	−0.1	−0.2	0.1	0.1	−0.1
Business services	0.0	−0.6	−1.0	0.3	−0.5	−1.7
Personal services	0.0	−0.1	−0.1	0.4	0.0	0.2
Total	0.0	−5.5	−12.3	6.4	−0.3	−11.8
Panel (b). Efficient allocation vs. nonlinear pricing						
Agriculture	0.2	0.6	0.9	0.0	1.2	2.8
Mining	1.8	0.1	0.2	−0.4	0.2	1.9
Manufacturing	1.9	2.9	3.9	−0.5	4.1	12.1
Utilities	2.2	0.4	0.5	−0.3	0.8	3.6
Construction	0.8	1.1	1.2	−0.3	1.5	4.4
Retail and wholesale	8.7	4.8	7.7	−3.7	6.1	23.6
Transport and ICT	6.3	2.2	4.8	−1.8	6.3	17.9
Financial services	12.0	3.2	2.2	−3.8	5.5	19.1
Real estate services	0.6	0.5	0.3	−0.1	0.6	1.8
Business services	2.6	1.5	1.7	−0.5	2.2	7.5
Personal services	3.5	0.6	0.2	−0.9	0.4	3.8
Total	40.6	17.9	23.6	−12.2	28.7	98.5

Notes: Log points (×100). Each panel decomposes the change from the nonlinear-pricing status quo to the counterfactual, using Equation (16). Markup terms are assigned to the sector that charges them and variety terms to the sector of the entering firms. The change in final expenditure (0.0 in Panel (a), −80.9 in Panel (b)) is excluded, so totals equal the price-index component of the decomposition. Entries may not add up because of rounding.

E.2 Robustness of the Welfare Results

Table A16 reports how welfare and optimal policy change under alternative calibrations. We consider six exercises. First, as an alternative calibration, we replace $\sigma^u_{s'}$ with the value implied by the observed quantity discount, $1/d_{s'}$, in the five sectors with decreasing and convex price schedules. Second, we scale the labor elasticity α^ℓ_s of all firms by 1.1 and 1.2. This reduces the weight of intermediate inputs and hence the scope for markups to accumulate along the supply chain. Third, we scale all elasticities of substitution, $\sigma^u_{s'}$ and σ^r_s, by 1.1 and 1.2. The level of market power is hard to measure, and these scalings provide robustness to lower aggregate markups. Fourth, we re-estimate the screening elasticities from broader parts of the employment distribution, comparing the top 1% with the top 10% and 25% of firms, and recalibrate the full fee-correction fixed point.[52] Fifth, we replace the final-demand shares θ_s with household final consumption shares from the national accounts. Sixth, we set a common screening elasticity, $\xi = 1.33$, across all sectors and layers, holding the elasticities of substitution fixed. Allocative markups then no longer vary across buyer sectors for a given seller, which removes the buyer-sector dimension of marginal prices while keeping buyer-sector-specific flat fees.

Welfare levels move in the expected direction. Higher elasticities and a higher labor elasticity lower the welfare cost of market power under both regimes: welfare under uniform pricing rises from 74.5% of the efficient level to as much as 83.0%, and the log gain from nonlinear pricing falls from 11.8 to between 7.2 and 9.4 log points. The other exercises change welfare levels little. The share of the gap between uniform pricing and the efficient allocation closed by nonlinear pricing ranges from 35.2% to 42.9%; it is lowest with screening elasticities estimated from the broadest part of the distribution, which are closer to one and hence to uniform pricing. With a common screening elasticity, welfare without policy (84.5%) and the gap closed (42.5%) are close to the baseline, so heterogeneity in allocative markups across buyer sectors matters little for our results. The policy results are also robust. In every exercise, the optimal revenue subsidy under nonlinear pricing (23–30%) is well below its uniform-pricing counterpart (34–43%), the optimal entry subsidy is much larger (15–18% against 1–3%), and applying the uniform-pricing optimum to the nonlinear-pricing economy costs between 2.7 and 5.7 log points of welfare.

[52] When an alternative calibration implies a hold-up index above one in some sector-layer, so that lowest-type profits would be negative, we set the corresponding elasticity at the largest value consistent with the participation condition (Appendix C.7). This binds for the discount-implied σ^u in transport and ICT and in personal services, and for the retail elasticity of manufacturing in the recalibrations of ξ, where it changes the elasticity by less than 0.06.

Table A16: Robustness of Welfare and Policy Results

	Welfare (% of efficient)			Optimal (τ^y, τ^e)		Cost of uniform
Scenario	Uniform	Nonlinear	Gap closed	Nonlinear	Uniform	prescription
Baseline	74.5	83.9	40.1	(0.29, 0.17)	(0.43, 0.02)	5.0
Discount-implied σ^u, convex sectors	76.1	84.6	38.8	(0.28, 0.18)	(0.42, 0.03)	4.6
$\alpha \times 1.1$	79.1	86.9	40.1	(0.29, 0.16)	(0.42, 0.02)	3.6
$\alpha \times 1.2$	82.9	89.4	40.0	(0.29, 0.15)	(0.42, 0.02)	2.7
$(\sigma^u, \sigma^r) \times 1.1$	79.4	87.0	39.3	(0.26, 0.16)	(0.38, 0.01)	3.9
$(\sigma^u, \sigma^r) \times 1.2$	83.0	89.2	38.7	(0.23, 0.16)	(0.34, 0.01)	2.9
ξ: top 1% vs. top 10%	75.0	83.6	38.0	(0.30, 0.16)	(0.42, 0.02)	4.2
ξ: top 1% vs. top 25%	75.5	83.3	35.2	(0.30, 0.15)	(0.42, 0.02)	3.3
Final demand: national accounts	74.3	84.4	42.9	(0.28, 0.18)	(0.43, 0.02)	5.7
Common $\xi = 1.33$	74.5	84.5	42.5	(0.29, 0.16)	(0.43, 0.02)	4.5

Notes: Gap closed is the share (%) of the log welfare gap between uniform pricing and the efficient allocation closed by nonlinear pricing. Optimal (τ^y, τ^e) are the welfare-maximizing uniform revenue and entry subsidies under each pricing regime. Cost of uniform prescription is the log-point welfare loss from applying the uniform-pricing optimum to the nonlinear-pricing economy, relative to its own optimum. Discount-implied σ^u sets $\sigma^u_{s'} = 1/d_{s'}$ in the five sectors with decreasing and convex schedules. $\alpha \times k$ and $(\sigma^u, \sigma^r) \times k$ scale the labor elasticity and the elasticities of substitution of all firms. The ξ rows recalibrate the full fee-correction fixed point using screening elasticities estimated from the ratio of employment in the top 1% to the top 10% or 25% of firms (Table A9). Final demand: national accounts uses household final consumption shares by sector from the national accounts. Common $\xi = 1.33$ sets the same screening elasticity in all sectors and layers, holding the elasticities of substitution fixed (Section 5.5). Optima are computed on a grid with step 0.005; the baseline cost differs slightly from Section 5.4 (4.9) because the uniform-pricing entry subsidy is 0.015 on this grid.

E.3 Welfare across Subsidy Combinations

Figure A5 reports welfare, as a percent of the efficient allocation, for each combination of the revenue subsidy τ^y and the entry subsidy τ^e on the grid used in Section 5.4. Both panels use the baseline calibration and the same efficient benchmark, so welfare levels are comparable across the two pricing regimes.

Under uniform pricing, the optimum lies at a large revenue subsidy and an entry subsidy close to zero, consistent with entry being distorted only through markups. Under nonlinear pricing, the optimum moves toward a smaller revenue subsidy and a sizable entry subsidy, reflecting the entry distortion created by flat fees. Furthermore, for any given revenue subsidy τ^y, the welfare gain is smaller under nonlinear pricing than under uniform pricing. In both regimes, welfare falls steeply once the revenue subsidy exceeds its optimum. Under nonlinear pricing, revenue subsidies of 55% or more leave welfare below the no-policy level, so a prescription derived under uniform pricing,

if set too large, risks leaving the economy worse off than the status quo.

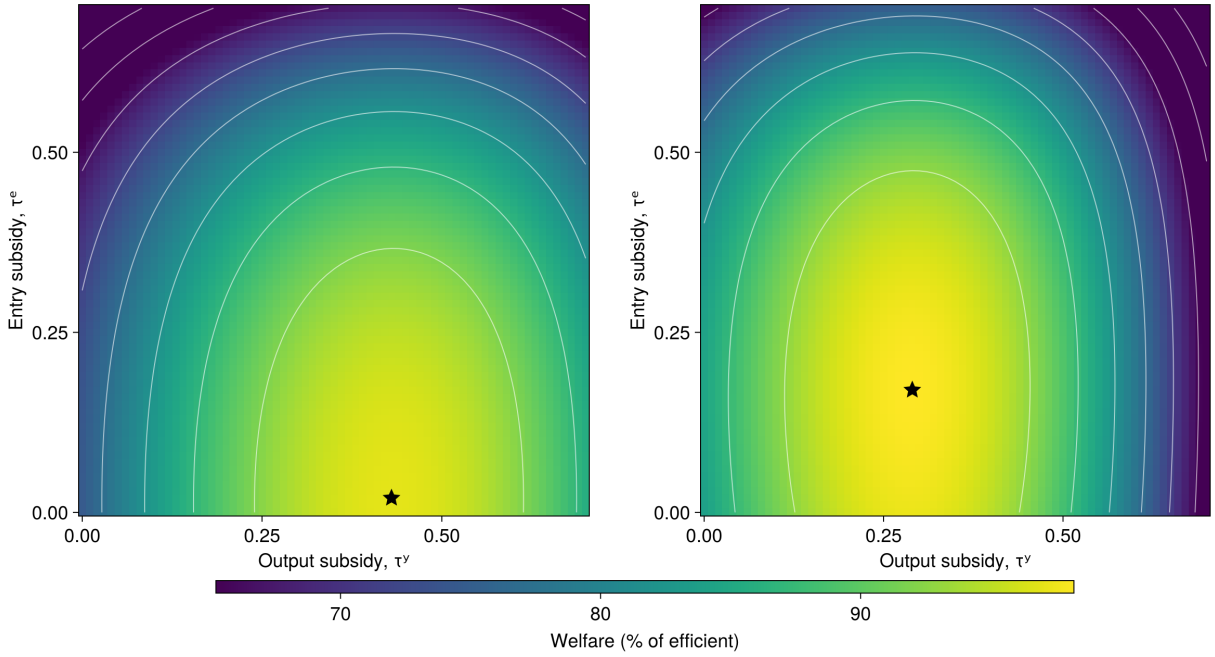


Figure A5: Welfare Effects of Revenue and Entry Subsidies

Notes: Welfare, as a percent of the efficient allocation, for each combination of the revenue subsidy τ^y and the entry subsidy τ^e, under uniform pricing (left) and nonlinear pricing (right). Both panels use the baseline calibration and the same efficient benchmark, so colors are comparable across panels. Stars mark the optimal policy, and white lines are iso-welfare contours. Policies below the no-policy level (74.5% under uniform pricing and 83.9% under nonlinear pricing) lower welfare. The color scale is clipped at the 5th percentile.

E.4 Aggregate Cost of Market Power

The main text holds the primitives fixed and changes only the pricing regime. Here we ask a different question: how large is the welfare cost of market power when a researcher calibrates a model that ignores nonlinear pricing to the same data? We calibrate the uniform-pricing model following the strategy in Section 5.1, but without correcting for flat fees. The elasticities of substitution are set from the raw revenue-to-profit ratios, $\hat{\sigma}^u_s$, and $\hat{\sigma}^r_s$, and the input-output elasticities from observed payments. All other parameters are unchanged. We then compare welfare under uniform pricing with the efficient allocation of this model. Table A17 reports the results.

Panel (a) shows that the uniform-pricing model infers much higher elasticities. The average upstream elasticity is 4.9, against 3.0 in our model, and the average retail elasticity is 5.6, against 3.1. The reason lies in how each model reads profits. By Result 4, a firm's profits equal a share $1/\sigma^\ell_s$ of revenue minus the flat fees it pays to its suppliers. A uniform-pricing model attributes the whole observed profit share to the CES margin. Since fees lower observed profits, it reads them as

weak market power.

Panel (b) shows the consequences for welfare. The uniform-pricing model attains 90.1% of efficient welfare, a loss of 10.5 log points. In our model, the status quo attains 83.9%, a loss of 17.6 log points. Ignoring nonlinear pricing therefore understates the welfare cost of market power by about 40%. The gap appears in both margins. With higher elasticities, markups are lower and compound less along the supply chain: the markup terms add up to 59.5 log points, against 82.0 in our model. The extensive margin is also much smaller, 3.4 against 16.5, mostly because upstream variety rises by only 9.5 log points in the efficient allocation, against 28.7 in our model. Lower markups depress upstream sales and entry less, so less correction is needed.

This result goes in the opposite direction from the one in the main text. There, for given primitives, nonlinear pricing reduces distortions relative to uniform pricing. Here, accounting for nonlinear pricing in the calibration reveals more market power than a uniform-pricing model infers.

Panel (c) shows that the same calibration error affects policy. The optimal policy in the uniform-pricing model is a revenue subsidy of 22% and an entry subsidy of 4%. Applied to our economy, it attains 96.8% of efficient welfare, 1.4 log points below our optimal policy. The loss is smaller than the 4.9 log points in Section 5.4 because two biases partly offset. The calibration error understates markups and calls for a smaller subsidy, while ignoring nonlinear pricing overstates allocative markups and calls for a larger one. The size of this loss therefore depends on how the two biases happen to combine at our calibration. In both cases, however, the entry subsidy is far below its optimal level of 17%.

Table A17: Welfare Cost of Market Power: Uniform vs. Nonlinear Pricing Model

	Uniform-pricing model	Our model
Panel (a). Calibrated elasticities (mean)		
Upstream, σ^u	4.86	2.98
Retail, σ^r	5.60	3.06
Panel (b). Efficient allocation vs. status quo (log points)		
Final expenditure	−52.4	−80.9
Retail markups	25.1	40.6
Retail-to-upstream markups	13.7	17.9
Upstream-to-upstream markups	20.7	23.6
Total intensive margin	59.5	82.0
Retail variety	−6.1	−12.2
Upstream variety	9.5	28.7
Total extensive margin	3.4	16.5
Total $\Delta \log \mathcal{W}$	10.5	17.6
Status quo (% of efficient)	90.1	83.9
Panel (c). Optimal policy		
(τ^y, τ^e)	(0.22, 0.04)	(0.29, 0.17)
Welfare in our economy (% of efficient)	96.8	98.2

Notes: The uniform-pricing model is calibrated to the same data without correcting for flat fees. Its status quo is uniform pricing, and ours is nonlinear pricing. Mean elasticities are unweighted sector averages. Panel (b) decomposes the log welfare change (×100) from each model's status quo to its efficient allocation using Equation (16). Panel (c) reports each model's optimal uniform revenue and entry subsidies and the welfare each attains when applied to our economy.